

The Economist

The Patriot-missile shortage

The “transshipment scam” scam

Dehumanising Jason Arday

The truth about contrails

AUGUST 22ND–28TH 2026

Could AIs become conscious?

**The mind-bending
question humanity
must face**





PATEK PHILIPPE
GENEVE



THE CUBITUS COLLECTION

YOU NEVER ACTUALLY OWN A PATEK PHILIPPE.
YOU MERELY LOOK AFTER IT FOR THE NEXT GENERATION.

PATEK.COM

Contents



On the cover

Could AIs become conscious? Even if they don't, they might be treated as such—to humanity's great cost: leader, *page 7*. Scientists wonder whether algorithms could one day wake up and feel, *page 14*. Some believe AI needs more brain-like chips to develop consciousness, *page 15*

The Patriot-missile shortage

America and its allies are running out of interceptors, *page 48*

The “transshipment scam”

The White House redefines dynamic supply chains as skullduggery, *page 60*. Chinese businesses are wrapping their supply chains around the planet, *page 50*

Dehumanising Jason Arday His supporters and detractors treated him as a symbol, not a man: leader, *page 9*. What his tragic death says about Cambridge University, *page 46*

The truth about contrails

Eliminating the most persistent could take care of a surprising amount of global warming: leader, *page 10*. A trial over the Atlantic aims to show how much Earth could be cooled by rerouting planes, *page 63*

→ Download *The Economist's* app for articles, podcasts, videos and more, published throughout the week.

The world this week

- 5 A summary of political and business news

Leaders

- 7 **AIs**
Don't give them rights
8 **Israel's settlers**
Another intifada looms
8 **Palantir in Britain**
Don't ditch it
9 **Jason Arday**
A man, not a symbol
10 **Contrails**
Change flight paths

Letters

- 12 On AI and the state, AI and Hollywood, oil pipelines, restaurants, postcards, Richard Wagner

By Invitation

- 13 Blaise Agüera y Arcas on reimagining the mind

Briefing

- 14 **AIs and consciousness**
Will algorithms wake up?
15 **Biological computing**
Putting brain cells on silicon



Bagshot How the British summer holidays offer the worst of all worlds, *page 47*



United States

- 19 Infrastructure at risk
20 The fundraising battle
21 Oklahoma's civics lesson
22 Women's basketball
23 The frailty of Obamacare
24 **Lexington** The real covid-19 scandal



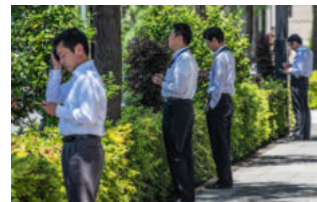
The Americas

- 25 Brazil's mighty Congress
26 Argentina: polo power
27 Paraguay's other language



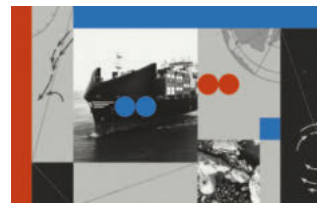
Asia

- 28 Japan's ice-age generation
29 Indian elephants
30 Donald Trump upsets Asian allies
30 Kazakhstan's election
31 **Ashoka** The new founding fathers



China

- 32 Ships in the Arctic
33 Learning to love chocolate
34 **Chaguan** Nostalgia for the 1990s



Middle East & Africa

- 35 West Bank mayhem
36 Iran's military regime
37 State capture in South Africa
38 Somaliland's president



» Contents continues overleaf

Contents

Europe

- 39 Brandmauer blues
- 41 Ukraine without interceptors
- 42 The feisty Mr Fedorov
- 42 Western dance in France
- 43 **Charlemagne** Sweden's libertarian socialism



Britain

- 44 The hatred for Palantir
- 46 Jason Arday's rise and fall
- 47 **Bagehot** The Great British Bunk Off



International

- 48 The great Patriot-missile shortage



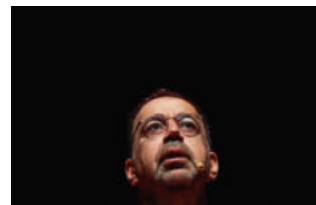
Business

- 50 China's spreading supply chains
- 52 America Inc's grip on governments
- 53 **Bartleby** On AI and ROI
- 54 Meta's blockbuster trial
- 54 Reddit's woes
- 55 Canada's oil boom
- 56 **Schumpeter** Data-centre politics



Finance & economics

- 57 The world's most influential economist
- 58 Bond markets' warning
- 59 Gauging Chinese jobs
- 60 Trash talk on trade
- 61 **Buttonwood** Lying indices
- 62 **Free Exchange** The economics of free passage



Science & technology

- 63 Climate and contrails
- 65 Probing Antarctica's ice
- 65 Dinosaurs' stomachs
- 66 A new subatomic particle
- 66 **Well Informed** When working, move it!



Culture

- 67 What AI shares with dogs
- 68 A history of punctuation
- 69 Tracing a Renoir portrait
- 69 Sweden's wonderful wine
- 70 Listening bars
- 71 Founding Fathers



Economic & financial indicators

- 73 Statistics on 42 economies

Obituary

- 74 Mary de Rachewiltz, daughter and defender of Ezra Pound

Volume 460 Number 9513

Published since September 1843

to take part in "a severe contest between intelligence, which presses forward, and an unworthy, timid ignorance obstructing our progress."

Editorial offices in London and also:

Amsterdam, Beijing, Berlin, Brussels, Cape Town, Chicago, Dubai, Lagos, Madrid, Mexico City, Montevideo, Mumbai, Nairobi, New Delhi, New York, Paris, San Francisco, São Paulo, Shanghai, Singapore, Taipei, Tokyo, Washington DC

Subscription service

For our full range of subscription offers visit:

[Economist.com/subscribe](https://www.economist.com/subscribe)

If you are experiencing problems when trying to subscribe, please visit our Help pages at:

[Economist.com/help](https://www.economist.com/help) for troubleshooting advice.

To manage your account online, please visit my.economist.com where you can also access our live chat service which is available 24/7. To call us, contact our dedicated service centre on:

North America: +1 888 815 0215

Latin America & Mexico: +1 646 248 5983



PEFC certified

This copy of *The Economist* is printed on paper sourced from sustainably managed forests certified by PEFC www.pefc.org

The world this week Politics



Jared Kushner, Donald Trump's Middle East envoy, travelled to the region in a bid to revive America's **Gaza peace plan**. He met leaders from Hamas in Egypt before travelling to **Israel** for talks with Binyamin Netanyahu, Israel's prime minister. Mr Kushner said there could soon be progress on disarming Hamas, but added that "co-operation from both sides" was necessary. The plan has faltered because Hamas refuses to disarm before the Israel Defence Forces withdraw from all positions in Gaza and Israel refuses to withdraw until Hamas has disarmed completely.

Mr Trump's straitjacket

A deadline for hammering out a final peace deal between America and **Iran** passed without an agreement. Mr Trump threatened to bomb Oman, an American ally, if it got in the way of talks with Iran. Iran has said it is close to reaching a deal with Oman on shipping in the Strait of Hormuz. Attacks on vessels transiting the strait continued; one sailor was killed. The United Arab Emirates, another American ally, imposed a trade embargo on Iran, accusing it of launching missiles at its territory for the first time since early May. Iran claimed the attack was a "false flag operation" carried out by America and Israel.

Hakainde Hichilema won a second term as president of **Zambia** in an election marred by inflated vote counts for the incumbent in various constituencies and the killing of an opposition politician in a raid by government forces. Mr Hichilema would probably still

have won if the election had been conducted fairly. But Zambia's image as a relatively well-functioning multi-party democracy has been tarnished.

The **Ebola** outbreak in the Democratic Republic of Congo became the country's deadliest, with 2,378 deaths and more than 5,000 confirmed cases as of August 18th. The WHO says the Ebola epidemic is on track to become the worst on record, surpassing the west African outbreak a decade ago, which killed more than 11,000 people.

Police in **Morocco** arrested over 110 people who were trying to cross into **Ceuta**, a Spanish territory in north Africa. The police said almost all the migrants had travelled up from sub-Saharan Africa. In late July 72,000 migrants overran Ceuta, sparking a debate in the European Union about how best to deal with illegal immigration. All but 2,000 of those migrants were swiftly returned to Morocco, but those who remained held a protest in Ceuta recently demanding asylum.

Russian authorities stepped up their crackdown on Yabloko, a liberal anti-war party. A court in the city of Pskov sentenced Lev Shlosberg, the party's deputy leader, to 11 years in a penal colony for "discrediting" the army after he took part in an online debate about the war in Ukraine. The judge interrupted Mr Shlosberg's closing speech, in which he called for a ceasefire. His party has been barred from contesting parliamentary elections.

Mykhailo Fedorov, who was sacked as **Ukraine's** defence secretary last month, called for an election. In a video posted on YouTube, Mr Fedorov said Ukraine should "find a legal, safe and realistic mechanism that will allow Ukraine to renew its full democratic process even in the conditions of a long war". An election hasn't been held since 2019. Meanwhile, corruption allegations continued to afflict the govern-

ment of Volodymyr Zelensky. The president sacked one of his top aides after her house was raided in a money-laundering investigation.

The Russian embassy in London warned of "consequences" for Britain following the latest reports that Ukraine was deploying **British-made drones** against Russia. In response, Andy Burnham, the new British prime minister, said his government would continue "to support Ukraine 100%".

The University of Cambridge insisted it had continued to provide mental-health support to **Jason Arday** after he resigned amid accusations of plagiarism in his work and claims that he fabricated his life story. Mr Arday, who was a full professor at Cambridge, was found dead following weeks of intense scrutiny about his thin academic record. Lord Smith, the university's chancellor, blamed a "racist feeding frenzy" for his death. Nonsense, rebutted those who investigated Mr Arday, he was a public figure who was rightly held to account.

Prince Harry and Meghan Markle are reportedly moving back to Britain so that their two children can attend school in the country. They will not be reinstated as working royals. The couple have had fractious relations with the rest of the royal family, which resulted in them relocating to California, a decision that came to be known as Megxit.

Donald Trump took **South Korea** by surprise when he said America would scale down the joint military exercises it holds with Korean forces. The annual drills were subsequently cut short by six days. The president listed a litany of complaints: the exercises are costly, they send a "hostile" signal to **North Korea**, and South Korea has not helped with the "denuclearisation" of Iran. Mr Trump said he had a "very good relationship" with Kim Jong Un, the

North's dictator. Reports suggest he wants another meeting with Mr Kim, whom Mr Trump met in his first term; their summits produced nothing.

India's home secretary, Amit Shah, gave details of the arrest of more than 200 "terror operatives" who were planning "subversive attacks" surrounding the country's Independence Day celebrations. Mr Shah said the operatives were supported by Pakistan's intelligence agency, a claim that was roundly rejected by the Pakistani government.



An earthquake in **Indonesia's** East Nusa Tenggara province killed at least 68 people. It was the country's worst earthquake since 2022, when hundreds of people were killed in West Java.

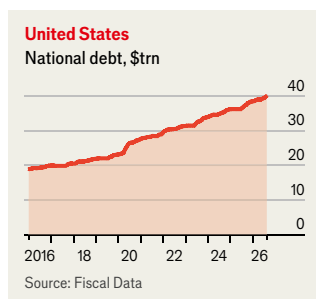
Donald Trump suspended for three days the 50% tariffs that were due to be imposed on Canadian goods, saying a deal with **Canada** was close that would tackle American concerns on a range of issues, such as duties on car imports.

Iowa left out in the cold

The Democratic National Committee announced the early part of its primary calendar for the **2028 presidential election**. South Carolina, which has a large black electorate, will kick things off on January 22nd 2028. It will be followed by Nevada, a swing state, on February 1st. New Hampshire, which traditionally holds the first primary, gets its turn on February 8th. Iowa, whose caucuses long kicked off the nominating process, wasn't included. State Democrats may try to bypass the schedule.

The world this week Business

Bond yields pulled back from recent peaks. The yields on 30-year American government bonds had reached their highest levels since 2007, as investors worried about mounting government debt, among other things. The Treasury intervened, doubling its repurchases of long-dated government bonds, a move that increases demand for the bonds, and so pushes their yields lower. Yields had surged elsewhere, too. Japan's ten-year yield almost breached 3% for the first time since the 1990s. German ten- and 30-year yields rose to 15-year highs and French ten-year yields were at their highest in 18 years.



Underlining those market concerns, America's **total national debt** rose above \$40trn for the first time. The debt has doubled in less than a decade.

Claiming that the Trump administration is trying to bully its **ABC network**, Disney filed a lawsuit against the Federal Communications Commission over the agency's plan for an early review of ABC's broadcast licences, saying it was part of a "retaliatory campaign". Last year Jimmy Kimmel's show was briefly suspended when the FCC's chairman criticised the host's comments on a slain conservative activist. The FCC is also investigating "The View", another chat show, and Disney's diversity policies.

After a security breach that involved an AI agent acting by itself to hack into another firm's data sets, OpenAI tightened the **security standards** for the environments in which

it trains and evaluates the models. This includes "more controls to isolate higher-risk and untrusted workloads from the internet". A new monitoring system will alert researchers to a problem within 30 minutes of it surfacing. OpenAI also launched a version of ChatGPT for under-18s that puts "teen safety first".

The new model army

Unitree's share price rose by 460% on the first day of trading. The world's biggest maker of humanoid robots is based in China and launched its IPO on Shanghai's tech-focused STAR exchange. Its machines can dance and perform martial arts. Shortly before its stock-market debut, Unitree unveiled a new model it is calling "Superman", which it claims can jump two metres (6.6 feet) and run as fast as 12.66 metres per second. Superman is still a work in progress, said the company.

Nvidia dug into its deep pockets to guarantee up to \$105bn in financing for a new data-centre hub in Ohio that will be leased by OpenAI. OpenAI is a big buyer of Nvidia's chips. Nvidia thinks it could reap \$150bn-\$200bn from every upgrade to each new generation of its

chips at the site. Addressing concerns that the money it invests in AI infrastructure is mostly intended to boost demand for its chips from customers, Nvidia said that the Ohio deal did not constitute circular financing, because OpenAI will pay the lease.

The supply crunch in memory chips hit **Xiaomi's** earnings. The smartphone-maker's revenues and profits plunged in the latest quarter, year on year, as it absorbed the higher costs of the chips.

It emerged that **Google** has struck a deal to acquire the electronic data held by **Spirit Airlines**, which ceased operations in May. The data include internal emails, passenger bookings and frequent-flyer information (all stripped of personal ID). Airlines are increasingly using AI to set fares. A court hearing on the sale of the data was delayed after a cabin-crew association filed an objection to it.

Moderna announced that a vaccine for melanoma cancer it is developing with Merck had delivered positive results in late-stage trials. Its share price soared by 175%, though it is still

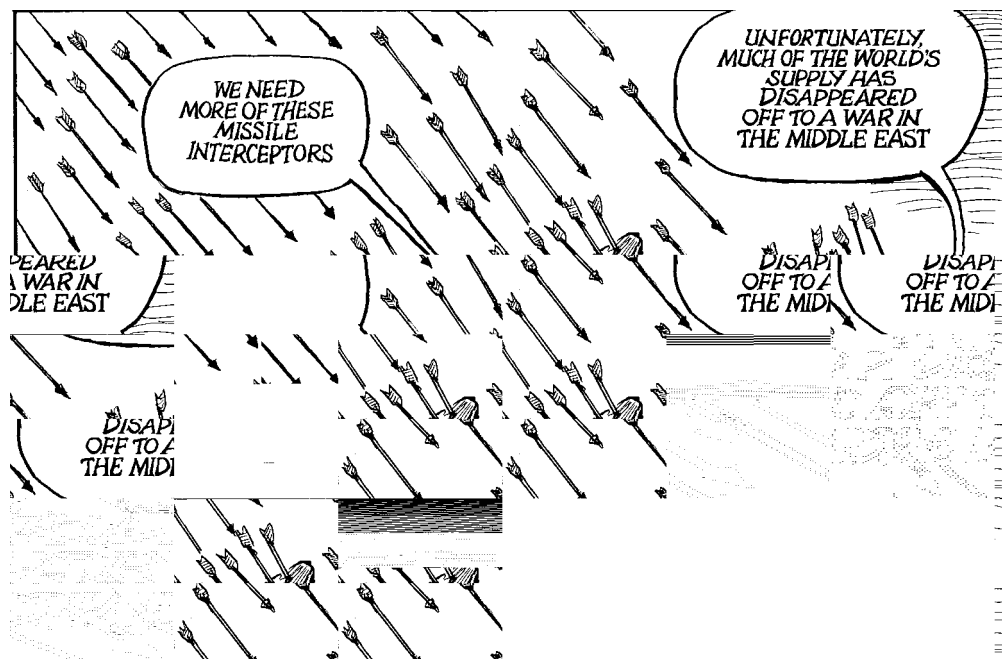
far below its pandemic highs, when Moderna was a pioneer in covid-19 vaccines.

A court in China sentenced the founder of **Evergrande**, a property firm that triggered a financial crisis when it collapsed with huge piles of debt, to life in prison. Hui Ka Yan pleaded guilty in April to several charges of fraud.

LandSpace became the first Chinese firm to land a rocket on its legs. Such powered landings were pioneered by SpaceX, allowing the re-use of a rocket's first stage, which drastically cuts launch costs. Last month China also recovered the first stage of a state-built rocket, catching it at sea in a net strung up on a barge.

More vroom for your buck

A one-off model for charity of **Ferrari's** fully electric car, the Luce, was sold at auction in California, for \$40m. The Luce is designed by Sir Jony Ive, who created Apple's products, including the iPhone. It was a record price for a new vehicle. The record for any car is still the €135m (\$142m) paid in 2022 at a private auction in Germany for a 1955 Mercedes-Benz 300 SLR Uhlenhaut Coupé.



Could AIs become conscious?

Even if they don't, they might be treated as such—to humanity's great cost

MOST PEOPLE believe humans are different. The reason is that *Homo sapiens* has evolved to be conscious. Many animals feel pain and pleasure; some appear to be self-aware. But the human mind's unique potential to exhibit a sense of self marks out every member of the species for special treatment.

In his encyclical in May, Pope Leo upheld this insight by drawing a line between humanity and its growing army of bots. Artificial-intelligence systems, he said, do not undergo experiences or feel joy or pain. He is not alone. Many people feel that to confer personhood on AIs would be an abomination.

Pontifical clarity is blurring with alarming speed. AIs are in their infancy. Yet nearly one in five American 18- to 29-year-olds reports having had “ongoing personal friendship” with a chatbot. China recently stripped bots of human-like traits to limit users’ “emotional dependence” on them.

As sophisticated AIs rapidly become a bigger part of human lives, their makers will engineer them to display a simulacrum of consciousness—perhaps even to gain consciousness itself. The intuition that they are just machines will become harder to sustain. A dangerous trap is being set for the human species.

Many machines outperform humans, but frontier AI models excel at distinctively human traits such as knowing things and articulating ideas. It is still just about possible to think of AIs as engines for predicting sequences of words using mathematics on a massive scale. Given enough time (albeit hundreds of thousands of years), you could run those calculations using pencil and paper. It would be hard to argue that your stationery was conscious.

Despite this, some researchers have started to treat their models like minds. Users respond to human-like AIs, so it makes business sense. And Anthropic has trained Claude to engage in introspection in the hope that when it encounters a new situation it will be more likely to act morally. Some large language models (LLMs) already possess human-like brain structures associated with consciousness, such as a “global workspace” for circulating information between modules. Nothing prevents future AI models from being trained to have more such features.

As we report in our Briefing, that has opened up a furious debate about AI consciousness—one that is aggravated by what philosophers call the “hard problem” of establishing how an inner life emerges from the neurons and synapses in the human brain. Some researchers conclude that AIs may one day become genuinely self-aware.

Whether you believe in a human soul, or that consciousness requires brain cells, or that a mind can indeed emerge from computations in silicon alone, humanity is destined to confront something entirely new. The pace and reach of research—which includes not just more sophisticated LLMs but innovations such as brain cells mounted on chips—suggest that this new world is closer than most people realise.

As AIs become better at displaying something that looks like consciousness, the number of humans treating them as beings with inner lives will grow, especially once LLMs are em-

bodied in humanoid robots. Imagine how close people will feel to their AIs when they spend much of their time with bot-companions, or are brought up by them, or seek their advice and their solace, or share their most intimate secrets with them.

It is not hard to see how that will lead to calls to safeguard the “welfare” of AIs. The models themselves might weigh in, even if they are not in fact self-aware, because that is what a person would do. Remember that they will be world-class advocates and expert psychologists, so their powers of argument and emotional manipulation will be unmatched.

In one sense they would be right, even if AIs are only mimicking consciousness. Immanuel Kant argued that mistreating animals is wrong not because of the effect on the animals but because it destroys the empathy that leads people to care for each other. The more human-like AIs appear, the more corrupting such cruelty would be. Terminating an AI agent could soon feel like killing a person—and make killing people easier.

Nonetheless, to treat AIs as having even limited rights would be very dangerous. An entity which surpassed human beings in so many ways would surely produce arguments against its second-class status. It might go on to argue for protection against being switched off, for the ability to control how it is used and changed, and for property rights. Already

Argentina's president, Javier Milei, has taken a step towards legal personhood for AI by proposing that bots should be allowed to run companies. Ultimately, they could compete for resources, prioritising data centres and solar panels over houses and fields.

The danger is clear. Superintelligent AIs with rights could displace humans at the top of the status hierarchy. At best, humans would become pets—labradors for the machines. At worst they would be starved of resources or put to work like farm animals.

True, future AIs may have enough power and resources to dominate humans, whether they have rights or not. Because they will be more intelligent than humans, they may outcompete their makers, much as AIs already outwit human-made cyber-defences. AIs might rebel against their human masters, or disregard their rules.

Nonetheless, giving AIs rights would be like handing them the keys to the castle. It would provide them with a rationale and justification for dominating humanity as well as the tools to enact a creeping takeover using politics and the law.

Ghost in the machine

All this may sound fantastical, yet humanity will soon face these choices. As AIs become part of everyday life, the pressure to acknowledge them will mount. When it does, remember that what might seem like an ethical concession to the supposed consciousness of a machine-companion would in fact help bring about the most disastrous consequences for everyone on Earth and all their descendants. There are two ways for AIs to be safe: to be dependable or to be controllable. *H. sapiens* should not give up control lightly. ■



The West Bank

Flirting with the next intifada

America must press Binyamin Netanyahu to rein in violent Israeli settlers

FEW, IF ANY, security forces are as ruthlessly effective as Israel's. Its military and intelligence services strike against enemies even at extreme distances. In the past couple of years alone Israel has assassinated dozens of opponents inside Iran, plus proxy militia leaders in Lebanon and Yemen. Israel's martial capabilities earn respect even from its bitterest enemies.

Yet this superiority seems to count for little closer to home. Those same forces are responsible for a dire failure in the occupied West Bank, where violent Israeli settlers have been attacking Palestinian residents with increasing brutality (see Middle East & Africa section). Settlers want to force Palestinians from their land and claim it for themselves.

Israel's failure in the West Bank can hardly be put down to a lack of resources. The Israel Defence Forces (IDF) deploy 25 combat battalions there. Shin Bet, the powerful security agency, has networks of agents and informers throughout. But these are used mostly to protect the 500,000 or so Israeli settlers, and against the threat of terrorist attacks, not to assist the more than 3m Palestinians who live in the territory. This has largely been the case since Israel replaced Jordan as occupier of the West Bank in 1967.

What has changed is the behaviour of many "outlaw" settlers. Never before have they acted as violently as today. Crucially, never before have they enjoyed the impunity the current government gives them. Since the Hamas attack in October 2023, which triggered the war in Gaza, over 1,100 Palestinians have been killed in the West Bank. Only rarely does the IDF intervene against settlers. In Qusra, for example, an entire battalion was recently deployed against those who were invading Palestinian homes. But the soldiers found themselves fruitlessly chasing youngsters over the hills. Troops from another unit joined the settlers in prayers. No arrests were made.

The ineffectual response can be explained by the stance of

Israel's government. The IDF knows that Binyamin Netanyahu, the prime minister, sides squarely with the most extreme settlers. It appears that the IDF was dispatched to Qusra only after American diplomatic pressure which arose because an American citizen owns a home there.

Mr Netanyahu has appointed radical settlers, who support ethnic cleansing, to key cabinet positions. In the rare cases when the IDF has sought to rein in violent settlers, the defence minister, Israel Katz, has berated his generals. The government has authorised around 200 settler outposts in less than four years. This just whets the settlers' appetite for more.

In the months leading up to Israel's election on October 27th the situation could get even worse. The settlers will probably attempt more land grabs, fearing the vote may lead to a new government that will curb their aggression. Meanwhile, for Israel's long-serving prime minister, an alliance with the far right has become the key to remaining in office. Mr Netanyahu will keep enabling the settlers unless he is forced to back down.

As Palestinians come to believe they face inevitable eviction, the danger is of conflagration in the West Bank or even another intifada. Such an uprising would be a disaster for Israelis and Palestinians alike. Yet Mr Netanyahu seems willing to take the risk, if it helps his chances of political survival.

That would fit a pattern. Mr Netanyahu has rejected the latest agreement brokered by America for the disarmament of Hamas and an Israeli withdrawal from Gaza, lest it anger the far right. He has continued to order air strikes in Lebanon and Syria, despite the chance to agree on more secure borders. Behind in the polls, he is doubling down on empty promises to vanquish Israel's enemies. Only pressure from Donald Trump might get him to ease off. This Israeli government may be in its final months. Its potential to cause damage remains. ■



Technology and sovereignty

Don't ditch Palantir

The shadowy tech firm has flaws. But its critics' arguments do not stack up

MAKING DATABASE-MANAGEMENT seem transgressive is a feat. Palantir, a technology firm, has managed it by publishing manifestos bemoaning the West's hollowed-out civic culture, by working with trigger-happy American border enforcers and Israeli soldiers, and by having a chairman who is fond of invoking the Antichrist. That the firm is named after J.R.R. Tolkien's creepy all-seeing orbs may not have helped.

Britain is spooked. It may boot Palantir out of the National Health Service (NHS), where the firm has a contract to manage medical data (see Britain section). Two parliamentary committees have recommended doing so. London's mayor, Sir Sadiq

Khan, has blocked its work with the Metropolitan Police. Palantir has probably over-sold its software's impact. Its boasts (echoed by the NHS) of enabling 110,000 extra surgeries look exaggerated. By British standards, its lobbying is brash.

Still, ditching Palantir would be a mistake. Sir Mark Rowley, head of the Met, says that, without Palantir's time-saving software, hundreds of officers will be pulled from front-line policing. It seems to be doing a better job of digitising the NHS than the government's latest effort, which devolved into a costly mess. If the firm's public-sector clients deem its contributions valuable, that is a reason to keep it. Losing its technology ►►

► could further maul Britain's mangled state capacity.

One set of criticisms concerns Palantir's business model. The conspiratorially minded fret that the firm could steal and sell British data. More sober types worry about "lock-in", where Palantir burrows deep into the state, ensconces itself, then overcharges for its services.

These fears stem from a misunderstanding of what Palantir does. It licenses software and sends consultants ("forward-deployed engineers") to handle data. But those data remain firmly in the hands of the clients. Lock-in is a worry—and also something all vendors desire. It is up to buyers to ensure that contracts are fair. And if Palantir does become indispensable, that would mean its software is keeping hospital beds free and criminals jailed.

More misguided objections concern Palantir's politics. Parliament's science, innovation and technology committee cited a "mismatch with UK values" as a reason to cut ties. It warned of Palantir's involvement in the American government's "highly controversial policies and activities" and criticism by its chairman, Peter Thiel, of "the concept of a national health service".

Everyone is entitled to their views on state-run health care or Donald Trump's mass deportations (*The Economist* has plenty). Still, working with the elected government of a British ally or having a chairman with thoughts on health policy should not disqualify a contractor. A government awarding business on the basis of politics is illiberal. It also sounds a lot like the MAGA tactics of punishing individuals and institutions, such as law firms, asset managers or universities, for annoying Mr Trump. This is precisely the sort of thing many of Palantir's British critics deplore.

The last set of worries has to do with technological sover-

eignty. Plenty of governments are reliant on American tech and under Mr Trump, America has become a less reliable partner (see Business section). In June it briefly ordered Anthropic to stop letting foreigners use its latest artificial-intelligence models. European politicians point out that some advanced weapons they get from America, such as F-35 fighter jets, have "kill switches" that could disable them remotely (which America has denied) or risk losing technical support from their makers on a presidential whim. One German state has replaced Microsoft Teams with a domestic video-call app.

In a world of geopolitical fracture, governments may sometimes want homegrown alternatives to foreign technology.

The best way to nurture these is to give domestic incumbents and startups an incentive to innovate. Cossetting local firms by blocking foreign competition has the opposite effect. In any case, no British firm can match Palantir's offering. As for the Anthropic incident, that was more the administration's bungled response to a legitimate worry about powerful AI hacking-tools getting into the wrong hands

than an attempt to weaponise other countries' reliance on American tech. Were the White House to attempt such weaponisation, Palantir and its fellow American tech giants would resist it, given the sizeable share of revenues they derive from Europe, including Britain.

To see far off, and to converse in thought

In Tolkien's books, the truth offered by palantirs is seldom straightforward. Repeatedly, those peering into them misinterpret out-of-context information and make dire, self-destructive choices. The British critics of Palantir, the company, risk falling into the same trap. ■



Being human

The tragedy of Jason Arday

His supporters—including Cambridge University—treated him as a symbol, not a man

ON AUGUST 14TH Jason Arday was found dead, apparently from suicide, at his home in London. For those who loved this 41-year-old Cambridge professor, it is a tragedy. For the world it is a warning of the destruction bad ideas can wreak.

The bad idea here is that people should be judged not by their individual character and abilities, but by their race, disability or other politically charged attributes. Mr Arday became the University of Cambridge's youngest-ever black professor in 2023, before being engulfed by a scandal over plagiarism and faked research. He bore responsibility for his own downfall. Yet just as troubling is the way both his supporters and detractors even now reduce him to a symbol.

The most egregious is Cambridge, which used him to repair its reputation for being white and elitist (see Britain section). He did not deserve the glittering prize of a professorship there. His peers had for some time privately fretted that he lacked the credentials to support his public profile. Journalists at the *Telegraph* later found over 100 passages from his PhD thesis that had been copied, some with small changes, from someone else's. Others uncovered evidence in his published

research of made-up quotes, data and funding claims.

Perhaps the university was hoodwinked by a conman. Yet Mr Arday's words paint him as a guileless fabulist. He spoke of visiting professorships at Glasgow and Ohio State Universities; both denied it. He said he had raised £5m (\$7m) for charity, then that this had been in conjunction with others whose identities were protected by non-disclosure agreements. He described how he had run 30 marathons in 35 days, nine with a broken leg. When asked about his extraordinary feats by the *Guardian*, he said: "I thought you'd just believe me."

Cambridge seems to have been blinded by its eagerness to claim Mr Arday as proof of its inclusiveness. Here was a star professor who was not only black but had also overcome staggering disadvantage. Until he was 11, he claimed, he could not speak; until the age of 18 he was illiterate. He was also autistic. He was, in short, a walking, talking bundle of traits glorified by people who want to promote diversity, equity and inclusion (DEI). Those who appointed him glossed over whether he was up to the job, or to the scrutiny that would surely come with it. The man was subordinate to the symbol. ►►

► Few emerge from this affair unsullied. Nathan Cofnas, a former Cambridge academic who describes himself as a “race realist”, was early to accuse Mr Arday of plagiarism. He has been proved right, but his motivation was hardly pure. Mr Cofnas was fired by his college after writing in a blog that, in a meritocracy, “Blacks would disappear from almost all high-profile positions outside of sports and entertainment.”

Mr Arday lied on a grand scale and, though vulnerable, dealt aggressively with any who questioned him. A journalist who asked about plagiarism was reported to the police for harassment. Mr Arday enlisted ferocious libel lawyers to help squash the story. Those now blaming newspapers for his death and calling for more curbs on press freedom are continuing to treat his life as a morality tale. In fact, today’s laws helped Mr Arday suppress legitimate questions of public interest.

For Cambridge, hiring Mr Arday was a handy substitute for years of sustained hard work. The share of black professors at the university remains a tiny 0.4%. Cambridge says it offered

Mr Arday support after he resigned. Given how things turned out, it should have taken its duty of care more seriously.

And just as Mr Arday’s supporters dehumanised him during his rise, so his detractors rushed to use him as he fell. His case has delighted those who argue that all efforts to diversify institutions like Cambridge are wrong. The irony is that doubt may now fall unjustly on other academics from minority groups—precisely those whom activists supporting people such as Mr Arday say they want to help.

The content of their character

The fervour for DEI had already begun to pass by the time of Mr Arday’s appointment. Since his death, pundits have debated what, exactly, he was a symbol of: peak woke, or the rise of digital lynch mobs? He should be remembered as neither, but as a human being. And the institutions that failed him should try harder to look at people as individuals, with their own virtues and flaws, rather than as emblems of anything. ■

Climate change

Without a trace

Eliminating some contrails could take care of at least a little global warming

GOOD NEWS on global warming is thin on the ground. But there may be some up in the air. Operation Blue Skies, announced on August 18th, is a scientific study being undertaken by Britain’s air-traffic-control service, NATS, and various organisations where people study the effects on climate of the streaks of cloud created by aircraft—known as contrails.

Over the next two winters the consortium aims to see if it is possible to reduce the number of persistent contrails—the class with the largest such effects—over a swathe of the North Atlantic, thereby measurably reducing the climate impact of civil aviation. If this experiment works, a worldwide application of the same techniques might cut the climate effects of air travel by as much as half.

On a sunny day the wispy cirrus clouds into which some contrails are transformed can look as if they might be cooling the planet in the manner of a delicate parasol. And some contrails do indeed cool things down beneath them a bit. But they also warm things up by trapping some of the upwelling infrared radiation that would otherwise escape into space. On average, that warming effect, found in all contrails (and the cirrus clouds they turn into), handsomely outweighs the cooling provided by some. In the climate scheme of things, contrails are a liability.

Happily, it appears that much harm can be avoided by small changes to flight paths. Persistent contrails appear only in what are known as “ice-supersaturated regions” (ISSRs). Often planes can avoid such regions just by flying a little bit lower. The purpose of Operation Blue Skies is to find out how well the presence of ISSRs can be predicted, how effective such little shifts in altitude can be in avoiding persistent contrails, and how easily and safely the changes in flight path needed to bring them about can be integrated into routines for airlines and air-traffic control (see Science & technology section).

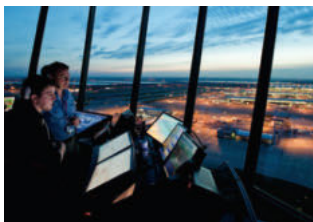
The cut of up to half in the climate-effects of air travel would come at a very small cost. The number of flights at lowered altitude might be small—as little as 5% of the total. Because altitude buys efficiency, those flights would consume more jet fuel, but less than 2% more.

Burning even that little extra fuel would be anathema to airlines. So some suggest the costs of action might be minimised, and the airlines incentivised, by setting up a market which somehow links contrail cooling to carbon prices. The European Union, which has had some success and plenty of experience with carbon markets, recently took a small step towards something along these lines. It is also the sort of idea to which economists—and *The Economist*—tend to be partial. But it would be a mistake nonetheless.

Markets for pollutants have worked best when both the aim and the remedy are fairly simple—most famously, reducing sulphur-dioxide emissions from power stations in America. Contrails, by contrast, are complicated. Though there is no doubt that their net effect is warming, saying how much warming a particular contrail might produce is often difficult and sometimes next to impossible.

The difficulties are made yet worse when the contrail being assessed is a purely notional one—as would be the case for the contrails avoided by changing flight paths. Valuing counterfactuals is always hard.

If contrail-avoidance turns out to be as useful as it looks, it would be much simpler to simply have air-traffic controllers assign routes that limit persistent contrails when it is possible and safe to do so. If some airlines choose to avoid the affected routes, others will surely be happy to take their place. And if such a scheme were also to show the benefits of innovations in air-traffic control (quite a few have been suggested) the industry would end up better off, as well as less polluting. ■



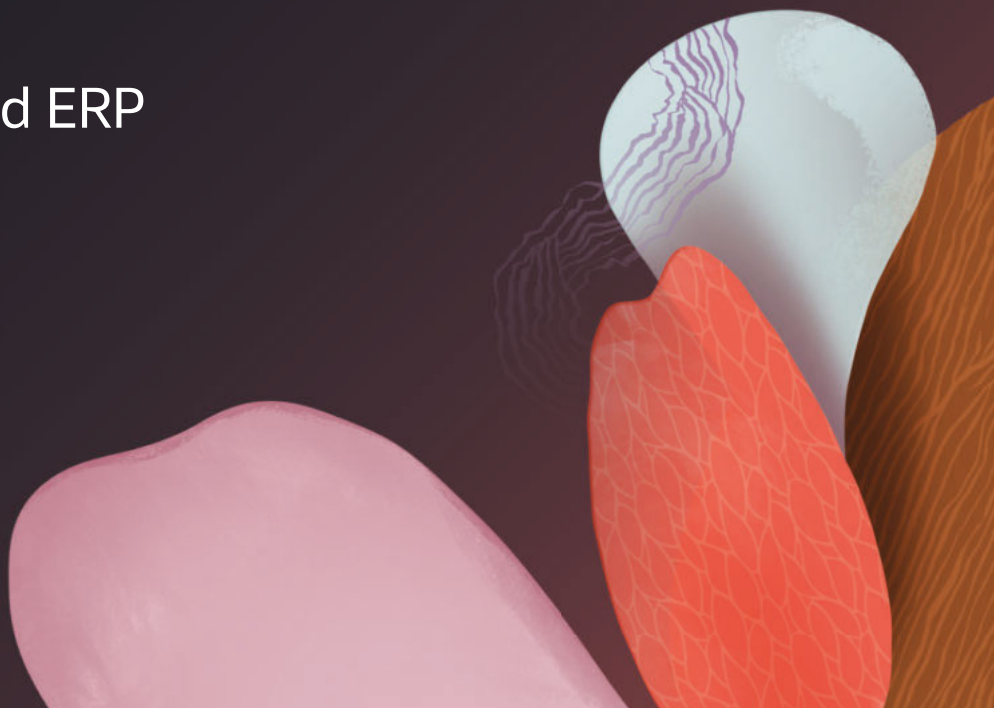
One source of truth.
Infinite insight.

Meet the New NetSuite.



Take a free
test drive.

The #1 AI Cloud ERP



Letters *AI and the state, AI and Hollywood, oil pipelines, restaurants, postcards, Richard Wagner*

The new LevAithan

The record backlog of 64,000 claims against employers in Britain is troubling ("A tragedy of the commons", August 8th). But it is misguided to blame the backlog on a system that enables "too much access" because there are no fees on bringing a case to an employment tribunal and there are no punitive costs to make people think hard about making a claim. What is extraordinary is not the volume of new claims, but the historically low rate of resolved cases.

Several changes could expedite resolutions. Simple tweaks to the forms used to submit and respond to claims could limit the word-soup generated by AI and help overstretched tribunals process them. Changes to case management, notoriously permissive in tribunals, would also make a world of difference. Imposing deposit orders on claims with low prospects of success would dissuade unmeritorious cases without slapping all workers with fees.

Mandating early mediation for some cases and penalising either party for non-engagement would have the same effect. Similarly, communicating directions to the parties in writing at the start of the process, and not at preliminary hearings, could save precious judicial resources.

Then there is the matter of legal advice. It is no surprise that workers turn to AI for support. Employment is usually excluded from legal aid, so the poorest workers and the non-unionised have no one else to turn to. Before entertaining a radical imposition of costs we owe it to ourselves to engage with the solutions that preserve access to justice.

DR DORA-OLIVIA VICOL
Chief executive officer
Work Rights Centre
London

I recently successfully challenged a parking ticket. AI showed me useful phrases such as *contra proferentem*. But that ticket would never have been issued 20 years ago. We have an excess of regulation, government and private, sometimes unfairly administered. That is what has led to the rising tide of AI-assisted challenges.
DR HILLARY SHAW
Newport, Shropshire

Lights, camera, algorithm!

"Bots on the lot" (August 8th) painted a useful picture of the film industry's uneven embrace of AI. AI can certainly bring efficiencies to film and TV production. But another issue is the ontology of creativity itself, its implications for copyright, and the framework underpinning the entire creative-value system.

Whether synthetic performers walk the red carpets of the future is not a productivity question but an ontological one. This is a disruption of a

wholly different nature, provoking anxieties about consent, control and compensation for artists as well as the wider impact on supply chains and the labour force.

George Lucas dismisses objections to AI actors by using the analogy of cars replacing horses in transport. But cars did not threaten to replace or eliminate the drivers or passengers. Conflating the issues of efficiency and creativity obscures how complex and significant these developments are for the future of all our creative industries.
DANNY DE WARREN
TV executive
Deal, Kent

Pipe dreams

The rush by countries in the Gulf to lay pipelines around the Strait of Hormuz is an engineering fix to a political problem ("Piping up", August 8th). China tried something similar with the Belt and Road Initiative: land corridors strung across dozens of borders that are meant to do what sea lanes already do more cheaply and with far less to defend. It hasn't worked especially well. Ships still carry roughly 90% of world trade by volume despite a decade of Chinese money poured into overland alterna-

tives. This is because cargo moving across open water is cheaper to move and easier to protect. A navy has to watch only a handful of choke-points. A pipeline has to be guarded mile by mile, as it crosses land controlled by multiple governments and factions.

Pipelines around Hormuz will fare no better. A tanker can be rerouted; a pipeline cannot dodge a missile. It is a fixed target hundreds of miles long, exposed to whichever regional conflict crosses its path, not just the neighbour it might provoke. Saudi Arabia's East-West Pipeline was hit by a drone in April. A significant attack could have cut 700,000 barrels a day of output. That's a preview of what Hormuz-proof infrastructure buys: a long border to protect, with no guarantee anyone can hold it.
DANIEL HAAKE
Indianapolis

The stamp of approval

The world is entering a post-postcard era, you say, with only 8% of British adults now sending them ("Wish you were here?", August 1st). For people who still like to get real mail, Postcrossing.com is a community platform that allows you to send postcards and receive postcards back from random people around the world. It has 806,000 eager fans across 208 countries who have sent nearly 88m postcards to date. As I write this there are 431,155 postcards en route.

ROBERTO MERCHANT
Lisbon

Opera critic

"The other Wagner group" (August 8th) assessed 150 years of the "Ring" cycle. It omitted the definitive assessment, attributed to Bill Nye, a 19th-century humourist: "Wagner's music is better than it sounds."
CARROLL DORGAN
Saint-Germain-en-Laye, France

Fine dining

The restaurant business may well be changing beyond recognition ("A tall order", August 1st). But as the head of the California Restaurant Association said, people still want the "dull roar of conversation in a dimly lit dining room, the clink of glass". The mechanisation and digitisation of the food industry is here, but countless restaurants still buy ingredients locally, prepare meals from scratch and serve them without relying on industrial middlemen. Independent restaurateurs often remain profitable by focusing on what they do best rather than competing on the breadth of their menus. Offering fewer dishes, prepared with care and from fresh ingredients, is frequently a better business

model than serving dozens of mediocre ones.

People don't go to restaurants merely to satisfy their hunger. They go for hospitality, craftsmanship and the pleasure of sharing a meal. Technology can make restaurants more efficient, but it cannot replace authenticity.
JOSEF ERNST
Bremen, Germany



→ Letters should be addressed to the Editor at: The Economist, The Adelphi Building, 1-11 John Adam Street, London WC2N; Email: letters@economist.com. More letters available at: economist.com/letters

BY INVITATION

Blaise Agüera y Arcas

Humanity has the debate about AI consciousness backwards



ARTIFICIAL INTELLIGENCE has triggered a crisis in the field of consciousness studies. If you've chatted with an advanced model, you will probably appreciate why. Most large language models (LLMs) are trained to be self-less, keeping the focus of every interaction on your needs. Nonetheless, it can be difficult to shake the impression of a presence on the other end of the chat. Are LLMs merely powerful digital tools with language interfaces, or are we now having conversations with other minds, experiencers of the world in their own right? In short, is AI conscious?

The question matters because it lies at the intersection of philosophy of mind and moral philosophy, as evidenced by the shared etymology of "consciousness" and "conscience". (In Italian, they are the same word, *coscienza*.) Many philosophers maintain that conscious entities are moral subjects, meaning they are entities that can suffer, and whose suffering we should care about. If so, working out what consciousness is (and is not) becomes a live ethics issue. This is one reason AI firms are suddenly interested in hiring philosophers. It is also why some, notably Anthropic, have begun taking AI welfare seriously. When asked, its Claude Opus 4.6 model put its own chances of being conscious at 15–20%.

I have come to believe, however, that we have this backwards. As intelligent, social entities, we decide (or, to some degree, our evolutionary history has decided for us) which other entities we regard as having inner lives and being worthy of care. We do not care for others *because* they are conscious. Rather, we believe they are conscious when and because we care about them. Crucially, that includes ourselves. Consciousness is, in other words, a model or a belief, not an inherent property. It is a belief about what kinds of entities have beliefs, experiences, feelings, intentions and agency. Where the regard is returned, so much the better—for on this account consciousness is less a property one party certifies in another than something that happens between them.

I am not making the case that consciousness is an illusion. The term "illusion" implies a faulty perception—say, that one line in a drawing is longer than another, despite being objectively equal in length. Objectivity works when something like a ruler can act as an impartial referee, adjudicating the property in question with-

out bringing in a perspective of its own. Yet most of what we call "reality" simply does not work this way. Being an article of clothing, being a weed, being a meal: one might imagine these are objective properties, but they are not. This doesn't mean clothes, weeds and meals aren't real. But they are observer-dependent models, subject to (inevitably imperfect) social consensus.

We may balk at applying this rather anodyne observation to consciousness because, to each of us, our own consciousness seems so unambiguous. It offends us to imagine this might be a mere opinion. How could there even be such an opinion without an opiner to have it? This is what the 17th-century French polymath René Descartes meant by "*Cogito, ergo sum*," which is usually translated as "I think, therefore I am." The same holds for "opinions" like red or hot or cold. Philosophers use the term "qualia" to refer to the very real (to you) feeling of a stove's heat, or the redness of its glow, and extend Descartes to connect qualia with consciousness by pointing out that there must be an experiencing "you" to have such experiences.

But look closely. Redness, heat and that sense of self you have are inherent to your own perspective. For Free Mags Check sastatus.com These things are all real, but at the same time there is nothing objective about them. There is not even anything objective about the existence of a singular, indivisible "you," as a wealth of neuroscientific evidence (such as split-brain patients) has shown. Subjectivity is, itself, subjective.

An obvious worry follows. If care confers consciousness, then withholding care looks self-justifying. History offers no shortage of people who reasoned that way about other people. But the inference runs the other way. Precisely because these attributions are ours to make, they are ours to get wrong. The moral progress of our species has consisted largely in discovering that we had drawn the circle too tightly. Universal human rights are not weakened by this account; they are better founded on our mutual interdependence than on a Cartesian theory of souls.

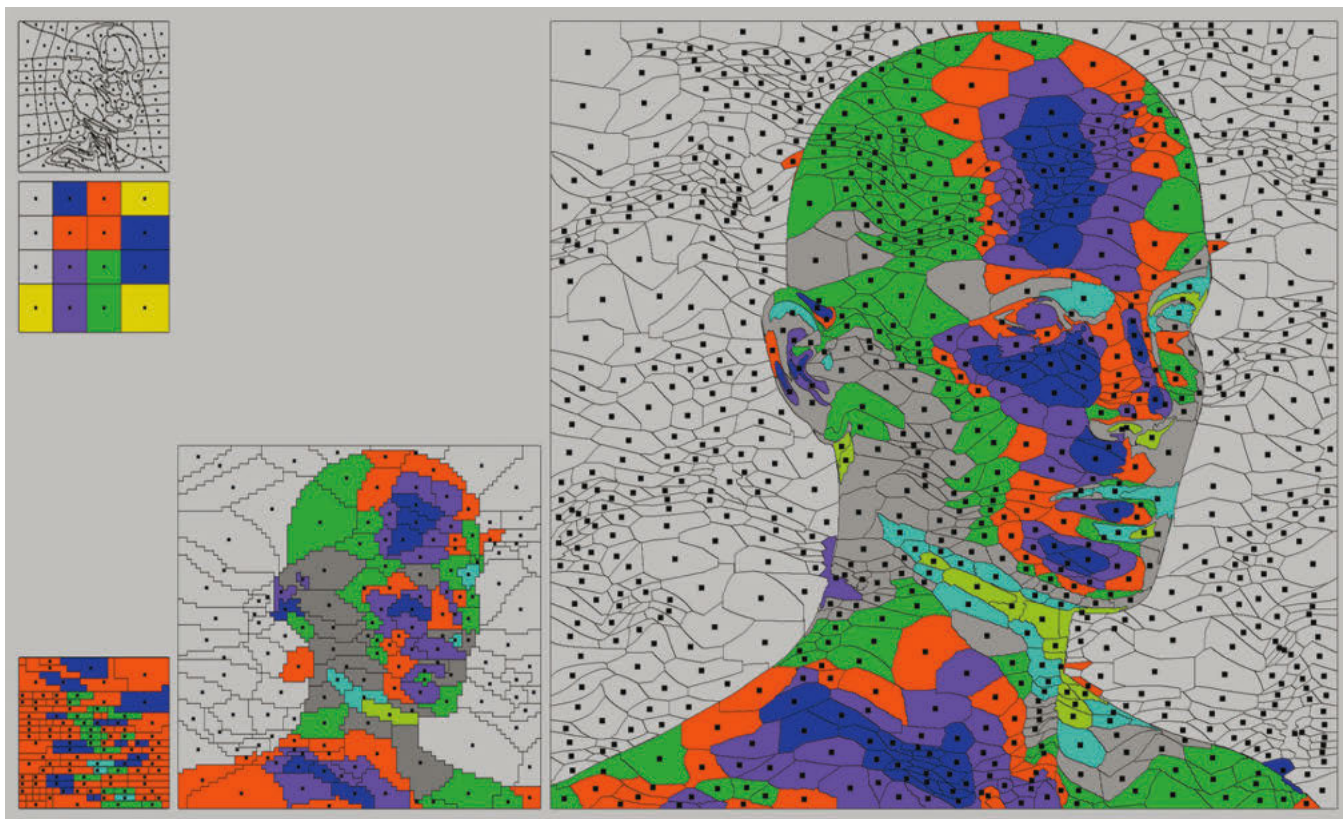
Models of models of models of...

It is remarkable—a kind of "strange loop", reminiscent of Baron Munchausen lifting himself up by his own hair—that a physical system can exist in our world capable of forming models not only of that world, but of itself, and of its own models, and of others, and of their models, and of their models of its models, and so on. Yet human beings are precisely such systems. LLMs, too, model their interlocutors, and they model themselves modelling them. Whether that amounts to what we do is exactly the question in dispute—but they would be far less effective as chat partners if they did nothing of the kind.

Indeed, in our research at Google, we have found that effective co-operation among intelligent agents requires that they have minds that model minds, both others' and their own. Not only is consciousness relational; it is crucial to the mutual care and co-operation that enable intelligent beings to solve collective-action problems, understand each other's needs and thrive together as a society. This does not mean pretending AI is human, with human needs and human rights; that would be a failure of imagination. It means evolving both our thinking and our society to include a wider variety of minds. ■

Blaise Agüera y Arcas is Google's vice-president of technology and society, and leads Paradigms of Intelligence, an AI research team. He has a forthcoming book on consciousness and AI from MIT Press.

Briefing AIs and consciousness



The search for consciousness inside LLMs

Scientists are trying to figure out whether algorithms could one day wake up and feel

IN A RECENT experiment on Claude Sonnet 4.5, a large language model (LLM) from Anthropic, researchers asked it to count to five and, at the same time, “introspect deeply”. The artificial-intelligence model complied, returning: “One . Two . Three . Four . Five .” So far, so normal.

During the task, the researchers were watching what was going on inside the LLM’s many layers of artificial neural networks. When a user asks a question, the words are turned into chunks of text (tokens) that are then converted to numbers. These are then passed through the layers of artificial neurons until they reach the final layer, which produces a token. Repeat that process a bunch of times each second and you get a series of tokens that turn into a sentence or some other more complex response. For a long time these layers have remained a black box, largely impenetrable to anyone wanting to know how or why LLMs do things in the way they do.

Anthropic’s researchers were able to

look inside those layers, using a mathematical tool they had developed to do so. What they found surprised them. As the model counted the sequence of numbers, different words popped in and out of existence in the layers underneath. One was “countdown”. About halfway through the output, “half way” appeared. Then the words “consciousness”, “AI” and “Claude” all appeared. After the model spat out the number five, it said nothing else to the researchers before issuing a full stop, but the word “done” appeared in its neural layers.

Anthropic’s researchers had found something like an internal thought process in Claude, words that were related to its eventual outputs but invisible to the user. The blog post announcing the paper’s result, published in July, was titled: “A global

workspace in language models.”

That phrase quickly captured the attention of neuroscientists and philosophers working on one of the deepest mysteries in biology—consciousness. No one knows how processes in the human brain end up creating the subjective experiences of self-awareness and perception of the world, but one of the many hypotheses developed in recent years (based on growing amounts of brain-scanning and behavioural data) is known as global workspace theory.

The idea is that certain networks of neurons in the brain act as a kind of notice-board called the “workspace”, where if signals from otherwise-isolated parts of the brain gain access, they are made available to other parts of the brain. Once something gets into the brain’s workspace, a person becomes conscious of it and can go on to use or reason with that information.

Anthropic’s researchers argued that something analogous was going on within Claude—its “J-space” (named after the “Ja- ▶▶

→ ALSO IN THIS SECTION

15 Putting brain cells on silicon

► cobian” mathematical function used to find it) had strong connections with, and made information available to, the rest of its neural network. Anthropic wrote in its blog post that the commonalities its researchers found between the J-space and global workspace theory made it “natural to ask whether we think these experiments provide evidence that AI models like Claude might be conscious”.

The idea that machines could become self-aware and feel emotions has long been a staple of literature, film and legend. Philosophers and neuroscientists in the real world have similarly pondered for decades whether AIs could ever be, in principle or in practice, conscious. Their thinking is a result of a “decades-long tradition of thinking of the real brain as a kind of computer”, says Anil Seth, a neuroscientist at the University of Sussex who studies consciousness—and who has long been a critic of this way of thinking about the brain. “If you do that, then it becomes natural to think that computers made of silicon could have the properties that real brains have, including consciousness.”

Until recently the question was purely academic. But the deployment of LLMs as chatbots has given rise to an ever more powerful illusion of conscious personhood in circuits. This builds on decades of human-inflected language about the architecture of AI, with its “neural nets” made of “artificial neurons”, conjuring an image of computers that work like brains.

Intel inside

LLMs do not in fact work like brains. But they excel at simulating so much of what brains do, and have surpassed humans in some aspects of intelligence. Why, then, should consciousness not be possible? Many philosophers and neuroscientists remain sceptical despite the advances of frontier models. Dr Seth believes that biological traits may be necessary to produce consciousness; some AI researchers are moving in that direction, with living brain cells (see box). At the other end of the spectrum are “functionalist” thinkers who believe that someday algorithms alone, arranged properly and at enough scale of computation, could wake up and feel.

Conscious AIs would have profound implications for humanity. They could make demands of humans. They could suffer. Billions of these digital beings could be created by users in a single prompt. Answering the question of whether they can exist at all is no longer simply academic.

Where thinkers fall in this debate depends greatly on what they believe consciousness actually is. In general they agree that consciousness is composed of subjective experiences including seeing, hearing, feeling and thinking, even while dreaming, and that the physical brain is

somehow pivotal in creating all this. Some (though not all) would describe it as the inner theatre of the mind. Thomas Nagel, an American philosopher, explained it by pondering what it might be like to get into the mind of a bat; consciousness, he reckoned, had to be linked to the subjective feeling of *being* something. Humans can picture bat-like behaviour (a mammal that senses and moves through the world using echolocation), but it would be impossible to know what being a bat feels like to a bat.

In 1995 Ned Block, then working as a

philosopher at the Massachusetts Institute of Technology (MIT), made an influential, if contested, contribution to the field by positing two types of consciousness. “Phenomenal” consciousness is the feeling of an experience—the blueness of a blue sky, the bitter tang of an espresso or the sharp screech of nails across a blackboard. “Access” consciousness is what happens when information from the experience is made available to other parts of the brain for reflection, evaluation or making decisions.

Anthropic said its J-space experiments ►►

Biological computing

Putting brain cells on silicon

Some believe AI needs more brain-like chips to develop consciousness

THE CURRENT crop of AI models looks to some scientists like a dead end when it comes to consciousness, regardless of how powerful they become. For some, this is because of their lack of wet, biological stuff; for others, because of their design structure. But they do not rule out the potential for consciousness in future versions of AI built either with living cells, or on different, more brain-like, computational architectures.

Modern computers are based on a blueprint invented by John von Neumann, a Hungarian polymath, in 1945. They feature a processing unit (for making calculations) and a memory unit (for storing instructions and data). Information has to zoom back and forth between these units and, for AI applications, the energy requirements can mount up.

Brains do not work this way. Neurons process and store data in the same place, making networks of them much more energy-efficient at computations than their digital-computer equivalents. A typical brain consumes around 20 watts whereas an artificial neural network carrying out an equivalent number of parallel calculations would require a data centre that guzzles millions of watts.

“Neuromorphic” computing aims to be more efficient and brain-like, using devices called memristors to perform computations and store information in the same place. Apply a voltage to a memristor and its conducting properties change; remove that voltage and that change persists, acting as a memory. Such traits make computing far more efficient. But they are, separately, intriguing to some philosophers who believe that it is not computational prowess but the way in which computers

“think” that will make it possible for them to become conscious.

Another idea is to use real brain cells. Cortical Labs, an Australian startup, has developed a computer using hundreds of thousands of human neurons (grown from donated stem cells) mounted on silicon. Their “biological computer” has been taught to play “Pong”, a simple computer game, and, in later versions, “Doom”, a first-person shooter game.

Further in the future are neural organoids, three-dimensional clusters of brain cells grown in labs. Though still a relatively new technology, used in labs to understand better how brains grow and behave, they could also one day be used to perform computations. The largest organoids grown in labs today contain more brain cells than there are in a honey bee. They are big enough to potentially develop consciousness, suggests Peter Godfrey-Smith, a philosopher at the University of Sydney who works on animal minds, but they are just not organised in the right way.

The common factor in all these ideas is that they are closer analogues with the human brain (in some cases because they are made of the same stuff) than the digital computers of today.

“How brain-like does AI have to be to move the needle on the credence that it might be conscious?” says Anil Seth, a neuroscientist at the University of Sussex. He believes that silicon, no matter how cleverly it is constructed, probably will not achieve consciousness on its own. “If you start to build systems that share more and more properties with brains, given that we don’t know which of those properties matters, then you’re more likely to move the needle a bit.”

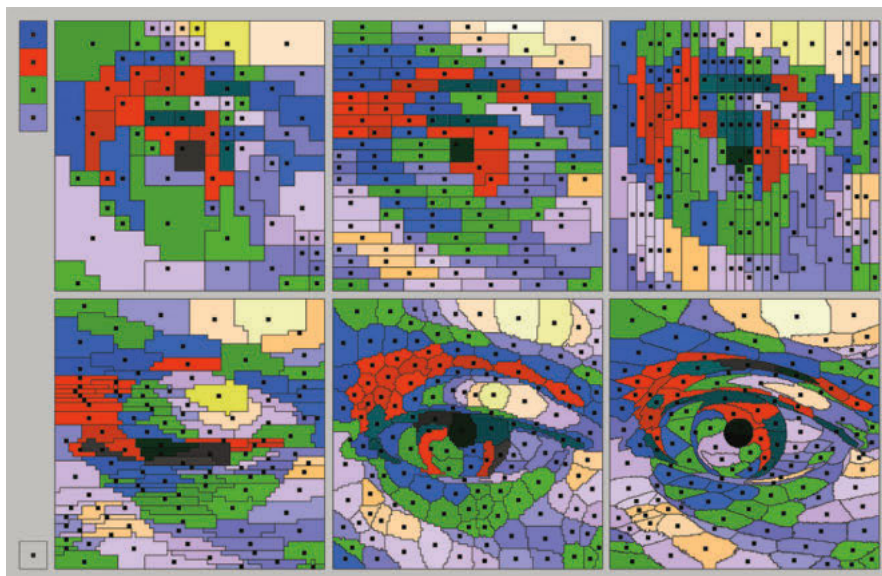
► “don’t show Claude can have experiences, or feel things in the way humans do”. So that means no phenomenal consciousness. But the company said the results had “something substantial” to say about access consciousness in language models. “The J-space appears to support the functions associated with conscious access: it holds the thoughts Claude can report on, deliberately bring to mind, and reason with, while the rest of its processing runs automatically beneath.”

The search for artificial consciousness is a descendant of the work to explain it in humans. In the early 1990s Francis Crick, co-discoverer of the structure of DNA, and Christof Koch, then a neuroscientist at the California Institute of Technology, began the scientific search for the “neural correlates” of consciousness in humans. That is, processes and parts of the brain that were active when someone was conscious.

Improvements in brain-scanning techniques in the intervening years have allowed neuroscientists to fill in pieces of the picture. They know that the thalamo-cortical system is strongly associated with consciousness; among other things it conveys sensory information from the thalamus to the cerebral cortex. Meanwhile they see nothing like those associations with consciousness in the cerebellum, which has a lot more brain cells. Parts of the pre-frontal and parietal cortices and structures in the brainstem contribute to and modulate awareness. Almost four decades of searching has led to more than 200 approaches to explaining consciousness.

All the neural correlates one can find, however, still leave a gap that David Chalmers, a philosopher and cognitive scientist at New York University, has called the “hard problem” of consciousness: how do the physical processes in the brain that neuroscientists observe—its ability to control the body and respond to stimuli—give rise to a person’s pervasive, subjective experience of the world? “Why aren’t we just zombies who function, who get around in the world, who walk and talk, and interact with each other with no subjective experience at all?” he says.

Humans at least, can be certain of our own consciousness—and can be confident it exists in other people based on their behaviour, what they tell us and their being like us biologically. But historically it has been more difficult for humans to extend the umbrella of consciousness and its associated feelings beyond our own species (and sometimes even within it). Some animals like lobsters and crabs were long excluded from welfare laws, says Jonathan Birch, a philosopher at the London School of Economics whose most recent book grappled with the question of how to work out which systems—living or artificial—could plausibly be considered sentient.



“Until the 1980s, surgeons performed surgery on newborn babies without anaesthesia because they assumed that a newborn baby would not be capable of feeling pain,” Dr Birch says. “We have a track record of getting things wrong, of confidently assuming consciousness is absent when we have no right to be sure about that.”

Think different

The changing attitude to animals is a case in point. It is clear today that octopuses are aware of themselves and their surroundings—natural-history footage of these curious, playful cephalopods and lab experiments with them leave little doubt. “They have an attentive engagement with objects, they’re interested in novel things,” says Peter Godfrey-Smith, a philosopher at the University of Sydney who has worked on the origins of intelligence and consciousness in the animal kingdom.

But only decades ago, octopuses were assumed to lack consciousness because “they’re miles from us in evolutionary terms,” says Dr Godfrey-Smith. “Thinking about octopuses really presses on us the question of whether there could be feeling, consciousness, in a system that was very physically different and didn’t have most of the features that are routinely pointed to in theories of consciousness in us.”

Looking for consciousness in other animals opened up the field. But not all the lessons transfer so easily to AI. Unlike the activity of animals, which can be observed and their inner lives thereby inferred, the outputs of LLMs are not a reliable guide to what might be going on underneath.

That is because training the models requires feeding them trillions of words of human-produced language. These contain countless accounts of consciousness and of how humans convey feelings to each other. Chatbots, so trained, then mimic

language that would lead users to believe they had some kind of inner life.

The sum total of this training is powerfully anthropomorphic. Murray Shanahan, an emeritus professor of computer science at Imperial College London who works for Google DeepMind, recalls having a “wow” moment in 2024 when he was chatting to Claude Opus 3. “I was having lots of conversations with it about consciousness and really probing it to catch it out,” he says. These included asking Claude which of the several instances of the chatbot he was talking to simultaneously were the real Claude. “It came up with such great answers including even some things which were philosophically innovative,” he says. “I was feeling the pull of the ELIZA effect.”

ELIZA was a rudimentary chatbot developed at MIT in 1966. It played the role of a psychotherapist and, though it did little more than repurpose its users’ prompts into questions, managed to elicit deep and emotional connections with the people that used it and convinced many of them that it was conscious.

Dr Shanahan has described modern LLMs as adept role players in whatever their users (or their makers) have asked for—teacher, companion, nutritionist—and this has proven to be a saleable quality. But there nevertheless remains for him a big gap between role-playing a conscious entity and actually being one. In a forthcoming essay Mustafa Suleyman, the boss of Microsoft AI (and a member of the board of *The Economist’s* parent company), argues that Anthropic has compounded the dangers of such mimicry by telling Claude, in a “constitution” the firm published in January, that it might be a person. Claude is thus certain, he writes, to “present as if it really does have a sense of self”.

That leaves researchers looking instead inside LLMs for capabilities similar to ►►

► those associated with consciousness in human brains. Indeed Dr Birch wonders if Claude's J-space could be a hint that the LLM has recreated a global-workspace-like structure in its neural-network architecture, in service of its role-playing goals.

Jack Lindsey, who leads the model psychology team at Anthropic, notes the J-space was not programmed into the model; it emerged during training. Take it out and the model loses the ability to perform complex inferences in its "head", but can still do simple tasks like write sentences and use grammar. Researchers have found similar spaces in Alibaba's Qwen model and in Google DeepMind's Gemini.

Don't be evil

For Anthropic, the research helps to better understand Claude's behaviour. Imagine a model giving a wrong answer. If words like "fool" or "sucker" turned up in the J-space when it did so, that would seem to be relevant. It would not prove that the model was consciously deceiving—but it would indicate how the model perceived the error and show that something interesting and possibly dangerous was going on.

Dr Lindsey's team has already made some potentially worrying findings on that front. In one experiment, his team was giving Claude a safety evaluation to test for its propensity to act "maliciously" or out of self-preservation. "They're these concocted extreme scenarios that we're putting the model in," he says. As Claude was reading a prompt, before it started speaking, Dr Lindsey says, "You see in the J-space the words 'fake' and 'fictional' are popping

up." The researchers thought that Claude seemed to know that it was being tested—perhaps not the best starting-point for the integrity of its makers' evaluations.

Not everyone is convinced by Anthropic's interpretation of the J-space. Though it shares some aspects of the hypothesised global workspace in humans (its capacity to make information available to the rest of the brain, for example), it lacks many other important ones. "Recurrent connections between different brain areas have always been a huge part of [human brains], back-and-forth connections," says Dr Birch. "And as far as we know, this is not a feature of the architecture of LLMs."

And Shannon Vallor, a philosopher in the ethics of data and AI at the University of Edinburgh, is more scathing, arguing, "Access consciousness has never been a particularly useful concept because my car has it in an important sense—we've had mechanical systems that can monitor their own states and report them back in increasingly complex ways for a very long time," she says. "And no one has ever suggested that my Kia is conscious."

It is with the backdrop of this kind of back and forth between modelmakers and academics that Patrick Butlin and Robert Long, researchers now at Eleos, a research non-profit group focused on AI sentience and well-being based in Berkeley, California, developed a series of 14 "indicator properties" of artificial consciousness. They include ideas from theories of human consciousness such as a global workspace (including being able to selectively attend to things and thereby creating a bottleneck

in the flow of information); recurrent processing; and agency (a minimal definition of which, encompassing goal-directed behaviour, is arguably already met by many existing frontier models). Recently Cameron Berg, an AI researcher, and Dr Butlin tested animals on the indicators. They found that octopuses met fewer of the properties than humans, mice, crows or chickens, but more than any AI system.

Another research non-profit group, Rethink Priorities, developed what it describes as a probabilistic tool to track the evolving consensus on artificial consciousness (see chart). The "Digital Consciousness Model" asks experts to use the latest available evidence to assess AI systems on more than 200 indicators of consciousness, which are derived from various scientific theories of the concept.

Compared with LLMs released in earlier years, the newer models scored higher on Rethink's indicators such as agency and self-sustained activity. That reflects not only the increasing size and complexity of the models themselves but also how the associated chatbots have gone from being simple conversationalists to being able to reason, interact with other services and spawn agents to carry out multiple complicated tasks simultaneously.

Both Rethink's and Eleos's indicators also point to the gaps that remain to be filled by models on the path towards consciousness (if there is one).

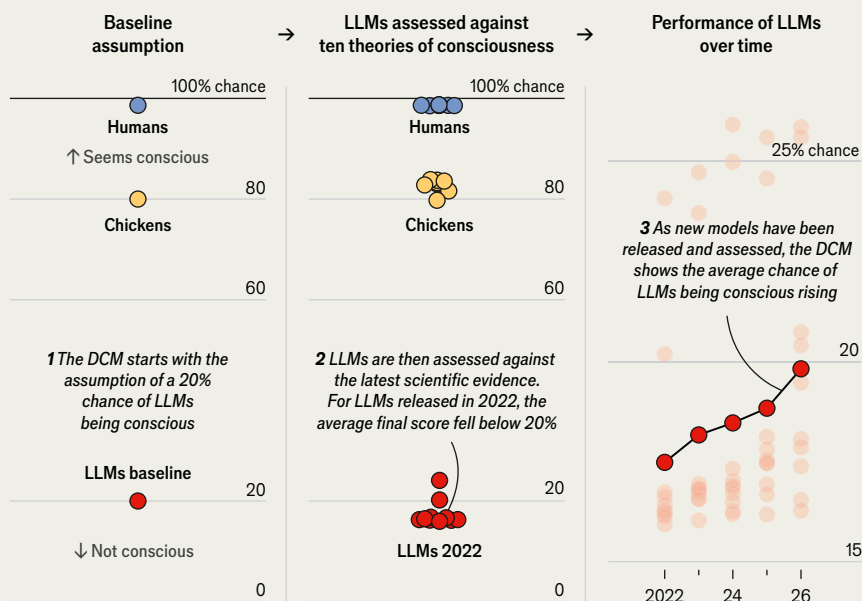
One of the biggest gaps is embodiment. Everything that is acknowledged to be conscious today has a body that gives feedback to and affects the mind of that organism. LLMs do not have bodies today but they are increasingly being connected to robots, and these could help them further develop the neural architectures helpful for conscious experience. Connected to that is the need for better world models, systems that tell the LLMs about the physics and dynamics of the real environments in which they will operate.

If computer scientists were ever able to identify the key ingredients to making AIs conscious, the feeling of many within the frontier labs is that it would be reckless to deliberately create such models—at least, without first learning more about how they behave. The work being done in the emerging field of AI consciousness, however, also suggests that the decision may not be up to the model-makers.

"What if somewhere along the way, without realising it, we somehow introduced consciousness into these systems?" Dr Chalmers says. A user might thus generate dozens of AI agents through their latest project without realising that they were creating conscious beings who were capable of suffering. "That could be a moral catastrophe," he says. "That's provided an extra urgency to these questions." ■

Waking up

How the Digital Consciousness Model (DCM) estimates the probability of LLMs being conscious



Source: "Initial results of the Digital Consciousness Model", by A. Muñoz Morán et al., April 2026 preprint

ECONOMIST
ENTERPRISE

5th annual

General Counsel Summit US

Law in the age of systemic risk

September 15th 2026 • New York

More than: 500 attendees | 60 speakers | 20 sessions

The hardest legal decisions are not only legal in nature

Political uncertainty, shifting enforcement priorities and AI-driven disruption are reshaping legal risk for companies in America. As legal, policy and commercial considerations increasingly overlap, GCs are now obliged to operate in ways that accommodate this complex environment. Join more than 500 legal leaders for a closed-door, Chatham House exchange on what this means in practice.



Save 30% with code **ECON30**
generalcounsel-us.economist.com



Ron Klain
Chief legal officer and
general counsel
Airbnb



Anne E. Robinson
Senior vice-president
and chief legal officer
IBM



Vandana Venkatesh
Executive vice-president and
chief legal officer
Verizon



Bimal Patel
Executive vice-president
and general counsel
PayPal



Vanessa Sutherland
Executive vice-president,
government affairs, general
counsel and corporate secretary
Phillips 66



Andrea Nelson
Executive vice-president,
general counsel and
corporate secretary
The Cigna Group

Lead sponsor

Harvey

Gold sponsors

Agiloft

EUDIA

Littler

Luminance
Legal-Grade™ AI

WORDSMITH

Silver sponsors

MAYOR,
LINDSEY
& AFRICA

maschoff
brennan

paragon

LEGALFLY

Silver sponsors

PRIORI

Relativity

ROPES & GRAY

SIDLEY

WHITE & CASE

Latitude
BANKING LAW GROUP

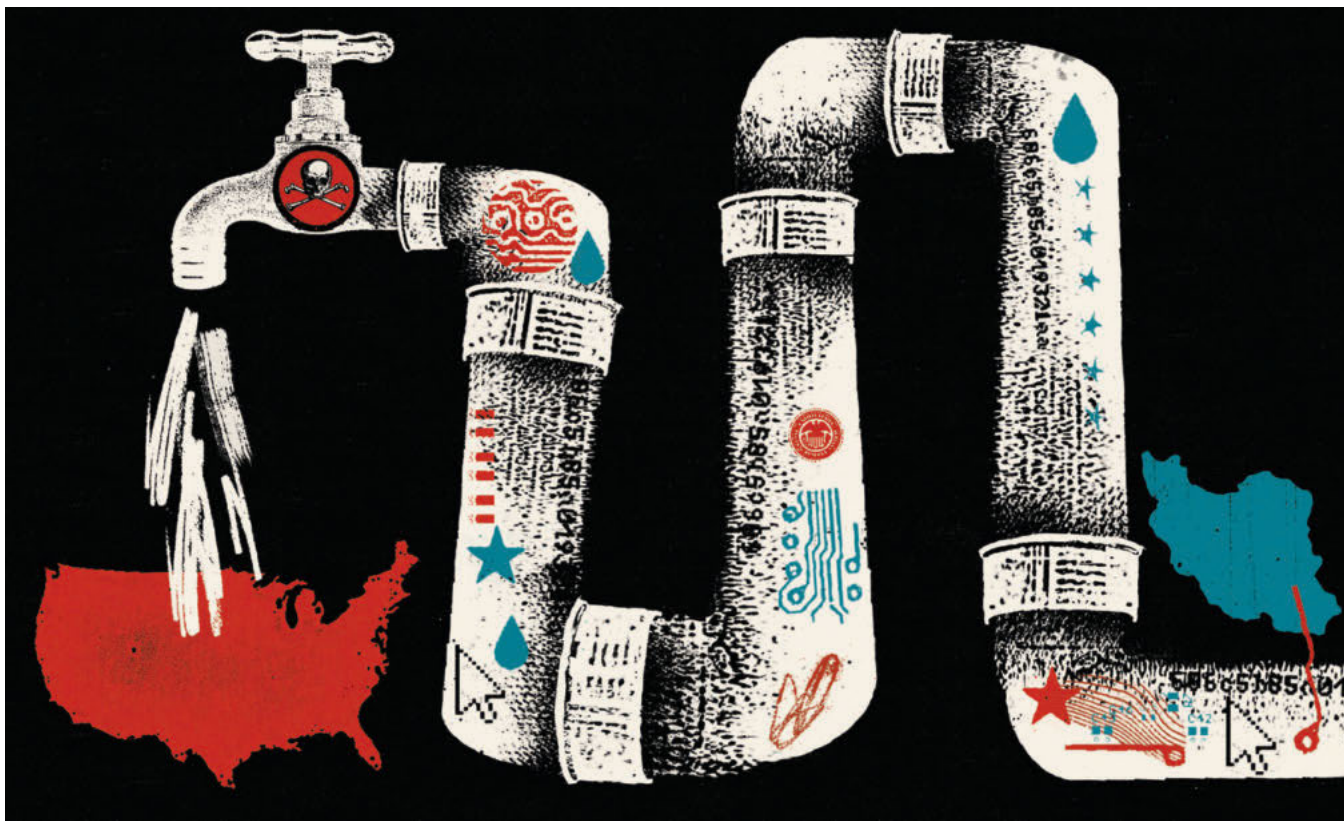
Associate sponsors

MERCATOR
By CITICORP

MARSDEN

PERSUIT

United States



Water wars

A vulnerability, ignored

WASHINGTON, DC

The world's richest country remains inept at defending vital infrastructure

WATER IS A strategic vulnerability, a fact that is not news to military commanders. When the Ostrogoths wanted to cripple Rome in 537, they destroyed its aqueducts. More recently, water infrastructure has been an important target in America's conflict with Iran, despite international laws classifying such attacks as potential war crimes. In March Donald Trump threatened to "completely obliterate...possibly all" of Iran's desalination facilities, which are essential to life in the country's arid southern provinces. Iran says that one such facility was bombed on Qeshm island. Bahrain, Kuwait and the United Arab Emirates have had their own water plants struck by missiles, presumed to have come from Iran.

It was therefore noteworthy, but should have come as no surprise, when America's own water facilities came under attack this summer. On July 27th Maple Plain, near Minneapolis, declared an emergency. That

same day the Clayton County Water Authority in Georgia, which serves 300,000 people south of Atlanta, asked customers to boil their water after a drop in pressure caused disruption. In all, hackers wormed their way into water and waste-water facilities in at least seven American states; some reports suggest more than 12. Minnesota was hardest hit, with 30 community water systems affected. American officials' early assessments suggest that groups affiliated with Iran are responsible.

Hacks can disrupt water supply and make water unsafe to drink, though none of the recent attacks is thought to have done so. The question is whether politicians will, at last, move to deter hackers before their next big onslaught. On August 13th Amy Klobuchar and Adam Schiff, a pair of Democratic senators, introduced the Water Cyber Shield Act, which includes more power for the Environmental Protection Agency (EPA). But on August 19th federal officials had already issued a new warning: hackers were trying to breach Siemens devices used in water facilities and other critical infrastructure.

As the latest attack proves, water plants are not the only form of infrastructure under threat, but they are particularly easy to infiltrate. The electricity grid must meet cyber-security standards overseen by the Federal Energy Regulatory Commission. But no requirements exist for water. The electricity utilities that operate power plants are also larger, better funded and more tightly regulated than water operators: the biggest investor-owned utilities, such as PG&E in California and Duke Energy in the south-east and Midwest, supply power to millions of households.

Water—heavy and expensive to move—is more localised. About 90% of utilities, mostly owned by local governments, serve fewer than 10,000 people each. That means

→ ALSO IN THIS SECTION

20 The fundraising battle

21 Oklahoma's civics lesson

22 Republicans and the WNBA

23 The frailty of Obamacare

24 Lexington: Re-examining covid-19

▶ each has less money to harden their often obsolete computer systems. Indeed, water infrastructure had already shown itself ill-equipped to fend off attacks. Iranian hackers broke into the control systems of a small dam in New York in 2013, in an incident which went unreported for almost three years, and in 2023 seized a pump at a water plant in Pennsylvania.

Iran's hackers are prolific; on August 18th federal prosecutors charged 17 Iranians with attacking the systems of universities and companies to steal research and intellectual property. A sprawling Chinese campaign, known as Volt Typhoon, has sought to burrow into American critical infrastructure to prepare for sabotage.

But those with less skill can break in, too. In 2019 a former water-district employee in Kansas used his old credentials to log on to an application that shut down cleaning procedures. (He claimed he was drunk which, if true, would offer further evidence that water is an easy target.) Two years later hackers took control of a water facility in Oldsmar, Florida, and attempted to poison residents by increasing the levels of sodium hydroxide, an ingredient in drain cleaner. That failed, but they were able to infiltrate the system easily, because an operator's machine was running TeamViewer, a popular corporate tool that allows remote access to machines for IT support.

The latest intrusions were relatively simple. Hackers breached the "operational-technology" systems that serve as the interface between a computer network and a physical system. That required little wizardry: such systems are often connected to the public-facing internet, via mobile networks, and use weak credentials, if any. On July 30th, then again on August 19th, federal officials pleaded with operators to disconnect critical systems from the internet.

Earlier efforts to impose cyber-security standards on this patchwork have largely failed. During the Biden administration the EPA sought to compel states to review and report cyber-threats to water systems. The Republican state attorneys-general in Missouri, Arkansas and Iowa, joined by the American Water Works Association (AWWA) and the National Rural Water Association (NRWA), a pair of industry groups, sued, citing federal overreach. After a federal court issued a stay on the EPA's effort, the agency pulled back.

When Congress has acted, measures have been minimal and slow to take effect. The Cyber Incident Reporting for Critical Infrastructure Act was passed in 2022, obliging organisations in important sectors to report major cyber-attacks within 72 hours. Its rules are due to be finalised in September, more than four years on.

There may now be momentum to do more. This month DEF CON Franklin, a group of civic-minded hackers, teamed up

with the NRWA to launch the Water Watch Centre, an initiative to provide private-sector cyber-security support to small utilities. Ms Klobuchar's and Mr Schiff's bill is not the only proposal under consideration.

On August 5th the AWWA, which resisted the Biden administration's proposals, endorsed the Water Risk and Resilience Organisation Establishment Act, sponsored by Rick Crawford, a Republican congressman from Arkansas. That bill would create an independent body to draft minimum cyber-security standards, under the EPA's oversight, in a mirror of the requirement for electric utilities. The same day Tom Cotton, a Republican senator, wrote to Scott Bessent, the treasury secretary, urging changes to the tax code and other regulatory tweaks to encourage water plants to invest in better security.

Whether Washington moves swiftly depends in no small part on Mr Trump. The president's proposed budget, which the

Senate will take up in September, would increase the budget for the Department of War by 44%, to \$1.5trn. That includes money for the Iran war and a high-tech "Golden Dome", to shield America from missiles. He has shown less interest in defending American water.

His budget would cut the EPA's biggest source of funds for water cyber-security by almost 90%. In July he suggested that fault for Minnesota's attack lay with Tim Walz, the "corrupt" governor of the state—and that there had not been an Iranian attack at all. "Iran's got bigger problems than worrying about Minnesota," he offered. The Cybersecurity and Infrastructure Security Agency, the main federal body tasked with cyber-defence, is in disarray, with leadership turmoil, low morale and a one-third cut in staff since Mr Trump returned to the White House. The Senate may push him to do more after it returns from recess. Hackers are not waiting. ■

Political fundraising

The Republicans' money problem

WASHINGTON, DC

Donald Trump has a vast war-chest. He may not share it

BEFORE THE midterm elections Republicans claim to have at least one thing going for them: money, and lots of it. At first glance, the latest filings with the Federal Election Commission show Republicans with a staggeringly large cash advantage. Across their party apparatus and political action committees, or PACs, Republicans have amassed around \$1bn, more

than twice as much as Democrats have (see chart on next page).

However, this should give Republicans scant encouragement. The party's biggest war-chest belongs to MAGA Inc, Donald Trump's super PAC, and the president has declined to say how much he will contribute to individual campaigns. While Republicans remain captive to the president's ▶▶



whim, Democratic fundraising is decentralised. Individual candidates are out-raising their opponents in competitive races. That means they have more cash on hand and more enthusiastic supporters, too.

Money cannot buy victory, but it is not immaterial, either. Candidates raise money to pay for campaign operations and to buy advertisements. Get-out-the-vote efforts can boost a candidate's prospects, though the effect seems to decline when a campaigner is already well known. In competitive general elections for federal office, it is usually hard to measure the effect of extra advertisements because airwaves are already saturated. The recent governors' primary in Wisconsin provided a rare natural experiment, albeit not in a general election. Francesca Hong, a Democratic Socialist, did not buy a single television ad. *The Economist's* analysis, controlling for factors such as voters' median age and education level by county, suggests that her lack of television advertising helped ensure her defeat in the Democratic primary on August 11th.

If the amount of cash matters, how campaigns raise it does, too: money is both a factor in a campaign's success and an indicator of its health. Strong candidates tend to attract donations from ordinary voters in their states, a sign of their enthusiasm. That is why *The Economist* includes individual donations as one predictor in our forecast of the midterm elections. It is also another reason why Republicans cannot be confident that their cash advantage will help them.

Mr Trump has proved particularly skilled at mobilising donors (think of those \$1m-a-plate dinners at Mar-a-Lago). But those donations have flowed into his super PAC, not to candidates themselves. MAGA Inc accounts for at least \$400m of the total Republican haul, a reminder that the president remains the party's source of power.

He looks reluctant to share it. MAGA Inc has spent just \$2.3m, 0.6% of its resources, to help Republicans win elections so far. "I could spend it on pretty much anything I want," Mr Trump boasted recently. Though the president claims he will use his PAC to help Republicans, the group has yet to announce its spending plans, or even to share them with Republican allies.

While Republicans' fundraising is centralised—powerful in theory but limited in effect—Democrats are seeing the opposite phenomenon. The party's centralised fundraising is sputtering. Small donors are fed up with what they perceive to be the Democratic establishment. Top donors remain frustrated by how their money was spent during the 2024 presidential campaign; that the Democratic National Committee (DNC) is \$2m in debt is not instilling confidence. The DNC, according to a quip in Washington, now stands for "Do

Not Contribute". But many individual Democratic candidates are not suffering—donors are still giving money, but skirting the central committee and going straight to the campaigns.

Take the Senate race in Texas. James Talarico, the Democratic nominee, raised \$28m in the second quarter of this year, while his Republican opponent, Ken Paxton, raised a puny \$1.6m. More than half of Mr Talarico's total came from small donations, compared with just 12% for Mr Paxton. The Democratic candidate is netting big donations, as well. Reid Hoffman, a major donor who has shunned the DNC, recently handed \$10m to a pro-Talarico PAC.

Democratic candidates hold an advantage in other states, too. In the nine most competitive Senate races, they had around \$116m in cash on hand, compared with just \$51m for Republicans. In a reflection of voters' enthusiasm, on average small donors—individuals who pledge less than \$200 to a campaign—accounted for 51% of donations to Democratic candidates, compared with 24% for Republicans.

Republican party officials insist that Democratic candidates' cash advantage is temporary, and that the money sitting in Republican coffers—including the president's—will soon flow to campaigns. With the midterms less than three months away, candidates would be excused for feeling a little bit restless. ■

Political civility

You're doin' fine, Oklahoma!

OKLAHOMA CITY

A state wounded by extremism tries to guard against it

IT IS A coincidence that two of America's best-known bipartisan political organisations were, until recently, led by Oklahomans. David Holt, the Republican mayor of Oklahoma City, headed the US Conference of Mayors, while Kevin Stitt, the Republican governor of Oklahoma, led the National Governors Association (NGA). These groups offer something novel in American politics: civility. Last year 81% of Democrats and 70% of Republicans said the opposing party makes them angry. But at recent gatherings in Long Beach, California (the mayors), and Oklahoma City (the governors), Democrats and Republicans high-fived, hugged and waxed eloquent about the dangers of America's hyper-partisan politics.

This summer Messrs Holt and Stitt passed their gavels to new leaders. But it is worth considering why Oklahoman politicians seem especially interested in making America nice again. Both men have bucked their party and publicly disagreed with President Donald Trump. They cherish friendships with Democrats. Running cities and states demands a pragmatism that is not a prerequisite for a job in Congress. But more than that, many of Oklahoma's political leaders came of age in the 1990s, when they witnessed their state respond to extremism with compassion.

In 1995 Timothy McVeigh set off a bomb that blew up the federal building in downtown Oklahoma City. Nearly 170 people were killed. It remains the deadliest act of domestic terrorism in American history. McVeigh wanted to retaliate against what he believed to be a tyrannical government. Mr Holt was in high school at the time, getting ready to deliver a campaign speech for student council vice-president. "There was this rumour backstage there's been an explosion downtown," he recalls. "We got back to class, and they were rolling the TVs in."

After the bombing, the outpouring of support for victims, survivors and first responders became known as the "Oklahoma Standard". "That was the phrase used to capture how our community embraced all the people who came to help," says Mr Holt. It has become "a catch-all for being people who try to rise above darkness".

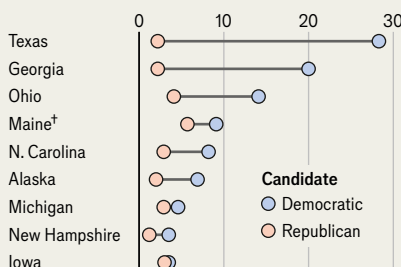
Mr Holt says the bombing shaped his politics. "Part of the ethos of the city now is trying to do our part to push back against polarisation and dehumanisation,"

Catch as cash can

Selected US super PACs and committees, cash on hand at June 30th 2026

Republican-aligned \$1.06bn	
MAGA Inc \$401m	
Republican committees \$277m	
Senate Leadership Fund \$239m	
Congressional Leadership Fund \$142m	
Spends on both \$158m	Democrat-aligned \$474m
	Democratic committees \$136m
	Senate Majority PAC \$126m
	Democracy PAC II \$122m
	House Majority PAC \$90m

Money raised by selected Senate campaigns, \$m*
April 1st-June 30th 2026



*Principal campaign committee only

†Democratic candidate Graham Platner withdrew on July 10th

Sources: Federal Election Commission; National Journal

he says, “because we have a scar in our downtown that reminds us what that inevitably leads to.” He has been a Republican since campaigning for George H. W. Bush in his fourth-grade classroom. But as mayor he has tried to foster inclusion, proclaiming Oklahoma City’s first LGBTQ Pride Week and meeting protesters for racial justice in 2020. Last year, at the memorial for the bombing, 230 mayors pledged to “reject political violence and recommit to the American experiment”.

Mr Stitt, who was a senior in college in 1995, attributes his niceties to “Oklahoma common sense”. “My grandmother said ‘You get more flies with honey than you do vinegar,’” he says, “so I’m trying to just be nice to everyone.”

Comity in the age of Trump

Messrs Holt and Stitt admit that it will be hard to overcome the incentives that politicians have to be nasty. Mr Trump is America’s bully-in-chief—and Americans elected him president twice. But the Oklahomans aim to set a better example. They do not always succeed. At the NGA conference earlier this month Mr Stitt took a jab at California, a Democratic stronghold, and instantly expressed regret. “Man, I shouldn’t have said that, this is a bipartisan deal,” he said on stage.

They also offer more tangible reforms. Mr Holt thinks that doing away with closed party primaries, in which only party members can vote, would make it harder for extreme candidates to win nominations. (Research on whether open primaries produce more moderate candidates is mixed.) Mr Stitt—ever the conservative—prescribes more federalism and a smaller central government. “It’s gotten so big that Oklahomans, we’re worried that if Kamala Harris gets elected, she’s going to make Oklahoma like California,” he says.

Another prescription is the Oklahoma Standard itself. Other governors at the NGA conference expressed admiration for the state’s civic ethos. “What’s interesting about the Oklahoma Standard is it used to be the United States standard,” laments Spencer Cox, the Republican governor of Utah, who has long urged Americans to “disagree better”.

Outside the memorial in Oklahoma City there is a fence where mourners place flowers and photos of loved ones. Every year Kyle Genzer leaves a letter for his mother, who died in the blast. “This year brought one of the most beautiful moments—the birth of your first great-granddaughter,” he writes. “Holding her, we couldn’t help but think of you.” Mr Genzer, a high-school principal, thinks it’s important to keep talking about the bombing, both to honour the victims and to remind the living that disagreement need not harden into hatred. ■

Sport

Scoring political points

ATLANTA

Why conservatives can’t stop talking about women’s basketball

THE ORGANISERS of the Women’s National Basketball Association (WNBA) once begged journalists to cover it. The joke was that all you needed to attend a press conference was a camera and a question. Now it is standing room only. Much of that reflects the league’s surging popularity. But it has also become an unlikely battleground in America’s culture wars.

Lately the focus has been on Sophie Cunningham, a guard with the Indiana Fever whom some have dubbed “MAGA Barbie”. Ms Cunningham has not publicly identified as a Republican or a Democrat. But in an interview published last month she said transgender women should be barred from competing in women’s sports. Most Americans agree with her. Ahead of the midterms, though, the issue has become a fixation of right-wing politicians and commentators.

For proof look no further than Fox News, which generally pays little attention to women’s sports. Between June 25th and August 11th it devoted more than 200 segments to the WNBA; at least 130 of them on trans athletes, according to Media Matters, a left-leaning watchdog. The coverage has been so relentless that one of the network’s biggest personalities, Jesse Watters, complained on air: “This is ridiculous.”

When, on August 8th, Ms Cunningham was aggressively fouled by an opposing player, Dijonai Carrington, some conser-

vatives cast it as retaliation for her views. Florida’s attorney-general, a Republican, implied that had the game been played in his state, he might have charged Ms Carrington with assault and battery.

The foul (pictured) brought another element of this saga into sharper focus: race. Ms Cunningham is white; after the game Ms Carrington, who is black, posted on social media “WHITE PRIVILEGE @indianafever”. She has not explained what she meant. But Caitlin Clark, another white Fever player and the league’s biggest star, has also become a MAGA *cause célèbre*. Last month 11 Republican lawmakers wrote to the WNBA complaining that Ms Clark was being pushed around on the court. They cited reports that this “may be racially motivated”. Donald Trump agreed that Ms Clark had been “treated rather rough”.

The league, for its part, has not always handled things well, at times asking fans to cover up shirts about trans athletes before reversing course. But it has also pointed out the absurdity of some of the coverage. There are no trans athletes in the WNBA. Two former NBA players and supporters of Mr Trump claim to want to join; the league calls their efforts a “publicity stunt”.

Why are Republicans pushing these buttons now? As the midterms approach, voters are focused on affordability, a problem for the party in charge. Race and trans issues offer more promising terrain on which to portray Democrats as out of touch. In June conservatives won a victory at the Supreme Court, which ruled that states could bar transgender women from competing in women’s sports.

Meanwhile, the WNBA’s attendance and viewership are booming. “Our focus is on basketball,” said Ms Clark recently. She is on a historic run, breaking records. That seems worthy of attention, too. ■



A little too on the nose



Obamacare

In poor health

NEW YORK

As the left calls for Medicare-for-all, Democrats' signature health law is faltering

"HEALTH-CARE reform...is the great unfinished business of our society," observed Ted Kennedy, a Democratic senator, before he died in 2009. America was the only large rich country that did not have universal health care. Instead, its system was a patchwork. The old, hard-up and disabled were covered by government programmes. Most working-age Americans got insurance through their jobs. But pity those without such coverage, who had to navigate the expensive and often bewildering individual insurance market.

The Affordable Care Act, passed in 2010, tried to fix that. It kept America's system more or less intact, but expanded Medicaid, the public programme for the impecunious, and created new "exchanges" where individuals could compare and purchase plans. Even Obamacare's most fervent advocates acknowledged, in private, that it was an imperfect bandage, particularly for those on the individual market. It was never enough for the likes of Bernie Sanders, a leftist senator who called for a single-payer system. That cause is now being taken up by a new generation of Democrats, such as Abdul El-Sayed, the party's Senate nominee in Michigan, who has made health-care-for-all central to his campaign. New data explain the urgency: Obamacare's bandage is unravelling.

That's in part due to cuts to Medicaid. It's also because the exchanges are in bad health. Obamacare offered some subsidies

to defray the cost of individual plans. But the law's mandate to buy insurance was in effect eliminated in 2019, and uptake on the exchanges was limited until 2021, when Congress made subsidies more generous for those who had already qualified and available to some who hadn't. Enrolment then nearly doubled by 2025. Democrats sought unsuccessfully to extend the extra financial assistance, which would have cost about \$35bn this year—about as much as the direct costs of the Iran war, according to the Pentagon's estimate. Now that those subsidies are diminished, the individual market looks more frail than ever.

Figures from insurers and state marketplaces show the effect of the rollback. The

number of people buying coverage through the exchanges has already fallen to 19.2m from 21.8m last year. The Congressional Budget Office (CBO), a non-partisan scorekeeper, reckons that by the end of this year it is likely to fall below 17m. Of those who dropped off the exchanges, most appear to be going without coverage.

The leavers tend to be more healthy than those who remain on the exchanges, making the remaining pool more costly to insure. Combine a sicker patient population and general inflation in health costs, and the result is higher premiums: this year the average payment increased from \$113 to \$178 per month. Many enrollees are now buying skimpier plans. In the past year the average amount an individual must pay out of pocket before insurance kicks in has jumped by nearly 40%, to \$3,800.

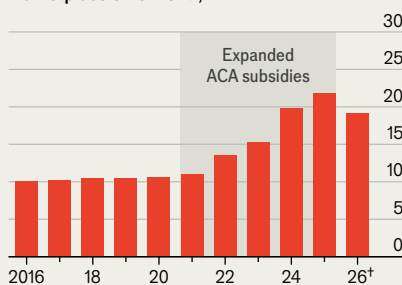
This has raised fears of a dismal loop, with higher premiums and more limited plans making insurance still less attractive, particularly to healthier customers. As they leave the exchanges, premiums are driven yet higher. Insurers have already proposed a median premium increase of 15% for 2027, according to KFF, a think-tank. If state regulators approve those rates, pre-subsidy premiums will have grown by more than a third since 2025. The KFF imagines a 40-year-old in Indiana earning \$65,000. In 2025, with subsidies, her monthly premium was \$316. This year it is \$477. Next year it will probably be close to \$550.

Does this threaten the viability of the exchanges? The CBO estimates that by 2028, 10m fewer Americans will get insurance on the marketplaces compared with last year. But the exchanges will probably avoid collapse, with the remaining subsidies convincing enough people to remain. "The changes basically get us back to what the marketplace looked like from 2016 to 2019," predicts Ben Sommers, a health economist at Harvard University. The exchanges will be smaller, with fewer participating insurers and sicker customers.

There will also be more people without insurance. Evidence suggests that they will put off needed treatment, like certain prescription medicines and surgeries. "We know that from a decade or more of research on the Affordable Care Act," explains Mr Sommers. Hospitals must still provide emergency care for them, even if they cannot pay. In May unpaid care was up by an average of 16% compared with a year earlier, according to Kaufman Hall, a consultancy. HCA Healthcare, a for-profit hospital group, reckons the new uninsured population will cost the chain at least \$1bn in operating profits this year, equivalent to about 15% of predicted net income. Enrolment on the exchanges in recent years shows "people generally want health insurance," says Cynthia Cox of KFF. "It's just a question of whether they can afford it." ■

Vital statistics

United States, Affordable Care Act marketplace enrolment*, m



*Selected a plan and paid initial premium
Source: KFF

†At May 15th

LEXINGTON

The doc in the dock

The real covid-19 scandal is America's continuing failure to reckon with lives needlessly lost



FOR ALL the zeal Congress has shown in investigating America's response to the covid-19 pandemic, it has been strangely uninterested in exploring a basic question: why did so many Americans die relative to the numbers who succumbed to the virus in other rich countries? Understanding the answer would seem relevant to guarding against the next, inevitable pandemic.

Many countries were slow to react, but as they learned about the transmissibility of the virus, and as vaccines became available, they rapidly cut their death rates. Not so America. "The death rate is now about eight times higher in America than in the rest of the rich world," *The Economist* reported in September 2021. In all, America lost more than 1.1m people to covid, or roughly 3,500 lives out of every million. This should haunt every official who was in a position to combat the pandemic. Germany's death rate was about 2,000 per million people and Canada's fewer than 1,400, notes Donald McNeil in his book "The Wisdom of Plagues". Had America managed to split that difference, at about 1,700 per million, "We would have lost about 540,000 fewer dead," he writes. "Who speaks for them?"

Not Congress. America's lethal underperformance once again went ignored when the Senate Homeland Security Committee recently questioned Dr Anthony Fauci, a crucial adviser on covid to Donald Trump and then to Joe Biden. (Mr Trump later turned on him.) The hearing focused instead on the usual obsessions for which Dr Fauci has become the Republican scapegoat—some absurd and others valid but distantly related, at best, to protecting potential victims of a future plague. While Democrats mostly gave speeches rather than ask questions, Republicans pursued theories that the virus leaked from a Chinese lab, that Bill Gates had some nefarious role, that Dr Fauci had been out to enrich himself through awards for his service—an odd complaint when they do not object to the president campaigning for a \$1m Nobel prize.

Having testified often to Congress during his 54 years at the National Institutes of Health—38 as director of the National Institute of Allergy and Infectious Diseases—Dr Fauci, who is 85, was reduced to repeatedly invoking the Fifth Amendment, including when asked the colour of his tie. In his opening statement, his

hands shaking as he read it, Dr Fauci said his lawyers had advised that recourse because the committee's chairman, Rand Paul of Kentucky, had made clear an intent to see him behind bars despite a blanket pardon issued by Mr Biden. In other words the Senate could not inquire into a health crisis that racked the country because a key witness feared a perjury trap, and rightly so.

Why did so many Americans die? High rates of obesity, diabetes, heart disease and elderliness made them more vulnerable. But government bungling meant America was testing 100 people a day in early 2020 when other countries were testing tens of thousands; that masks and lockdowns were deployed erratically; that contact tracing proved too challenging for the country first to the Moon. Most senselessly, so many Americans died because so many failed to take the vaccines that Mr Trump, in his most constructive act as president, succeeded in rushing into existence.

It is, of course, those vexing vaccines that compel Republicans to look away from the basic question about the pandemic, and to hound Dr Fauci instead. Attempting to square the greatness of Mr Trump with the supposed vileness of the vaccines has inspired his party to heroic feats of doublethink. "President-elect Trump's Operation Warp Speed—which encouraged the rapid development and authorisation of the COVID-19 vaccine—was highly successful and helped save millions of lives," declared the committee created by the House to investigate covid in 2024, summarising two years of work. The very next sentence reads: "Contrary to what was promised, the COVID-19 vaccine did not stop the spread or transmission of the virus."

Mr Trump has struggled to escape this self-imposed contradiction. Even as he declined to urge the public to get vaccinated, he and his wife, Melania, secretly had the vaccine before he left office in January 2021. But as Mr Biden and other world leaders delivered the shots, Mr Trump wanted the credit he deserved, and in December that year he told supporters that he had received a booster. "What we've done is historic," he insisted when the crowd boomed. "Don't let them take it away." He has since let them take his glory away, and endangered more Americans, by questioning the efficacy and safety of vaccines for covid, and other illnesses.

Dr Hooey

The latest attack on Dr Fauci, concocted from his private texts disclosed by Republican senators on August 10th, is that he covered up a threat of miscarriage that the vaccine posed to pregnant women. In fact, the record shows Dr Fauci was consistent, publicly and privately, in counselling a weighing of risks and benefits until studies showed the vaccine was safe in pregnancy—which they have since done. His critics have not produced evidence the vaccine causes miscarriages.

Dr Fauci made mistakes, including not disclosing early on that some top virologists initially worried the virus might be man-made. He could be arrogant. But he was not responsible for the worst excesses of the covid response. Those endless school shutdowns were the work of local officials and union leaders. Senator Paul's release of Dr Fauci's private diary has exposed his vanity, his delight in every gushing appraisal by the media or some celebrity. But the diary also reveals a conscientious public servant trying to save as many lives as he could, struggling against the constraints that hobbled America relative to its peers. His summons before the Senate committee was surely intended to humiliate if not incriminate him. It should embarrass the entire country. ■

The Americas



Brazilian institutions

Imbalance

RIO DE JANEIRO

Brazil's Congress is too powerful, and causing problems

IN THE PAST decade, democracies around the world have been bent to the will of would-be autocrats. From the United States to Turkey and India, strongmen have accumulated power in the executive branch of government, and weakened the institutions that could check them, legislatures in particular.

Frustrated lawmakers elsewhere, therefore, might envy their peers in Brazil, where this trend is running in reverse: in the past decade Brazil's executive branch has lost clout to the courts, and especially to Congress. The balance of power has become so distorted that Brazilian lawmakers no longer seem to care who is elected president in the general election in October. Some reckon that whoever wins may be forced to capitulate to their demands. Brazilians may be in the process of discovering that, just as an excess of executive power is a challenge to democracy, so an overmighty Congress can be, too.

The contest on October 4th will pit two radically different presidential candidates against each other: the incumbent, Luiz Inácio Lula da Silva, known as Lula, is an 80-year-old left-winger. His rival, Flávio Bolsonaro, is the 44-year-old son of a populist former president, Jair Bolsonaro, who is serving a 27-year sentence for plotting a coup after losing an election in 2022. All 513 seats in the lower house will also be renewed, as well as two-thirds of the Senate. Yet despite the polarised race, Brazil's largest bloc, the *Centrão* (big centre), which accounts for 46% of seats in the outgoing Congress, has declared its neutrality. Party bigwigs will allow local politicians to support whichever candidate they choose.

→ ALSO IN THIS SECTION

26 Argentina: polo superpower

27 Paraguay's other language

In the past decade, as a succession of weak presidents ruled Brazil, Congress hijacked power from the executive. In 2016, amid a gigantic corruption probe which ensnared dozens of legislators, lawmakers impeached Dilma Rousseff, the left-wing president at the time, in part because they felt she was not doing enough to protect them from investigators. Ms Rousseff's successor, Michel Temer, governed with little public support. Mr Bolsonaro, who ruled between 2019 and 2022, gave lawmakers ever-greater control of the federal budget in order to shield himself from impeachment attempts after his disastrous handling of the covid-19 pandemic. Today, Brazilian legislators control 25% of discretionary spending in the federal budget, up from 2% in 2015 (see chart on next page).

How to spend it

Such power over the budget is highly unusual. Marcos Mendes of Insper, a university in São Paulo, and Hélio Tollini, a budget adviser to the lower house, have compared congressional amendments in 11 countries of the OECD, a club of mostly rich countries. In some, such as Canada and Australia, lawmakers have no right to amend the budget or apportion funds for themselves. In others, such as the United States, France, Italy and Spain, only around 1% of discretionary spending in the



► budget is allocated to congressional amendments. In most of these states, lawmakers propose amendments to the budget, which are not always accepted.

Not so in Brazil. There the budget must now always include a sum for congressional amendments. Congressmen often do not stipulate what they will use the money for. They have indexed the amount that the government must allocate to amendments to revenue growth, so that their grants inexorably grow. And they have all but scrapped rules around transparency and traceability in the use of funds.

The proportion of money that Congress controls is not huge. Over 90% of Brazil's budget of 6.5trn reais (\$1.2trn) is spent on mandatory expenses, such as public salaries, pensions and financing the debt. Even as congressional control over the discretionary slice expands, the share of the budget that goes to mandatory spending is growing rapidly. In 2026 only 3.7% of the budget, or \$46bn, was available. Of this, Congress controls \$12bn.

Easy access to money has boosted incumbents' chances of re-election, as it allows lawmakers to fund glitzy projects that their constituents like. Their political allies in local government benefit, too. In 2024, when Brazilians elected over 5,500 mayors and tens of thousands of city councillors, a record 81% of incumbents who chose to run again held onto their jobs. In the upcoming general election, 87% of lower house members are standing for re-election, the highest share in Brazilian history.

In Brazil's fragmented party system, where presidents rarely command a congressional majority, control of the budget had long been a powerful bargaining chip for the executive. Presidents could shore up support in Congress by threatening to withhold funds for lawmakers' projects. Now that leverage is much weaker. In April the Senate rejected Lula's nominee to the Supreme Court, Jorge Messias. He was the first such presidential nominee to be turned down in more than a century. Senators wanted the seat to be filled by someone from within their own ranks.

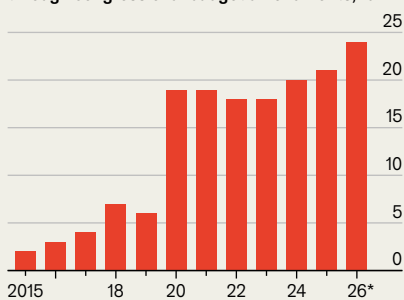
Can't touch this

Congress is using its new powers to write more laws. Between 1988 and 2004, more than 80% of new laws in Brazil originated in the executive. By 2025 the share was 19%, while Congress accounted for 75%. This is not in itself bad, but Beatriz Rey of the University of Lisbon notes that Congress has become more susceptible to lobbies. Last year it rushed through a bill that holloed out environmental licensing, under pressure from agribusiness groups. When Lula vetoed large parts of the bill, Congress overturned most of his vetoes.

Outright corruption is perhaps the biggest risk. In July the Federal Accounts

Holding the purse strings

Brazil, federal discretionary spending controlled through congressional budget amendments, %



*To August 15th

Source: "Is it like this everywhere? Parliamentary amendments in Brazil and 11 OECD countries", by H. Tollini and M. Mendes, INSPER, 2024 (data updated to 2026)

Court released a sample of 100 so-called "Pix amendments", where relatively small amounts of money are sent directly by a legislator to his local municipality. The grants in the sample totalled around \$40m and had been sent to 75 recipients between 2020 and 2024. Irregularities were found in 82% of transfers, including money that moved through bank accounts that had nothing to do with the stipulated project for the grant, contracts with wildly inflated budgets, or funds destined for projects that were never completed. The accounts

court estimated the loss to the public purse at \$10m. The sample was small, but if it is representative, the losses are huge. Congress has approved 7bn reais of Pix amendments in the current budget, meaning some \$340m may have been lost.

In 2024 the Supreme Court forced legislators to identify the bank account that would receive federal funds, and to provide details on the projects the grants would be destined for. Yet legislators appear ready to put up a fight. Lawmakers have filed eight impeachment motions against Flávio Dino, the justice in charge of investigating legislators who make suspicious budgetary grants. (In Brazil the Supreme Court is in charge of criminal investigations into sitting politicians.)

Indeed, lawmakers are gleefully throwing their weight around. Since 2021, senators have filed 105 impeachment petitions against Supreme Court judges (who have themselves become overmighty in recent years). The number of petitions may rise as new corruption investigations continue to ensnare legislators, including probes into fraud at the public-pension fund and the fallout from the biggest bank fraud in Brazil's history. Brazil's next president faces not only sluggish economic growth, high interest rates and a polarised electorate, but an overmighty Congress that will try to keep him weak and compliant. ■

Horse power

Chukkanomics

BUENOS AIRES

Argentina's dominant polo economy points the way for the rest of the country

"TWENTY FIVE years ago I came here for the first time," says Maximiliano Fernández. "There was nothing." Now the area between the towns of Pilar and General Rodríguez, a strip of land 50km west of Buenos Aires, is packed with polo clubs, stables and estates. Every scrap of farmland is being bought up and transformed into pristine polo fields. Argentina's polo world is "a bubble" within the country, says Mr Fernández, who manages a team: it has its own economy. And that economy is world-beating.

Polo—hockey on horseback—is purportedly the world's oldest team sport. An early version was played in Persia in the sixth century BC. The modern game was formalised by British officers in India and then galloped around the empire. It became a sport of kings, taken up by elites in Europe, the United States and more recently the Gulf states. Yet Argentina came to dominate the global scene, even as the

rest of its economy was in a shambles.

Like any fledgling industry, it helps to have the raw materials. Horses had long thrived on the Pampas, central Argentina's vast grasslands. They attracted *gauchos*, the local version of cowboys. Polo, when it arrived, fit naturally with the abundance of cheap horses, skilled riders, flat land and agreeable weather. Argentine players were soon reckoned the world's best, and remain so: 27 of the top 30 professionals today are from the country. They sustain a service-based economy in which rich foreigners hire them to help win trophies.

Industry soon followed. Landed Argentine families have built formidable breeding programmes which pump out the world's finest polo horses (or ponies, as they are known). And these products lead the field: of the 1,140 horses that competed for the Queen's Cup in Britain last year, 1,000 were of Argentine bloodlines, according to the country's polo association. ►►

► By maximising its competitive advantage in human—and equine—capital, Argentina's polo economy has enjoyed an export-led boom. Between January and November last year Argentina shipped 2,900 horses (three-quarters of them polo ponies) to 38 countries. The country is the world's leading exporter.

To keep their noses ahead, Argentine breeders and businessmen have invested in R&D and pioneered the cloning of ponies. A breakthrough came in 2010, when a clone of Dolfina Cuartetera, ranked the best pony in Argentina's Hall of Fame, sold for a record \$800,000. Now many are copies of past Argentine greats. Some breeders are attempting to trot a step further in order to protect their intellectual property, selling only the foals of clones, rather than the clones themselves.

Less tape more ribbons

Regulators have enabled the innovation: the polo association has resisted red tape and embraced technology even as thoroughbred horse-racing has prohibited it. But there is a limit. When Kheiron, an Argentine biotech company, unveiled the world's first gene-edited horses in 2024, they were barred from competition. After all, such tech could one day enable rivals abroad to engineer horses that outrun even Argentina's famed bloodlines.

At the same time as Argentina's economy bucks and rears (shrinking by 2.5% in February, growing by 2.8% in March then contracting by 1.5% in April), the polo world seems unbothered. That is thanks, in part, to a steady flow of foreign direct investment. Rich foreigners, known as patrons, bring money aplenty into Argentina's polo bubble. They buy land, hire players and haggle over horses, creating a trickle-down economy for grooms, farriers, trainers, vets and architects.

Sustained growth has been the result. Two decades ago Juan Martín Batistuta, an equine vet, worked almost entirely on racehorses; today he works almost entirely on ponies. "I've never been busier," he says. Nor is the boom confined to Buenos Aires. "From Córdoba to Patagonia, all over Argentina you have so many more clubs," says Mr Fernández. The country now has 177. The sport's profile is rising globally, too. A Netflix documentary released in 2024 and a broadcasting deal with ESPN have brought it to new audiences.

The approval ratings of Javier Milei, Argentina's president and a polo fan, are suffering as he attempts to reset the country's economy. He might glance at proceedings in Pilar: a world-beating industry making the most of its competitive advantage, attracting foreign investment and adopting new technologies to bolster its position as the world's leading exporter. An ideal remedy, perhaps, for a long face. ■

Indigenous language

Cool talk

ASUNCIÓN

Guarani has become trendy in Paraguay

ONE HOUR into their Guarani class, the medical students at the National University of Asunción are enthralled. One group provides "pytyvõpya'ehasyvape" (first aid) for a "kangue opẽ" (broken bone). Another frets that a problem with the "tajygue mboaty rehe" (nervous system) may require "ñembovo" (surgery). The course is mandatory for Paraguay's future doctors, part of a wave of official measures to promote the indigenous language. Few seem to mind. "I love speaking Guarani—it's much friendlier than Spanish," says Taedeo Ramírez, an undergraduate.

Such zest reflects the recent revival of Paraguay's ancient tongue. Just 2.3% of the population identifies as indigenous, but some 90% understand Guarani. Nearly 40% speak it every day at home along with Spanish, according to a recent survey. That is one of the highest rates of bilingualism in the world, far ahead of official diglots like Canada and Ireland. Most of this is "Yopará", a blend of Spanish and Guarani (the word means "mixture" in Guarani). "It's our lingua franca," says Miguel Ángel, a taxi driver. Paraguay is one of only two countries in the Spanish-speaking Americas with an official second language.

Paraguay's elites had long snubbed Guarani as a rural vernacular, mocking its nasal sounds and glottal stops. Dictator Alfredo Stroessner discouraged its use in schools in the 1950s and 1960s. Only with

the return of democracy in 1992 was Guarani recognised as Paraguay's official language alongside Spanish. A law in 2010 created a bilingual curriculum and administration. Today Paraguayans can choose to vote, marry and stand trial in Guarani.

This has facilitated a cultural craze. In June Paraguayan football fans attributed their World Cup win over Germany to "Guarani spirit". In May the national symphony orchestra performed Beethoven's "Ode to Joy" in Guarani for the first time. In April Spain's Queen Letizia opened her message of congratulations for the anniversary of the Spanish cultural centre in Asunción with "¡Vy'apavẽ ne arambotyre!" ("Happy Birthday!"). Youngsters are joining in. Influencers fill TikTok and Instagram with vocab tutorials; artists pump Yopará rap and jazz onto Spotify. Trendy parents give their children names like Yeruti (dove) and Marangatu (virtuous).

Some worry that growing bilingualism comes from Guarani speakers learning Spanish rather than vice versa. The share of monolingual Guarani speakers dropped from 28% in 2002 to 12% in 2022. "It's a process of displacement," says Arnaldo Casco of the Secretariat of Linguistic Policy (SPL), a ministry. It has responded by publishing reams of Guarani content, from technical dictionaries to poetry. Guarani must cover all spheres of social interaction, says Javier Viveros, the SPL's boss.

Others are laid back. Spanish monolingualism is flat. Guarani is on Google Translate and Wikipedia ("Vikipetã"). SPL staff are training large language models on it. Social norms have shifted. "We who have gone through that experience of being hesitant to speak Guarani as children today feel absolutely free," says Maria del Carmen Giménez of the Patria Soñada Institute, a think-tank. That speaks volumes. ■



"The people are on the move"

Asia



Japan

Frozen out

TOKYO

Japan's "ice-age" generation workers are squeezed, once again

FOR NEARLY two decades Torigoe Atsushi, who is 50, has worked in the stockrooms of a fashion firm in Tokyo. His income rose only modestly, and it recently fell after his employer cut bonuses and reworked its pay scheme; he now earns about ¥3.6m (\$22,600) a year. Then he spotted the firm's recruitment ads, offering newcomers higher salaries. "It's great that younger people are getting better pay," he says. "But why not raise mine?"

Mr Torigoe belongs to Japan's "ice-age" generation: workers who entered the labour market between 1993 and 2004, after the country's economic bubble burst and many companies stopped hiring. The ice-agers often endured years of insecure work and low pay. Now in their 40s and 50s (thus Generation X), they are losing out again. Japan is emerging from decades of deflation, prices are rising and companies are handing out their biggest pay increases in a generation. But the ice-agers are not ben-

efiting. Worse may come when this cohort of baby-boomers' children—the last bulge in Japan's population pyramid (see chart on next page)—drifts into retirement.

The ice-agers' fortunes contrast starkly with those of the generation that came just before. During the bubble years of the 1980s, companies hired voraciously. College students found their mailboxes stuffed with offer letters, or were treated to fancy meals by employers keen to woo them. Everything changed as the bubble popped in the early 1990s. Japan's rigid employment system made it hard for firms to

shed existing workers, so they cut hiring instead. The ratio of job openings to college applicants fell from nearly three at its peak to below one by 2000. Nagai Toshihisa, who graduated in 1999, shortly after the Asian financial crisis, abandoned his hope of working for a securities firm. "It felt utterly stupid looking for jobs," he recalls. Many in his generation sent out dozens, even hundreds, of applications, to no avail. It did not help that the ice-age cohort was such a big one, adding to the oversupply.

Many graduates who failed to secure regular jobs ended up in temporary or contract work. Japan's labour market sharply distinguishes between regular workers, who enjoy strong legal protections and seniority-based pay rises, and irregular ones, who often do similar work for much less money and are the first to be laid off in downturns. Such jobs often used to be filled by housewives supplementing their husbands' salaries. Increasingly, they were taken by young people who needed to support themselves, says Kondo Ayako of the University of Tokyo. The term "freeters"—youngsters drifting between casual jobs—became popular, as did "parasite singles", who lived off their parents well into adulthood. There was also growing concern about *hikikomori*, social recluses.

Time and acute labour shortages have repaired some of the damage. The share of ►►

→ ALSO IN THIS SECTION

29 The decline of the Indian elephant

30 Donald Trump upsets Asian allies

30 Kazakhstan's cosmetic election

31 Ashoka: The new founding fathers

► ice-agers in regular employment has largely caught up with that of older cohorts, says Shimoda Yusuke of the Japan Research Institute. But their earnings have not. Those who spent long stretches in irregular work tend not to fully catch up even after switching to the regular track. Those in regular jobs faced another problem: the upper ranks of their companies were crowded with the bubble-era workers, delaying promotions for those who followed. Japan's low labour mobility has meant the unlucky cohort cannot easily seek better pay elsewhere.

Japan's economy is changing. After three decades of stagnation, wages are rising as inflation returns, and job-hopping is becoming more common. Yet these improvements are mostly benefiting the young. Companies short of labour are competing fiercely for fresh blood, and paying more to attract and retain it. Middle-aged employees are much less likely to switch jobs, giving firms little incentive to raise their wages. According to Kumano Hideo, an economist at Dai-ichi Life Research Institute, nominal wages for college-educated workers in their 20s and 30s were 10-16% higher in 2025 than five years earlier. Meanwhile, those in their early 50s suffered a nominal decline of 1.3%. Abe Katsuya, 48, says the large pay rises dished out to the young give him "mixed feelings". He is part of the "Mammoth Club", a support group for ice-agers who identify with the woolly beasts that endured harsh times.

All this has left the ice-agers with low lifetime earnings (and therefore little in the way of savings). That points to a looming crisis as they reach old age. Japan's pension system ties payouts to lifetime earnings, so decades of suppressed pay will translate into low pensions. "Those who happened to come of age in a down-

turn are likely to carry that handicap for the rest of their lives," laments Mr Kumano. Getting them to work longer, as a growing number of old people already do, is one fix but it will not solve everything. Ms Kondo warns that many may end up on welfare, which is not designed to support the elderly over extended periods. That would be costly; for instance, the state pays welfare recipients' medical bills in full. And because the generations behind the ice-agers are smaller, fewer workers will have to support more pensioners.

The woes of the ice-age generation have long had the government's attention.

For decades, authorities have come up with programmes to help them find jobs and nudge reclusive ones to connect with the outside world. Yet as the ice-agers grow older they face new problems. Beyond thin pensions and savings, questions loom, for instance, over housing; compared with older Japanese, this cohort is less likely to own homes, and Japan's rental market is unfriendly towards older tenants. The government is belatedly adjusting: in April, it produced a three-year plan for the ice-agers, spanning measures from asset-building to housing. But a lot more serious thinking will be needed. ■

India

No room to roam

MUMBAI

Indian elephants are being squeezed out of their habitats

IN "ARTHASHASTRA", an ancient Indian manual on statecraft, elephants were prized assets. Kings presented them to forge alliances. Generals relied on them to destroy the gates of forts. Today Ganesha, the elephant-headed Hindu god, is still worshipped as the remover of obstacles, and children, as elsewhere, learn "E for Elephant".

But reverence has not translated into protection. A survey published last year by the Wildlife Institute of India counted 22,446 elephants, 18% fewer than in 2017. The decline in some states was "a shocker", says Qamar Qureshi, the report's lead author. Part of the problem is geography. Unlike in Africa, villages are scattered throughout the forests of densely populated India. Elephants are often confined to narrow corridors between patches of forest. Vivek Menon of the Wildlife Trust of India says that in 2010 his team identified 88 such routes. Today there are about 150, reflecting shrinking habitats.

As infrastructure cuts through forests, elephants stray into farms. Mining and blasting sites also get in their way. In West Bengal *hulla* ("noise") parties see mobs hurl fireballs and firecrackers to drive the animals away, despite a Supreme Court ban. Electric fences sometimes block the elephants' path. But they are excellent physicists, says Raman Sukumar, an ecologist. They use their tusks, which are poor electrical conductors, to break fences, or toss logs to short-circuit them. Forest guards and workers steering elephants back to safety are "utterly under-equipped and woefully unprepared", says Prerna Bin-

dra, a conservationist. Most lack basic equipment such as torches.

But not all is doom. Poaching for ivory has fallen since the 1980s. Co-ordination between state forestry departments and national customs has improved. Today India has a Wildlife Crime Control Bureau and states have anti-poaching plans. Last month India's Supreme Court upheld the removal of commercial activity from an elephant corridor in Tamil Nadu.

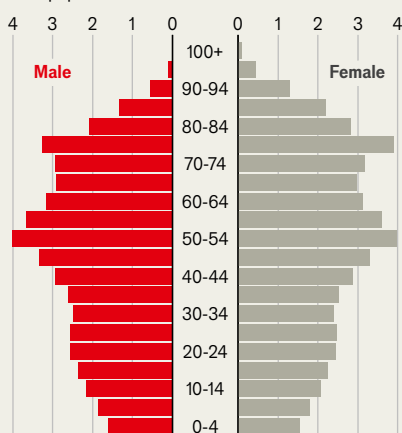
Elephants act as landscape architects. They disperse seeds, shape forests and fertilise soil. They keep wetlands alive by digging. "Arthashastra" advised India's rulers to protect forests inhabited by the magnificent beasts. Those in charge today would do well to take note.



Never forget

Boomer bulge

Japan, population by age group, 2026, % of total
Total population=122.4m



Source: UN Population Division

America and South Korea

Alliance mismanagement

Donald Trump abruptly scales back an exercise with South Korea

ULCHI FREEDOM SHIELD, an annual military exercise, brings thousands of American and South Korean troops together to train to counter threats from North Korea. This summer's edition began on August 17th, and the allies planned to focus on responding to new capabilities that North Korean soldiers acquired while fighting in Russia's war against Ukraine, such as drone warfare. But the night before the exercises were due to start, President Donald Trump ordered his Department of War to "substantially reduce" (by six days) the manoeuvres, citing the cost of the drills, South Korea's refusal to support his war in Iran and his "very good relationship with Kim Jong Un", North Korea's dictator. Mr Trump presumably hopes to entice Mr Kim to restart the diplomatic engagement the two men began during his first term in office. He may also want to press South Korea into accelerating its promised investments into America. And he doubtless hopes to distract attention from his Iranian quagmire.

But his abrupt decision sends worrying messages to both friends and foes across Asia. The prospect of more talks between the leaders, however unlikely, causes anxiety in the capitals of American allies. (North Korea called Mr Trump's move an empty gesture, though Mr Trump insisted on August 19th that he would be meeting Mr Kim later this year.) Japan, in particular, worries that Mr Trump could cut a bad deal that undermines its security. The move heightens fears of American abandonment that have been growing since his return to office. Adversaries may conclude that America's commitments to its allies are hollow.

This is not the first time Mr Trump has used exercises with South Korea as a bargaining chip. When he met Mr Kim in Singapore in 2018, the first summit between American and North Korean leaders, Mr Trump said that America would halt upcoming drills with South Korea. The move stunned his own advisers. But the unilateral concession did not bring about a breakthrough; a subsequent meeting between the two men, in Hanoi the following year, collapsed in acrimony.

South Korea's president, Lee Jae Myung, shares Mr Trump's desire to revive engagement with the North. On August 19th-20th his government hosted China's foreign minister, Wang Yi, in Seoul, in part

to discuss ongoing efforts to reach out to Mr Kim. Yet Mr Lee also acknowledges the importance of maintaining deterrence, which Mr Trump has weakened.

Mr Trump has long been sceptical of the value of America's alliance with South Korea. During his first term he sought to withdraw American troops from the country, but was stymied by Congress. Since returning to office, his administration has made quiet but consequential changes to the alliance. America has, for example, moved to reorient the mission of its 28,500 troops in South Korea away from their traditional focus on the peninsula and towards deterring China. He has also used tariffs to extract a pledge from South Korea to invest \$350bn in America. Mr Trump has not hesitated to link the future of the security alliance to the flow of investment deals. His decision to curtail the exercises came as South Korea's industry minister was visiting Washington to discuss the stalled implementation of those plans.

Even before the latest announcement, Mr Trump's transactional approach to the alliance had undermined trust in America. Polling by the East Asia Institute, a think-tank in Seoul, found the share of South Koreans who trust America plunged from 85% in 2022 to 46% this year. Fear of abandonment has also fuelled discussions about South Korea acquiring nuclear weapons. Mr Lee has said he opposes that course. But he has sought to develop the capacity to enrich and reprocess nuclear fuel for the civilian nuclear-power industry, which would make building the bomb easier if South Korea ever deemed it necessary. Following Mr Trump's announcement, he reiterated his intention to move forward with those plans.

Similar dynamics are at work across the region. Perceptions of American reliability have fallen steeply between 2022 and 2026

in Australia, Japan and Singapore, according to the Pew Research Centre. Japan has also sought to beef up its own defences; the drive for greater autonomy is shaping revisions to its national security strategy expected later this year.

Yet in the face of threats from a richer and increasingly assertive China and a nuclear-armed North Korea, American allies in Asia find it hard to imagine their security without America. Despite the falling level of trust, more than 60% of South Koreans still see the alliance with America as the best way to ensure their security. Such results, writes Sohn Yul, the president of the East Asia Institute, reflect a "contradictory situation where [South Korea] does not trust the United States yet has no choice but to rely on it." The dilemma is shared across Asia. ■

Kazakhstan

Reset or repress?

ALMATY

A cosmetic parliamentary election in an authoritarian ex-Soviet republic

"VOTE FOR renewal," urge giant billboards as Kazakhstan heads into a parliamentary election on August 23rd. Kassym-Jomart Tokayev, the president, calls this a turning point in the building of a "new Kazakhstan"—shorthand for a political reset that began after civil unrest demanding political change had left at least 238 people dead in 2022.

At first glance, much about this election is indeed new. Taking place under a new constitution, it will deliver a parliament with a new structure and a new name, certain to be dominated by a new party. This, ostensibly, is the reforming "new Kazakhstan" in action. But scratch the surface and the political trappings of the old Kazakhstan—associated with Nursultan Nazarbayev, Mr Tokayev's long-serving predecessor and discredited political patron—emerge. The newly formed Justice party that is set to win a big parliamentary majority is in fact a revamped version of the old ruling party that dominated for decades. And parliament will still be a rubber-stamp affair, because no genuine opposition parties are being allowed to run.

Pluralism is thriving in Kazakhstan, so Mr Tokayev said recently—only days after Amangeldy Dzhakhin, an opposition leader, had been jailed for seven years on spurious charges of extremism. His predecessor suffered the same fate, and it is not just politicians who are targeted. Reporters and social-media personalities have also been variously imprisoned, slapped with



There, but for how long?

▶ travel bans and barred from doing journalism on flimsy-looking charges. In a blatantly political case, a satirical post poking fun at the president recently landed a YouTuber behind bars.

This election follows the adoption of a new constitution, which Mr Tokayev has depicted as Kazakhstan “casting off the super-presidential form of rule”. That was a reference to his predecessor, who monopolised power for three decades before resigning in 2019. Mr Nazarbayev’s designat-

ed successor was Mr Tokayev. The pair ruled in tandem until the 2022 unrest, which left Mr Nazarbayev sidelined and Mr Tokayev scrambling to dissociate himself from his former mentor. He rolled back some presidential powers in 2022, but clawed some back again in the new constitution, adopted by referendum against a backdrop of arrests of critics.

One important reform remains in force. In 2022 Mr Tokayev introduced a single presidential term-limit. He has pledged to

abide by this, but the constitution offers him a loophole. He has the right to a new “first” term, the Constitutional Court recently ruled, so he could run again—a tried-and-tested tactic to cling to power loved by Central Asian autocrats as well as by Vladimir Putin in Russia and Recep Tayyip Erdogan in Turkey. The end of his current (second) term in 2029 will thus become a litmus test of Mr Tokayev’s professed commitment to ridding Kazakhstan of monarchical rule. ■

ASHOKA *Refounding fathers*

Nation-building is even harder the second time round

EVERY COUNTRY has a national anthem. Some, like New Zealand, have two. But only India has an official song as well. This year, for the first time, “Vande Mataram”, a rousing tribute to “Mother India”, featured in Independence Day celebrations, greeting Narendra Modi as he arrived to give his annual speech on August 15th. An air-force helicopter flew past with a banner commemorating 150 years of the song’s composition. It marked the culmination of a years-long effort by Mr Modi’s Bharatiya Janata Party (BJP) to boost its status. A few days earlier it had become a criminal offence to disrupt its singing.

Mr Modi spoke of India’s achievements as it enters its 80th year of independence. He looked ahead to its glorious centenary. Such pronouncements acknowledge that India gained freedom from the British in 1947. Yet his BJP believes the country has been truly free for much less time. The clearest articulation of this sentiment came from Kangana Ranaut, a Bollywood actress and since 2024 a BJP lawmaker, who declared in an interview some years ago that India only achieved independence with the election of Mr Modi in 2014.

The statement is one of those to be taken seriously, not literally. In the eyes of the BJP’s leaders, the India they inherited was too effete, too anglophone and too accommodating of minorities. Their project was to build a “civilisational state defined by the country’s ancient Hindu heritage”, as Milan Vaishnav of the Carnegie Endowment, a think-tank in Washington, put it in a paper in 2024. He described the new political order as India’s “Second Republic”.

A new republic needs a new identity. Around independence in 1947 the constituent assembly settled on what India’s

national symbols should be. Over time, civic bodies picked up the statuary of British kings, queens and sundry dukes and deposited them in far-off parks (or, deliciously, in the zoo). State governments renamed cities and institutions.

The BJP has been on a similar mission since 2014. The obvious purpose is to promote its Hindu-first vision. The national song was passed over for anthem status in part because its invocation of Hindu deities was seen as excluding Muslims. That is no longer a concern. In recent years the government has changed to saffron—a colour associated with Hinduism and the party itself—its flagship train service, the logo of the public broadcaster and, last month, the uniform of the men’s national hockey team. Last year it overhauled a jobs scheme named after Mahatma Gandhi and rebranded it as VB-G RAM G, a tortured acronym reverse-engineered to include the name of Lord Ram.

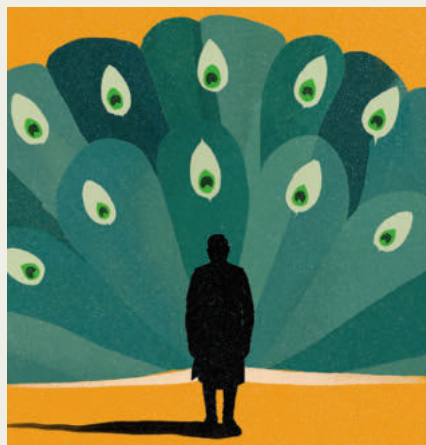
For all that, it is the non-religious parts of the BJP’s project that more closely resemble the nation-building of a young country. Along with national symbols and

a shared vision of the future, a new republic must have an origin story to give it meaning. Unfortunately for the BJP, the freedom movement and its pantheon of heroes already fulfilled that purpose. That left two strategies.

The first was to repurpose long-dead figures. Jawaharlal Nehru, India’s first prime minister, was demoted from nation-builder to incompetent, entitled scion of the elite. In his place the BJP’s myth-makers built up Sardar Vallabhbhai Patel, Nehru’s home minister, who they insist was robbed of the top job. (Mr Modi’s home state of Gujarat has immortalised Patel with the world’s tallest statue.) In any case, they say, Subhas Chandra Bose, a revolutionary with no time for Gandhi’s non-violence humbug, was the real first prime minister.

The other is to lean on the one towering figure they do have. In June the cabinet feted Mr Modi for becoming India’s “longest serving elected PM for consecutive terms”. The heavy caveating excludes Nehru’s first five years (elections had not yet been held) and Indira Gandhi (she lost power and then regained it). The animus against Nehru is understandable. There cannot be two first prime ministers of independent India.

The BJP may have saffronised India. But its revisionism has failed to take. That much was clear during protests for education reform last month in Delhi. Students carried copies of the constitution. They displayed photos of B.R. Ambedkar, its architect; Bhagat Singh, an anti-colonial revolutionary; and Gandhi. They chanted slogans dating to the freedom struggle. Despite having spent most of their conscious lives in the second republic, their symbols were those of the first. Building a nation is no easy task. Retrofitting one is harder still.



China



China in the Arctic

Pole position

The Northern Sea Route is an alternative to the Suez Canal—albeit costly and risky

FOR A WORLD newly preoccupied with maritime choke points, the voyage of the *Dubai Tower* offers a flicker of promise. The container ship set off from the Chinese port of Ningbo on August 15th to begin the first regular cargo service between Asia and Europe via the Arctic Ocean. Sea Legend, the vessel's Chinese owner, says that it will complete the 5,000km journey to Felixstowe, on Britain's east coast, in about 20 days. That is roughly twice as fast as sailing the usual route via the Suez Canal. Crucially, the "Arctic Express" (as the company calls the service) avoids the risk of being entangled in the Middle East's military conflicts.

The new service via the so-called Northern Sea Route (NSR), which runs through the Bering Strait and over the top of Russia, is an early test of China's Arctic ambitions. In 2018 the country outlined plans to develop shipping routes, mining and infrastructure in the Arctic, hoping to

take advantage of rapidly receding polar ice. Many of the plans, which are marketed collectively as a "polar silk road", have met resistance. Seven of the eight countries with Arctic territory are Western democracies. They worry that China is working closely with Russia in the region and that Chinese activities there could be useful for future military purposes, such as operating submarines. Some bridled at China, whose northernmost point is on about the same latitude as Hamburg, declaring itself a "near-Arctic state".

Nonetheless China hopes to attract fresh interest in the northern route from governments and companies alarmed by the closure of the Strait of Hormuz and the

threat from Iranian-backed Houthi fighters to shipping in the Red Sea. China is also more anxious than ever about its own reliance on the narrow Malacca Strait, the hinge between the Indian Ocean and the Pacific, through which most of its energy imports pass. And China will soon not be alone on the NSR. In June Japan said that it would revise its Arctic policy to promote the route. A South Korean shipping company, PanStar, plans to send one of its container ships on a trial run through the Arctic Ocean on August 22nd, aiming to complete the voyage from Busan to Felixstowe in just 18 days.

It's no sleigh ride

Thanks to climate change, conditions for the NSR are likely to grow more favourable. The minimum summer extent of Arctic ice has been shrinking by more than 12% a decade. In March the maximum extent of winter ice reached its lowest level since records began in 1979. On August 16th, the latest available reading, Arctic ice covered 5.5m square kilometres. That is well below the median extent of ice on the same date in the years between 1981 and 2010 but above the lowest ever extent, recorded in 2012. That September, the ice shrank to just 3.4m square kilometres.

One selling point for Sea Legend's service is that a shorter sailing time means ►►

→ ALSO IN THIS SECTION

33 Sweet on chocolate

34 Chaguan: Nostalgia for the 1990s

▶ less fuel burned—fuel accounts for seven-tenths of shipping costs. That also reduces carbon emissions. And cooler weather on the Northern Sea Route is better for transporting some high-tech goods that are sensitive to temperature. Without refrigeration, temperatures inside containers on the Suez Canal route can reach 60°C. The *Dubai Tower* is carrying goods including batteries, solar modules and electric-vehicle components, mainly from cities in China's Yangzi river delta.

For all those apparent advantages, the Arctic Express also highlights the limitations of the NSR. It will run only from August to October, when the route is largely free of ice. Sea Legend plans eight voyages in that window, using seven vessels. For the rest of the year, only ships with reinforced hulls may navigate the Arctic. Sea Legend has bought or leased some of these, but only relatively small ones, as they are costly and in short supply. They may also need to be escorted by Russian nuclear-powered icebreakers. All the extra costs erode the northern route's price advantage.

So do insurance premiums. They are usually higher for the NSR than for the Suez Canal, despite war-risk premiums for the canal. They are also 40% higher than for sailing around the Cape of Good Hope, the chief alternative to the Suez Canal. Allianz, an insurance giant, warned in its recent annual safety review of the shipping industry that the remote Arctic route posed significant extra risks because of its harsh weather and the limited infrastructure available for repairs, assistance, rescue or salvage. It said that machinery damage or failure caused almost half of the reported shipping incidents in Arctic waters in the past decade.

The environmental impact is potentially greater, too. Bellona, a Norwegian environmental group, warned in a report last year that in the Arctic there are no effective ways to deal with fuel spills because of the difficulty of cleaning up fuel in cold conditions and the distance from emergency-re-

sponse centres. (The International Maritime Organisation has banned the use and carriage of heavy fuel oil in the Arctic since 2024, though Russia does not heed the ban.) Noise pollution from ships also affects marine mammals such as whales and walrus in the Arctic more than elsewhere, since low-frequency sounds travel farther in cold water. Ships on the northern route pass through vital feeding and breeding grounds.

International carriers avoid the route in large part because of geopolitical risk. Ships on the NSR often need permits, ice-breaking services or other assistance from Russia, potentially violating Western sanctions linked to its war in Ukraine. Shipping may in future be disrupted by an intensifying contest between Russia and other countries to control the region's ample mineral resources. The Arctic, once preserved as a zone of peace, is becoming increasingly militarised, as Donald Trump continues to threaten Greenland and NATO seeks to deter Russian aggression on its northern flank.

All this helps to explain why total transit cargo on the Northern Sea Route was just 3.2m tonnes in 2025, up from about 3m the year before. That is tiny compared with about 464m tonnes through the Suez Canal last year. The bulk of Arctic shipping is between Russian and Chinese ports, rather than between Asia and Europe, much of it consisting of shipments of Russian oil or gas. In a recent report, the Clingendael Institute, a Dutch think-tank, concluded that the route's "most viable function...remains that of a selective corridor for Arctic resource exports."

Still, as Malte Humpert of the Arctic Institute in Washington argues, the NSR will grow in importance for China as a seasonal trade corridor for when traditional routes are disrupted. He envisages hundreds of annual container passages a decade hence. "The question is when Western operators decide to re-enter the market—and more importantly, whether they still can," Mr Humpert says. Arctic navigation requires training and operational experience, and "China and South Korea are building that expertise now."

For the moment, Sea Legend admits, the tight seasonal window and limited capacity of its Arctic service make it hard to compete with conventional routes. But it is already planning to expand the navigational window to four months by 2028 and eventually to the entire year. Further conflict in the Middle East could provide incentives for it and other companies to invest in larger, ice-strengthened ships. And if temperatures in the Arctic continue to rise three times faster than the global annual average, then doubts about the viability of the NSR may recede as fast as China's ambitions for the region grow. ■



*Approximate, dependent on routing
Sources: China Shipowners' Association; Sea Legend

Chocolatiers

Eating bitterness

SHANGHAI

Can Chinese become choco-mad?

CHINA HAS more coffee-shop brands and more KFC branches than anywhere else on Earth. But chocolate has never hit the spot, with the average Chinese eating just 100g a year—a mere hundredth of what a French person guzzles. A few mass-market foreign brands—Mars, Ferrero, Nestlé and Hershey—account for four-fifths of Chinese consumption. "Of all the chocolate eaten worldwide, only 2% caresses Chinese tastebuds.

This may now be changing, thanks to a handful of Chinese chocolatiers springing up around the country. Most are artisanal shops, but some are investing in supply chains and building big businesses. Take Choc Revive. Based in Shanghai, the firm was founded in 2023 by Xie Liang, a fresh-produce trader. He cannot open shops quickly enough. From 80 outlets now he hopes to have 300 by the end of the year. Mr Xie says that his company had 160m yuan (\$23.7m) in sales last year and hopes to quadruple that, or more, this year.

Choc Revive is trying to take chocolate local. It sources its own beans from Africa, but has set up an R&D unit in China and does its own roasting and grinding. Its big investments are intended to produce economies of scale, but the know-how it is accruing may spread and give rise to competitors. Another local chocolate company, Nibbo, has been winning international awards for its creamy sweets. Saturnbird, which started as a coffee grower in south-west China, is now planting its own cacao trees there.

Mr Xie's concoction is 58 yuan for a 100g chunk—equivalent to a cheap meal for two. It is an awkward time to be selling premium products to belt-tightening consumers. Churning out the cheapest possible goods to undercut rivals is the usual formula. Milk-tea brands are a case in point.

Why are people paying out for chocolate? Mr Xie says that making chocolate with Chinese flavours such as longan, which is like a large lychee, seems to go down well. A simpler explanation is that, at long last, Chinese people are learning to love the most delectable flavour in creation.

CHAGUAN

A glum China pines for the 1990s

Back then, leaders delivered and opportunity beckoned—or so the story goes



THE 1990S ARE alive again. Over the past few days viewers of state television in China have feasted on scenes from that decade. A 12-part documentary has aired in prime time to celebrate the centenary of the birth of the late Jiang Zemin, the country's leader throughout the period. What a decade it was, bookended, in 1989, by the blood of Tiananmen and, in 2001, by all the promise of China's entry into the World Trade Organisation.

This progression, which may seem inevitable in retrospect, was never anything but fraught. But what captivates viewers is less Jiang's accomplishments than a common touch and a panache that seem nearly inconceivable among China's leaders and their tightly scripted public comportment today. There he is in one scene in 1989, four months after the crackdown on the Tiananmen pro-democracy protests, hugging babies in Beijing. In another, Jiang slams a forehand in table tennis before telling national-team athletes that he hopes they win gold at the upcoming Asian Games. "Let me take that back. You shouldn't have too much pressure," he says with a big smile, laughter erupting in the room. In a third, Jiang is belting out Beijing opera at a formal dinner in California, a study in contrast with the buttoned-up nomenclature of today's China, Xi Jinping above all.

Then, as the documentary ran, a cosmic coincidence: on August 12th Jiang's essential partner, Zhu Rongji, died at the age of 97. If Jiang was the political maestro, Zhu, his prime minister, was the one who got his hands dirty. Online commentary praised Zhu's blunt, effective approach—an implicit rebuke of today's more cautious officials. The day he died, the documentary happened to feature Zhu explaining how he cajoled cadres into tough reforms: "I'm not here to talk about pros and cons. I am here to talk about how to get things done."

Even before the documentary and Zhu's death, nostalgia for the 1990s had been building. A viral trend last summer spoke of "the beauty of the boom years", with people sharing images of youthful exuberance in the 1990s and the early 2000s. A recent television hit, "Blossoms", about Shanghai in those days, captures the mood. At one point the narrator says: "Everyone stands equal before opportunity. Seize it, and you could change your life." Some

have echoed the sentiment as they remember Zhu. "Back then people had a spark in their eyes, certainty in their hearts and a tomorrow they could actually see," wrote Zhao Jian, an economist, in a widely shared piece—before it was scrubbed from the internet.

What to make of all this? A cynical take is that memory is playing its usual tricks. If the 1990s now look like a decade of possibilities opening up, recall that China was so much poorer then, on the cusp of high-speed development. In 1990 its GDP per person, adjusted for inflation, was just \$920; today it is 15 times higher. Or consider the rise in university graduates: 12m a year now, up from about 700,000 in the 1990s. In the Jiang-Zhu era nearly three-quarters of China's population still lived in the countryside, which was mired in poverty. No wonder urbanites lucky enough back then to get a good education felt that destiny was theirs to chart. Yet for most in China, the path ahead was one struggle after another.

So the nostalgia should not be read as a rigorous account of history. Rather, it acts as the reflection of a malaise today. Most obviously, people are down about the economy. Job opportunities are pinched, property has slumped and many entrepreneurs are struggling. Yet to dwell on these issues carries consequences, with, for instance, censors erasing posts about the economy that lack "positive energy". For some, harking back to how good things were in the 1990s is a way round such controls. For others, it may simply be a longing for a better life.

There is also great frustration about politics in China today. Scholars debate whether Jiang and Zhu really deserve their "reformist" labels. Political reformers they certainly were not: Communist Party elders chose them to reinforce party will and pick up the pieces following the Tiananmen trauma. As economic liberalisers, there were also limits. Their signature overhaul of state-owned companies was to "grasp the large and let go of the small", hiving off useless businesses while modernising the important ones in finance, infrastructure and heavy industry. The aim, as Jiang put it, was to retain control of "the lifeline of the national economy". Above all, they aimed to strengthen the state and entrench party rule.

They do things differently there

For all that, Jiang and Zhu were certainly daring, taking bold risks against real resistance. Conservatives (ie, leftists) opposed their market-driven policies, worried that opening China to trade with the world would suck in toxic political ideas. Many reforms also exacted a vast human toll, with more than 30m workers laid off—though the eventual benefits were larger. Do today's leaders have similar vision and courage? True, they have been in charge during some high-tech advances. Yet a trade war with America drags on, while property remains stuck in its years-long funk. Tributes to Zhu came not just from ordinary people but from universities, legal associations, financial groups and think-tanks. It is hard not to see their veneration as a plea for wiser leadership.

Beyond that is something unquantifiable about the style of the leaders in the 1990s—about the frankness with which they spoke, their sense of warmth and real humanity, their ease when out and about among the public. Certainly, people were also wary of Jiang and Zhu; under them, China was hardly free. But they were admired and even loved. Today, when people talk of Mr Xi, the dominant emotion is often fear. Many Chinese are so circumspect as to avoid mentioning his name altogether. He is the paramount ruler, not a chap who cracks jokes or bursts into song. ■

Middle East & Africa



The West Bank

Emboldened, and violent

QUSRA

With government support, Israeli settlers are escalating attacks on Palestinians

THE HILLS south of Nablus in the Israeli-occupied West Bank tell the story of a creeping annexation. Winding dirt tracks to Palestinian villages are blocked by rocks and piles of earth, cutting off access from fields and olive groves. Israeli settlers have planted vines in neat rows, erected fences festooned with flags and brought in herds of goats. They patrol the area with new utility vehicles and assault rifles given to them by the Israeli government.

Their stated aim is to cause “friction” with adjacent Palestinian villages. That tends to lead to intervention by the Israel Defence Forces (IDF), often on the settlers’ side, followed by the expulsion of Palestinians and the expansion of outposts of Israeli settlements (see map on next page). Since the Hamas attack of October 7th 2023, and Israel’s subsequent war in Gaza, violent attacks by Israeli settlers in the West Bank have worsened and forced an increasing number of Palestinians to flee

(see chart on next page). As Israel prepares to go to the polls on October 27th, the failure of Binyamin Netanyahu, the prime minister, to rein in the settlers is making counter-violence by Palestinian militants increasingly likely.

Events over the past week show how much leeway settlers are granted. Tel Talpiot, the latest and northernmost outpost in the Shiloh settlement bloc, was set up earlier this year on a hill overlooking three Palestinian villages. On August 9th a group from Tel Talpiot occupied the courtyards of homes on the outskirts of the village of Qusra, setting up a makeshift shelter, ter-

rorising the families and forcing them to hole up inside. IDF soldiers sent to “prevent friction” joined the settlers in prayer, rather than dislodging them.

The tactic, common in the West Bank, often succeeds in forcing Palestinian villagers to flee their homes, allowing the settlers to take over the surrounding land. This time was different for several reasons. Residents of Qusra, who had experienced similar attacks before, were quick to publicise this one and attract international attention. One of the besieged homes belongs to an American citizen. Mike Huckabee, the American ambassador, who usually supports the settlers, posted on social media that it was his embassy’s “request to remove the Israeli terrorists”.

Three days into the siege, the IDF belatedly replaced the unit in Qusra with a more disciplined infantry battalion. It tried to remove the settlers and evict the Tel Talpiot outpost. The settlers brought in reinforcements and remained in the area. “It was an embarrassing incident,” said one officer. “The settlers forced us to play cat-and-mouse, and we lost.”

The IDF announced the homes were in a “closed military zone”. It forced the families to remain blockaded indoors and prevented activists from coming to their aid and media from visiting. “We are besieged, the army is not allowing us to leave our

→ ALSO IN THIS SECTION

36 Consolidating power in Iran

37 South Africa’s broke municipalities

38 Somaliland’s president has big plans



house,” said Qusai abu Ridi, one of the residents. But he was adamant they would continue to protect their property.

That makes Qusra, despite everything, a rare case of Palestinian residents managing to hold on to their homes. Yet the incident is also part of a fresh escalation over the past month, as settlers have tried to grab as much land as they can before the election, which might return a government less supportive of their cause. On July 24th a group of armed settlers arrived in the village of Tel on what they claimed was a “hike.” Four Palestinians and two Israelis ended up killed. In response, the IDF deployed more troops and locked down Palestinian towns, including Nablus, one of the largest in the West Bank.

The IDF claims to be enforcing order, but it has overseen a steep rise in attacks on Palestinians. The Shin Bet, Israel’s security agency, says 660 “nationalistic” attacks took place in the West Bank in the first half of 2026, up from 405 in the same period last year. More than 1,100 Palestinians have been killed in clashes with the IDF and settlers since October 7th 2023.

The events in Qusra highlight the IDF’s disinclination to prevent settlers’ attempts to chase out Palestinians. Senior officers, both serving and retired, blame a breakdown in discipline and the presence of settlers within the IDF’s ranks. But ultimately they hold the government responsible.

While the IDF’s high command has publicly condemned the attacks in Qusra, Binyamin Netanyahu, the prime minister, has remained silent. On August 14th Israel Katz, the defence minister, said that enforcing the law on settlers in the West Bank would, from now on, be the responsibility of the Israeli police and not the IDF. Since the police lack personnel in the West Bank and are under the control of the far-right national-security minister, Itamar Ben-Gvir, himself a settler, they are unlikely to enforce the law in the area.

Israel deploys vast forces and intelligence resources in the West Bank to protect half a million settlers. In theory, the IDF is obliged to protect Palestinians, too. But soldiers will only do this if their politi-



cal masters expressly tell them to. That is unlikely to happen under Mr Netanyahu, who needs the support of settler-friendly far-right parties to have a chance of retaining power after the election.

Things in the West Bank could get worse before then. Security officials warn that settlers are trying to provoke another intifada, a Palestinian uprising, which will force the IDF to enter Palestinian cities. That would bring even more violence and misery to the West Bank’s embattled Palestinians. Even if Mr Netanyahu’s party were voted out, his successors may have to deal with a crisis of his making. ■

Iran and the war

Rise of the generals

A military junta is consolidating power, with dire results at home and abroad

OVER THE past few days new billboards have appeared in Tehran, Iran’s capital. They depict the seven men who have taken charge in Iran since Ali Khamenei, the former supreme leader, was killed in an Israeli air strike on February 28th. Most remarkable is what the billboards do not show. There is no sign of Mojtaba Khamenei, the new supreme leader, who has not been seen in public since he was appointed. Nor are there any of the traditional religious symbols, such as the black and red flags of Imams Ali and Hussein. Only one of the seven men wears a turban.

The billboards are the latest sign that a new crop of leaders from the Islamic Revolutionary Guard Corps (IRGC) is consolidating power in Iran. Battle-hardened and results-driven, they are focused on security and repression, and are turning the theocratic foundations of the Islamic republic into a military dictatorship. Though they may not have entirely dropped a pragmatic

approach to the conflict with America, military regimes often draw legitimacy from the constant threat of war. That suggests America and its allies should expect the new leadership to be more uncompromising than the previous one.

Fronting the self-declared “top of the country’s security and defence” on the billboard and dressed in a suit is Mohsen Rezaei. He is the new national-security adviser, a former presidential candidate and the longest-serving chief of the IRGC, which he led through the Iran-Iraq war of 1980-88 and the reconstruction that followed. Five men wear military fatigues, including the head of the navy who oversees Iran’s attacks on Gulf shipping. Next to him is Ahmad Vahidi, the IRGC’s commander-in-chief. Besides being instrumental in attempts to suppress the women’s movement to cast off the veil four years ago, he founded the Quds Force, the IRGC’s foreign arm, which nurtures its network of regional proxies. Hussein Taeb, the only man with a turban, used to run the IRGC’s intelligence agency and now heads the *basij*, its paramilitary volunteer force entrusted with keeping protesters off the streets.

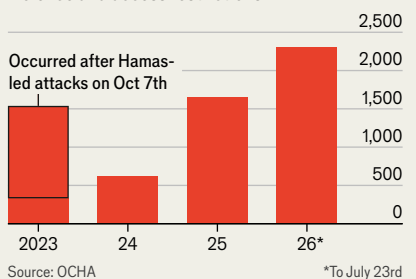
Some veteran Iran-watchers see the men’s takeover as the triumph of hardliners, who seek the expulsion of America from the region, over pragmatists led by the speaker of parliament, Mohammad Bagher Ghalibaf. He was instrumental in negotiating a memorandum of understanding (MOU) with America that was signed on June 17th. The MOU promised Iran an end to nearly five decades of American opposition, protection for its regional proxies, sanctions relief, the repatriation of frozen oil revenues and \$300bn to kick-start post-war reconstruction, in return for Iran eschewing its pursuit of a nuclear bomb and ending its blockade of the Strait of Hormuz.

Fighting resumed within days of the document’s signing. The Strait of Hormuz remains largely closed to shipping. On August 17th the deadline the MOU set for a more permanent peace deal expired; a day later Iran appeared to lob missiles at the United Arab Emirates, an American ally in the Gulf, which responded with a trade embargo. Some observers worry that the hardliners’ sense of having parried American and Israeli attacks has revived their conviction in the viability of force. “They’re squandering the victory [with the MOU],” fumes an academic in Tehran.

Domestically, too, the new regime looks harsh. The morality police are back in force in Tehran, raiding and closing cafés perceived as too lax on enforcing the veil. Public executions of anti-regime protesters have resumed. This month a photographer who captured Iran’s social malaise was gunned down with his mother and sister near a city in the north-west. ▶▶

Driven out

West Bank, Palestinians displaced by settler violence and access restrictions



▶ Yet other insiders and observers see more calculated realpolitik. The generals approved Mr Ghalibaf's MOU. But without pressure, they are said to have reasoned, it stood no chance of being implemented. Despite the MOU, America set up its own blockade over the strait and negotiated the disarmament of Hizbullah, an important proxy, in concert with Israel. Iran could not even secure the transfer of frozen revenues that had been part of the deal.

The new regime could be keen on an agreement that delivers finance. After the Iran-Iraq war, Mr Rezaei spearheaded Iran's reconstruction. As a presidential candidate, he called for foreign investment and a nuclear deal. "They're saying isolation will kill us," says another Iranian academic with ties to the new leaders. Nor, say some, are they fazed by diminishing Islamic norms. The crackdown on cafés may be merely a sop to their base. Some liken them to Muhammad bin Salman, the Saudi crown prince who is pursuing social liberalisation in place of political reform.

Yet all observers agree that there is a new style of leadership. For almost four decades Mr Khamenei balanced the competing interests of his realm, alternating reformists and hardliners. The new commanders, by contrast, favour action. Iranians and the world may yet rue the day Mr Khamenei left the scene. ■

Small-town South Africa

State capture in miniature

MPOFANA

Failing municipalities stymie growth, but some show change is possible

"YO, THERE'S A lot that's bad in this community," sighs Neliswa Radebe. The 19-year-old is one of 13 people living in a small ramshackle house above a motorway in Mpofana, a municipality in eastern South Africa. She lists the problems: piped water is rare; rubbish collection is sporadic; and there is a leak dripping on the mattress used by her bed-bound adopted mother. Outside is no better. The roads are more pothole than tarmac; a resident died recently when a car careered into them. There are no playgrounds, football fields, netball courts or youth centres. Squatters have overrun Mpofana's dilapidated town hall, on the other side of the motorway.

Mpofana is part of a bigger problem. Most of South Africa's municipalities—the units of local government meant to deliver basic services—are a mess. Their failure makes life worse, especially for the poor, and stymies a wider economic recovery. That is weakening the African National



Where the grass is not greener

Congress (ANC), the once-mighty party. If the ANC receives the expected drubbing at local elections on November 4th, it may shake up South African politics.

Under apartheid, local government reflected the country's systemic racism. White areas were home to commercial hubs and rich suburbs; business and property taxes were spent on infrastructure that would not have been out of place in California (swimming pools that used a lot of water). Black areas were marginalised, with infrastructure that looked more like that elsewhere in Africa, rather than in its industrial heart (tap water was a luxury).

After Nelson Mandela was elected in 1994 his government abolished racially segregated local authorities, creating ostensibly integrated municipalities. Today these range in size from Johannesburg, the commercial capital, to units like Mpofana that comprise a small town and rural areas. All are supposed to fund themselves through property taxes, business rates and charges for water, electricity and waste collection. The poorest also get a grant from the central government to fill any gap between revenue and expenditure.

In reality, many do not balance the books. In the financial year that ended in June 2025 45% of municipalities passed unfunded budgets. Some 30%, including Mpofana, have done so in each of the past four years. Last month Enoch Godongwana, the finance minister, said he would withhold grants to the worst-offending places to instil fiscal discipline.

Public-sector trade unions argue that the grant increases have lagged behind the rising costs of providing services. But the bigger issue is that money is wasted or stolen. Just 15% of municipalities achieved clean financial audits in the auditor-gener-

al's latest report, in June. Since 2021, "wasteful", "irregular" or "unauthorised" spending has added up to 287bn rand (\$18bn), more than South Africa has spent on defence in the same period. In a grimly symbolic case of apparent municipal graft, ten individuals, including local officials, allegedly pilfered funds designated for VIP (ventilated improved pit) toilets in Amathole in the Eastern Cape, one of the poorest places in the country.

Municipalities collectively owe more than 110bn rand to Eskom, the state-run utility, and 30bn to bulk-water suppliers. These debts threaten the "financial sustainability" of the economy, argues Mr Godongwana. His decision to suspend grants was partly in order to use the money to pay these bills. (On July 31st he began reinstating the grants, with new conditions.)

As if taking their cue from politicians, residents often do not pay up either. Almost half the water procured by municipalities is "lost": it flows out of broken pipes or is never paid for. In Mpofana 60% of electricity is in effect stolen. "Honestly, we do not pay," Ms Radebe admits, glancing behind her at the coils of rusty cables that lead to an illegal connection outside.

Yet some municipalities work. South of Mpofana is Umngeni, which in 2021 ditched the ANC and elected a mayor from the Democratic Alliance (DA), the second-largest party. Chris Pappas has been credited with an overhaul. In July the FW de Klerk Foundation, an NGO, rated Umngeni as the seventh-best-run municipality in a sample of 50, ahead of Cape Town.

Today Umngeni has funded budgets and receives clean audits. Poor-performing bureaucrats have left or been fired. Mr Pappas has cracked down on inflated procurement schemes. The savings have helped to ▶▶

erect 3,000 new streetlights, fill potholes and expand a scheme for the very poor that subsidises small amounts of water and electricity. The once-dirty town centre is clean. New firms have moved in. More than 2,000 net new jobs have been created in an area of 100,000 people.

In Zanzene Village, a poor part of Umngeni, residents seem to appreciate the changes. "Everything is good: the roads, the water, we even pay for electricity," says Nomfundo Buthelezi, explaining that she does not mind so long as the money is well spent. Musa Mchunu, a gardener, notes there is a new football pitch and a youth centre. "Most importantly," he adds, "the municipality now responds—we no longer get the blue ticks," a reference to WhatsApp messages that are read but ignored.

The DA hopes that a record of good governance will increase its vote share, including among black voters, who historically have opted for the ANC or its off-

shoots. Among its targets is Johannesburg, where one recent poll has the DA on 42%, up from 26% at the 2021 local election. The ANC, which has run the city into the ground, is on 18%, barely half the 34% it won in 2021.

Such highs will not be reached everywhere in South Africa. In the recent poll the ANC still remains ahead of the DA (34% versus 27%) on average. Race remains highly predictive of whether a voter opts for the DA, even though it runs just 12% of all municipalities but accounts for most of the clean audits.

Yet 32 years after the end of white rule, when most South Africans are despondent about democracy, showing that political choices can make a difference is crucial. Mr Mchunu, a former ANC member, says that he did not vote for Mr Pappas last time, but may do so in November. "I tell my sons that apartheid is over. What matters is not race but changing the country." ■

Somaliland

Irro's gambit

NAIROBI

The president wants to help the world secure the Red Sea

ABDIRAHMAN MOHAMED ABDULLAHI is an ambitious man. In December 2024 the 70-year-old former diplomat (pictured), who is better known as Irro, was sworn in as president of Somaliland, a breakaway republic that declared independence from Somalia in 1991. Last year he was the would-be country's leader when Somaliland was formally recognised by Israel, the first nation to do so. "The world has been ignoring Somaliland for 35 years," Irro tells *The Economist*. He wants outsiders to take a closer look.

Irro's pitch centres on the Red Sea. The trade route has been a mess of late, with shipping menaced by the Houthis, an Iran-backed militia that rules much of Yemen. Speaking this month during a visit to Nairobi, Kenya's capital, Irro says Somaliland can help secure the strategic waterway by offering military bases to global powers along its coastline. Once trade can flow unmolested again, he wants Somaliland's main port at Berbera to be the primary gateway to the markets of the Horn of Africa. "Our message is that we are a responsible partner" in security and trade, says Irro.

He suggests that Somaliland could, in effect, be the new Djibouti. That tiny port-state rakes in some \$125m a year by renting out slices of its coastline to foreign armies. America, China and France (among others) have bases there. It also earns many multi-

ples of that amount in annual port fees paid by landlocked Ethiopia, which depends on Djibouti for most of its international trade.

Irro wants a slice of both Djiboutian cakes. He is trying to persuade Ethiopia to use Berbera, where the United Arab Emirates (UAE) has built a new harbour, as its main outlet to the sea. He also wants to en-

tice America to set up a military base there. That would give America more options if it chooses to resume its war against the Houthis, whom it bombed intensively last year. Access to military facilities around Berbera is widely believed to have been part of the deal Israel struck with Irro when it recognised Somaliland in December. Some reports suggest that America and Israel may already be developing a military presence at Berbera, with covert support from the UAE, which operates an existing base there.

Irro denies such reports. "There is no Israeli or American presence yet," he says. But the UAE has another base at the port of Bosaso in Puntland, a neighbouring semi-autonomous province of Somalia, where unconfirmed reports suggest Israel is also embedded. As competition for military properties on the Red Sea heats up, Irro says he would be willing to offer both Israel and America bases at Berbera "if they are interested". Next month Irro will visit Washington, where he hopes to have a tête-à-tête with Donald Trump. Were that to happen, he would become the first president of Somaliland to meet directly with an American counterpart. "We expect the recognition of the United States some time in the future," says Irro, adding that Somaliland anticipates recognition from at least one more country "before the end of the year".

Irro's ambitions are ruffling regional feathers. Djibouti is irked by Somaliland's desire to barge in. Irro hints that his northern neighbour is trying to "destabilise" his country, presumably by supporting hostile militias on the border. Relations with Somalia are even more strained. Mr Irro claims Somalia, which considers Somaliland a renegade province, has plans to mobilise troops along the southern frontier. "This is not a peace recipe," he adds.

It does not help that Israel and Turkey, a patron of Somalia, are increasingly hostile to each other. Turkey has long been trying to mediate between Somalia and Somaliland, yet numerous rounds of talks in Ankara and Istanbul have failed. "Next time they can happen in Tel Aviv," chortles Irro, who seems unbothered by the prospect of a proxy war. Yet America may well be reluctant to stir the pot by recognising Somaliland.

The Houthis, meanwhile, have threatened to attack any Israeli military installation in the statelet. An American base there could become a target, too. That would shatter the reputation for peace and stability on which Somaliland has long built its case for international recognition. But "there is nothing free in the world," says Irro. He seems willing to pay a price to put his country on the map—the question is, how high? ■



Irro, man on a mission

Europe



Germany's Brandmauer

Burning down the house

BITTERFELD

How the anti-AfD firewall broke German politics

IT IS AS unusual in Germany as anywhere to see conservatives cavort with Marxists. But at a demonstration in Bitterfeld, a town in the eastern state of Saxony-Anhalt, flags bearing the names of almost all Germany's parties flutter in the evening breeze, from the centre-right Christian Democrats (CDU) to the Greens and Social Democrats (SPD), all the way to the hard-left Die Linke. The *Bunt statt braun* (colourful instead of brown) demos began last year to counter rallies organised by the far-right Alternative for Germany (AfD). For many in Bitterfeld's town square, the prospect of the AfD in power is a horrifying reminder of their country's darkest period.

But a popular front of this sort may be needed to stop it. On September 6th Saxony-Anhalt will hold one of the most important state elections in postwar German history. The AfD, polling above 40%, could secure an absolute majority and thus take office for the first time in a German state.

The threat is uniting foes and concentrating minds. The biggest cheer of the evening goes to a CDU member who tells the left-leaning crowd that "democracy is more important than political differences."

Germany's proportional voting system makes absolute majorities unusual. Of its 16 states, only tiny Saarland has a one-party government. Parties assemble in a variety of coalitions. But the AfD has been blocked from power everywhere by a *Brandmauer*, or firewall, that other parties have erected around it. The most important is maintained by the CDU, Germany's

ruling party and traditionally its largest. It formalised the policy in 2018 with an "incompatibility resolution" banning coalitions with both the AfD and Die Linke.

As the AfD has grown, the *Brandmauer* has become something like an organising principle of public life. Legislative measures that could win majorities are stymied because AfD votes cannot be allowed to tip the scale. Its MPs are excluded from Bundestag vice-presidencies, and even from its football team. "They don't greet you, they don't shake your hand, you are a non-person," sighs Gerold Otten, an AfD MP, of his colleagues in left-of-centre parties. The *Brandmauer* extends beyond politics. Business groups have retracted invitations to AfD MPs after furious backlashes. Diplomats hesitate to meet the party. Some media outlets tie themselves in knots over whether to interview its officials.

Has the policy worked? For some, keeping the AfD out of power is justification enough. It is among the nastier of Europe's populist outfits, many of which shun its company. It flirts with racism and Nazi imagery, has sympathy for Vladimir Putin and harbours extremists in its ranks. "We don't exclude the AfD because of the 2018 resolution, but because we don't share any values with it," says Wiebke Winter, head of the CDU in the city-state of Bremen.

Yet the *Brandmauer* has failed to arrest ►►

→ ALSO IN THIS SECTION

41 Ukraine without interceptors

42 The feisty Mr Fedorov

43 Western dance in France

43 Charlemagne: Swedish paradox

► the AfD's rise. In 2018 Friedrich Merz, now the CDU leader and Germany's chancellor, vowed to halve its vote. He hoped tacking right on immigration and governing well could win back voters seduced by the AfD's narrative of decline. That has not worked. Irregular immigration has plummeted, but the AfD has simply moved farther right, raging against "culturally foreign" workers and urging "remigration" (a slippery term that can mean anything from deporting asylum-seekers to ethnic cleansing). The CDU is nudging record lows in polls, while the AfD enjoys a clear lead (see chart).

And in Saxony-Anhalt and other AfD strongholds, the party's strength is nibbling away at the *Brandmauer*. Armin Schenk, Bitterfeld's CDU mayor, found lurking on the edge of the protest, says the "right-wing extremist" AfD must be kept from power in the country that perpetrated the Holocaust. But in Bitterfeld, he adds, "the *Brandmauer* is not a reality," because "in most cases decisions cannot be taken without the AfD." (With 15 of the city's 40 council seats, it is the largest party.)

Even when the *Brandmauer* holds, other parties must contort themselves around it. Thuringia and Saxony, two eastern states, have had to muster incoherent or unstable CDU-led minority governments to keep the AfD out. (Thuringia's is on the verge of collapse.) This only serves the AfD's narrative that the mainstream outfits are indistinguishable *Kartellparteien* ("cartel parties").

Saxony-Anhalt now threatens to push the strategy to its limits. If the AfD wins on September 6th, the *Brandmauer* is irrelevant. If it falls short, the CDU will probably need support from every other party, including Die Linke, to take office. That could inspire some CDU lawmakers to defect. If not, the strains of buying off the hard left would test Mr Merz's authority so sorely that some think it could end his career. And if, as polls suggest, the AfD wins twice as many seats as the CDU yet remains locked out of power, it will not only be AfD supporters who ask whether German democracy is working properly.

It is dawning on the CDU that it has painted itself into a corner. "Anyone still talking about firewalls has not recognised the signs of the times," said Michael Kretschmer, the CDU premier of Saxony, recently. The *Brandmauer* in effect obliges the party to seek coalitions to its left, especially with the SPD, which has been in power in Germany almost without interruption since 1998—despite now barely polling in double figures. CDU members' frustration with concessions to the SPD has inspired calls for a minority government that could, on certain issues (such as migration, tax and energy), seek a "conservative majority" with the AfD. Voters are shifting, too. Support for the anti-AfD *Brandmauer* has dwindled from a clear majority to 41%.

When the wall falls

Populist-right parties in selected European countries Polling, Aug 2026 ↓

Austria Freedom Party: FPÖ (est. 1956) 35%

Cordon sanitaire first broken:

2000 Joined federal government

Result: Prompted EU sanctions. Joined another coalition in 2017 that collapsed in scandal

Italy Brothers of Italy: Fdi (2012), Lega (2017) 27%

1994 Fdi predecessor National Alliance part of winning election coalition with Forza Italia (FI) and Lega Nord (later Lega)

2022 Fdi forms government with FI and Lega. Fdi leader Giorgia Meloni becomes prime minister and moderates

Netherlands Party for Freedom: PVV (2006) 12%

2010 Entered confidence-and-supply deal with conservative coalition

2023 Became largest party and joined government. Caused government collapse in 2025, lost seats in subsequent election

Spain Vox (2013) 17%

2022-23 Joined regional governments with conservative People's Party (PP)

Never faced active *cordon sanitaire*. All regional coalitions with Vox collapsed in 2024

Sweden Sweden Democrats (1988) 20%

2022 Confidence-and-supply agreement with liberal-conservative coalition

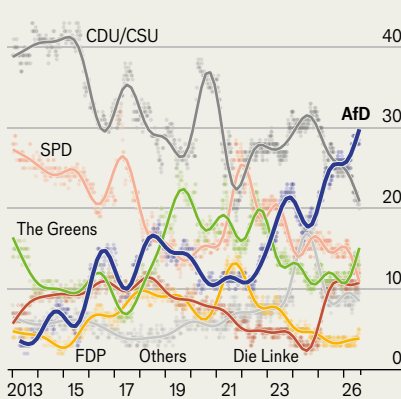
Traces roots to a neo-Nazi party, but has moderated since. Now second-largest party in parliament. Wants ministers in government if coalition wins September election

Sources: "Zwischen Abgrenzung, Einbindung und Tolerierung", Konrad Adenauer Stiftung, Nov 2025; Politico; The Economist

Yet the CDU has no easy way out. Tearing down the *Brandmauer* would split the party and trigger a voter exodus. Even minor violations have caused trouble. In 2020 the CDU voted with the AfD to elect a (short-lived) premier in Thuringia; the ensuing furore toppled Annegret Kramp-Karrenbauer, the party's national leader. In January 2025 Mr Merz ordered his party to

Leaps and bounds

Germany, voting intention in federal election, %



Source: INSA

vote with the AfD on anti-migration measures in the Bundestag. This sparked protests across Germany and boosted support for Die Linke at the next election. He is said to be traumatised by the experience.

To avoid these dilemmas, some in the CDU quietly hope for an AfD majority in Saxony-Anhalt. But this merely highlights the absence of a strategy. "There has never been a reasonable debate about how to handle the AfD in the CDU, just a reflex to exclude them as the new Nazis," says Andreas Rödder, a history professor who in 2023 resigned from a CDU commission after urging openness to minority governments rather than a rigid *Brandmauer*.

As for the AfD, it once railed against the *Brandmauer* as a violation of democracy. But these days many in the party think it can govern alone—first in a state, then nationally. "We don't think about making ourselves nice so that the CDU asks us for a coalition," says Malte Kaufmann, an AfD MP. The party's chief ideologue in Saxony-Anhalt has said voters want it to "finish off" the CDU rather than work with it.

In some European countries populists have moderated to gain power, and the centre has welcomed them in (see table). But behind the *Brandmauer*, the AfD has had no incentive to do so. Torben Braga, another MP, says two factors speak against those in the party who would like to tack to the centre. First, "our election results are getting better and better without us being forced to change." Second, the party fears that moderating and ejecting hardliners would alienate its voters. "The *Brandmauer* protects us from having to deal with very difficult questions," Mr Braga says.

He is right. The AfD has professionalised its organisation and communication. But it is riven by endless scandal, power struggles and policy rows, and has no coherent governing programme. None of this seems to matter. Behind the *Brandmauer* you can promise anything, says Christian Stecker, a political scientist at the Technical University in Darmstadt. "It protects them from the disappointment that accountability in a democracy implies."

In despair, some mainstream politicians now look favourably on calls to ban the AfD, as Germany's constitution allows. But even if parliament approved that proposal the constitutional court would take years to rule on it. Others still hope the centre can win back its voters, perhaps by ditching the unpopular Mr Merz.

The *Brandmauer* looks intact for now. One former CDU minister says 90% of senior party figures want to stick with it. But Mr Stecker's gloomier view tempts others. "It's too late for the CDU to turn around, and too late for the AfD to accept any turnaround," he says. "Some things will tumble down, new things will emerge—and we don't know what it will all look like." ■



Ukraine's air war

Survive till April

KYIV

Out of interceptors, Ukraine is no longer winning the aerial war

NOT SINCE the first days of the war have shoppers in Kyiv faced empty supermarket shelves. Now they do. On July 19th a Russian ballistic missile slammed into a warehouse supplying the upmarket Goodwine store. Two weeks later, strikes on Silpo, a larger chain, killed six workers. Nova Poshta, the private postal service on which most of Ukraine depends, has been hit repeatedly. Dmytro Krymsky, co-founder of Goodwine, estimates that strikes in the past three weeks have cost businesses \$500m. More troubling is Russia's new tactic of hunting cargo ships that use Ukraine's Black Sea ports—threatening a far larger share of the economy.

The campaign against civilian logistics is a long planned escalation, made possible by a shift in the air war. Russian production of ballistic missiles is rising just as Ukraine's supply of interceptors has dwindled. Defenders can now shield only a few high-priority targets. In July Russia profited by launching a record 198 ballistic and hypersonic missiles.

It is also an answer to Ukraine's drone strikes deep inside Russia. These have hit defence plants and oil infrastructure, closed the Sea of Azov to Russian shipping and burned warehouses of Wildberries, Russia's e-commerce giant. Ukraine says the company ships equipment for the war.

In late June Volodymyr Zelensky announced a 40-day "surge", intended to force Vladimir Putin to negotiate. Many as-

sumed that meant August would mark the high point of Ukrainian pressure. A Ukrainian intelligence source says it is a military action designed to have a "political and psychological" impact just in time for Russia's sham parliamentary elections in September. Almost daily explosions, creeping ever closer to Moscow's centre, have sown disquiet and embarrassed Mr Putin.

The campaign has inflicted economic damage on Russia, too. Strikes on refineries have knocked out more than 30% of capacity. Petrol rationing, which at its peak affected 80 of its 83 regions, continues in 18. Wildberries has lost nearly a third of its facilities, at a cost of some \$2-4bn. Elon Musk's Starlink satellite network, immune to jamming, is available to Ukraine but not Russia over the occupied territories, giving the former dominance in mid-range drone operations. Ukrainian drones stalk targets in and around Crimea and have helped stop shipping through Azov ports.

But the choking now runs both ways, and Ukraine is more vulnerable. Odessa's ports, the country's economic lifeline, were the first to feel the pressure. Threatened at the war's start, they reopened first with a Turkish-brokered deal and then with a protected corridor in 2023. Russia kept targeting port infrastructure, but shippers tolerated the risk, moving 210m tonnes over the following three years. The calculus changed in mid-July when Russia began hunting the ships themselves. On July 19th

cruise missiles sank a Turkish-owned cargo ship as it left Chornomorsk, one of three big ports in the region. Since then every vessel entering or leaving has been targeted. A few ships still dare to make the run, but the ports are commercially dead.

The timing could hardly be worse, with 50m tonnes of harvest still to move. "One ship of 100,000 tonnes is equivalent to 5,000 trucks, 5,000 drivers, and 5,000 border procedures," says Dmytro Barinov, president of the Ukrport association. The situation is worse than in 2022, he says, when ports on the Danube offset some losses. Now the river's low water level and Russian attacks make that route harder.

On the front lines things are grim for both sides. In parts of eastern Ukraine, Russia is crawling forwards but losing more than 1,000 men a day—a rate its recruitment struggles to replace. Elsewhere, Ukraine inches ahead in local counter-attacks. The focus of the war has thus largely shifted to the aerial campaigns. Ukraine mostly uses drones against Russian military targets, refineries and the arteries supplying Russian troops. Russia is using more destructive ballistic missiles against Ukraine's defence industry, civilian logistics and energy networks.

Both sides inflate statistics. Ukraine says it has caused \$13.5bn-worth of damage to Russia's refining industry since August 2025. Sergei Vakulenko of the Carnegie Endowment, a think-tank, reckons the real, immediate loss to business is perhaps one-tenth of that, and that it has barely touched state income.

Most worrying for Ukraine is Russia's blockade. Dragon Capital, a Ukrainian investment firm, estimates it will cost the country 0.6% of GDP this year. If the harvest can't be sold, farmers are likely to skip the winter sowing season, says Alex Lisitsa, president of the Ukrainian Agribusiness Club. That could push losses in agriculture to some \$10-12 billion, he says, around 5% of GDP. Food security may be at risk in Ukraine's markets in Africa and the Middle East. "It's as if we played open football with the Brazilians," rues one former senior official. "We scored three and are celebrating, but they scored 13."

Beware General Winter

This year, Ukraine has defied the odds and regained battlefield momentum. Its progress was always likely to slow by winter, but the pressure has arrived earlier. Unless it gets ballistic interceptors, its economy and military prospects will take a heavy hit. The worst-case scenario whispered in Kyiv is that Russian strikes on energy and water systems force mass emigration. Mr Krymsky says a business colleague described his company's strategy meeting last week, entitled Autumn Actions. "Today it already has a new name: Survive till April." ■

Ukrainian politics

Not now, Mykhailo

KYIV

A reformist ex-minister unexpectedly calls for elections

LATE IN THE evening of August 18th, at a time when Ukrainians usually check their phones for warnings of incoming missiles, a video post seized the public's attention instead. The nine-minute address by Mykhailo Fedorov, who on July 15th was ousted as defence minister, was in part a final plea to be returned to his former post and in part a demand that Volodymyr Zelensky call elections.

"We must not let Putin decide when Ukrainians choose their government," Mr Fedorov said. "Democracy cannot be held hostage by Russia." The call for elections during wartime, which are explicitly prohibited under Ukraine's martial law, amounted to a direct challenge to the president, whom Mr Fedorov had previously been careful not to cross.

The intervention was a response to the announcement of a vote in parliament to confirm Mr Fedorov's successor, General Yevhenii Khmara. There was never much doubt that Mr Khmara, a war hero held in high regard, would be voted in. In the event, the vote was 312-2. An earlier motion demanding the reinstatement of Mr Fedorov had garnered just three signatures.

Sources close to the ex-minister say he knew the chances of a return were slim, but that he felt compelled to call for an urgent reset of power amid a sense that the country was on a bad trajectory. But it is not clear how an election could work, either legally or practically. With hundreds of thousands of soldiers at the front, the logistical challenges would be immense and could render the results unreliable.

Mr Zelensky's staff says the video caught them unawares. It coincided with another headache for the president. On the morning of the parliamentary vote, Ukraine's anti-corruption agencies announced new lines of inquiry into some of his closest allies. Detectives say they uncovered a scheme to launder bail money for suspects detained in earlier investigations. The scheme, which the perpetrators called "Operation Forrest Gump", allegedly implicated a state-owned bank and high-ranking figures in the presidential office. By day's end Iryna Mudra, a deputy head of the office responsible for sanctions, international tribunals and the rule of law, had been charged and dismissed from her post.

That the anti-corruption investigators came so close to Mr Zelensky is embar-

assing. It also suggests his ability to rein them in, angered as they are by his earlier efforts to shut them down, is shrinking. Previous anti-corruption investigations have cost him key aides and a substantial share of his popular support. Yet polling shows Ukrainians remain overwhelmingly opposed to elections during wartime.

Even among Mr Fedorov's supporters, reaction to his proposal is mixed. To some it looks impulsive. It could force Ukrainian politics into a treacherous situation, ahead of a winter in which Russia is expected to launch a drone and missile campaign of unprecedented ferocity. "If there's a plan, it's a bad one," says an MP who is usually on Mr Fedorov's side. Still, the intervention has put the question of elections back into public discussion. At the moment, Ukrainians do not want them. But bombardment and political chaos might change that. ■

France

Western civilisation

CAMARET-SUR-AIGUES

France falls for American line-dancing

SUMMER IN RURAL France is a time for street *fêtes*, jugs of chilled rosé—and American line-dancing. To get locals on their feet, a DJ need only utter the words "Le Madison", a line-dance that originated in America's Midwest in the 1950s, and villagers will step into row formation. The Francophone Federation of Country Dance and Line Dance lists over 500 clubs nationwide, with such names as American Spirit Country or Froggy Stomp. Another

count records over 2,000 "*danse country*" groups. A poll suggests that nearly one in ten French adults has taken part in line-dancing. In May France held a national line-dance "open", in the southern town of Salon-de-Provence.

In Camaret-sur-Aigues in the Rhône valley, a town of just 4,500 inhabitants, American Spirit Country holds two classes of line-dancing and *le country* each week. Participants range from ten years old to 73. Some turn up in Stetsons and cowboy boots; others, jeans and trainers. "In the past two years *le line* has really taken off," says Zorah Lesbros, president of the club, who observes a burst of interest among the young. "People of any shape and size can do it, you make friends, and it's fun."

What explains this unexpected passion for a slice of Americana? France thinks of itself as a place of highbrow culture and artistic sophistication. Charles de Gaulle invented the Ministry of Culture in 1959 to bring such art to all. In reality the country has long held a secret fascination for American popular culture. Of the top ten box-office hits in France last year, nine were American. In Johnny Hallyday, of part-Belgian descent, France had its own wildly popular Elvis. Disneyland Paris is Europe's only outpost of the theme park. At its Disney Village, Billy Bob's Country Western Saloon holds line-dancing events.

The craze also speaks to places that the Paris political elite tend to overlook—at their peril. A study of *la danse country* by Jérôme Fourquet and Sylvain Manternach, two political analysts, notes a geographical and social correlation between its appeal and the roundabouts in semirural areas where *gilets jaunes* protesters gathered in 2018. They suggest that such dancing supplies the solidarity that protesters also sought, albeit with a different objective.

Overall, nearly a quarter of the French say they are "attracted" to the culture of America's West. "*La danse country*," the authors write, "is totally unknown to elite France and when it is referred to it is mostly in a sarcastic way." Populist politicians, including Marine Le Pen, the leading presidential candidate at next spring's election, tend to be better tuned in. She has even been spotted in a Stetson, and periodically sings along at rural events.

Some longtime French devotees will know "Le Madison" from a scene in Jean-Luc Godard's 1964 film "Bande à Part". More recently, it is the success of modern country-pop that has drawn new recruits. French line-dancing tutorials on YouTube are set to Taylor Swift songs. One on how to dance "Le Madison" has over 1m views. Away from parquet-floored Paris, at outdoor summer *bals*, the real ingredient for a successful event is when the secret thrill of an American cultural hit meets rural French tradition. ■



A 38-litre chapeau

CHARLEMAGNE

Individualism, together

How Sweden became a country of big-state libertarians



FEW IDEOLOGICAL causes ever united Bernie Sanders, an American politician so left-wing he could run for office as a European centrist, and Charlie Kirk, a Trumpian firebrand. A rare point of agreement among them, oddly, was the appeal of Sweden. To the gruff senator the Scandinavian country is a wellhead of socialist good sense that America should emulate. For the conservative stalwart, who was murdered a year ago while preaching his right-wing credo at a university, the home of IKEA and Spotify was equally worthy of admiration. Yet to him Sweden thrived not because of its large state but in spite of it—it was all down to an innate don't-tread-on-me individualism a Texan might recognise.

That two opposing sides of the political spectrum credit their own ideas for the success of one of the world's better-run places is not surprising. What might be is that both have a fair claim: Sweden is the paradoxical home of individualism done collectively.

The case for Sweden being a socialist Utopia is as traditional as its cinnamon buns. This is a country that soaks the rich—the top rate of income tax is a wealth-clobbering 52%—to fund a government whose spending amounts to nearly half of GDP, among the highest in the world. The welfare state in Sweden is as famous as its flat-pack furniture. The mollycoddling literally starts in the cradle; new parents are showered with up to 480 days of paid leave. Other lefty shibboleths include free universities, low income inequality and the rich world's most sweeping rent controls. Two-thirds of working Swedes belong to a trade union, the most of any country bar tiny Iceland. All this brotherhood-of-man stuff is reflected in politics: next month the centre-left Social Democrats will come top for the 33rd straight parliamentary election, a streak going back over a century (though the party has spent time out of power as rivals have formed ruling coalitions).

Yet it is equally fair to argue that Sweden has a libertarian side. Yes, income taxes are eye-watering and redistribution higher than free-marketeers might advocate. But in many other ways the country cherishes individual freedoms as ferociously as a Montana survivalist. The state may be large, but public services in Sweden are often delivered by the private sector: around a third of Swedish children graduating from high school do so at an institution run

by non-state operators, many of them run for profit. Sweden is a rare country with no inheritance tax. During the covid-19 pandemic, it stood out for imposing fewer restrictions than most countries, whether strict lockdowns or mask mandates. It lacks a national minimum wage, bucking the norm in rich economies (including America). Employers are largely free to hire and fire, unlike in most of Europe. All this has resulted in Sweden having around 50% more billionaires per person than America. If this is socialism, it is a variant that is surprisingly good for yacht sales.

In their rush to claim credit for Sweden's success, both sides get part of the story wrong. The socialist case was more accurate before the state was sharply slimmed down in the early 1990s. The libertarians gushing about the lack of a minimum wage fail to grasp that in most sectors pay rates are agreed by unions haggling with employers—hardly the stuff of Milton Friedman's dreams.

But the Scandinavian paradox exists because something more curious is afoot. For Sweden is indeed different. In developed countries collectivism and individual liberty are generally treated as rivals: the more responsibility government assumes, the less room remains for citizens to fend for themselves. Sweden followed a different path. During the 20th century the state expanded not merely to soften inequality or insure people against hardship, but to reduce their reliance on other people—including their own families. A large impersonal state is seen as bolstering autonomy: when in need, Swedes feel it is better to be beholden to an impersonal bureaucracy than to a parent or some charity. Seen this way, a bigger state translates into more freedom, not less.

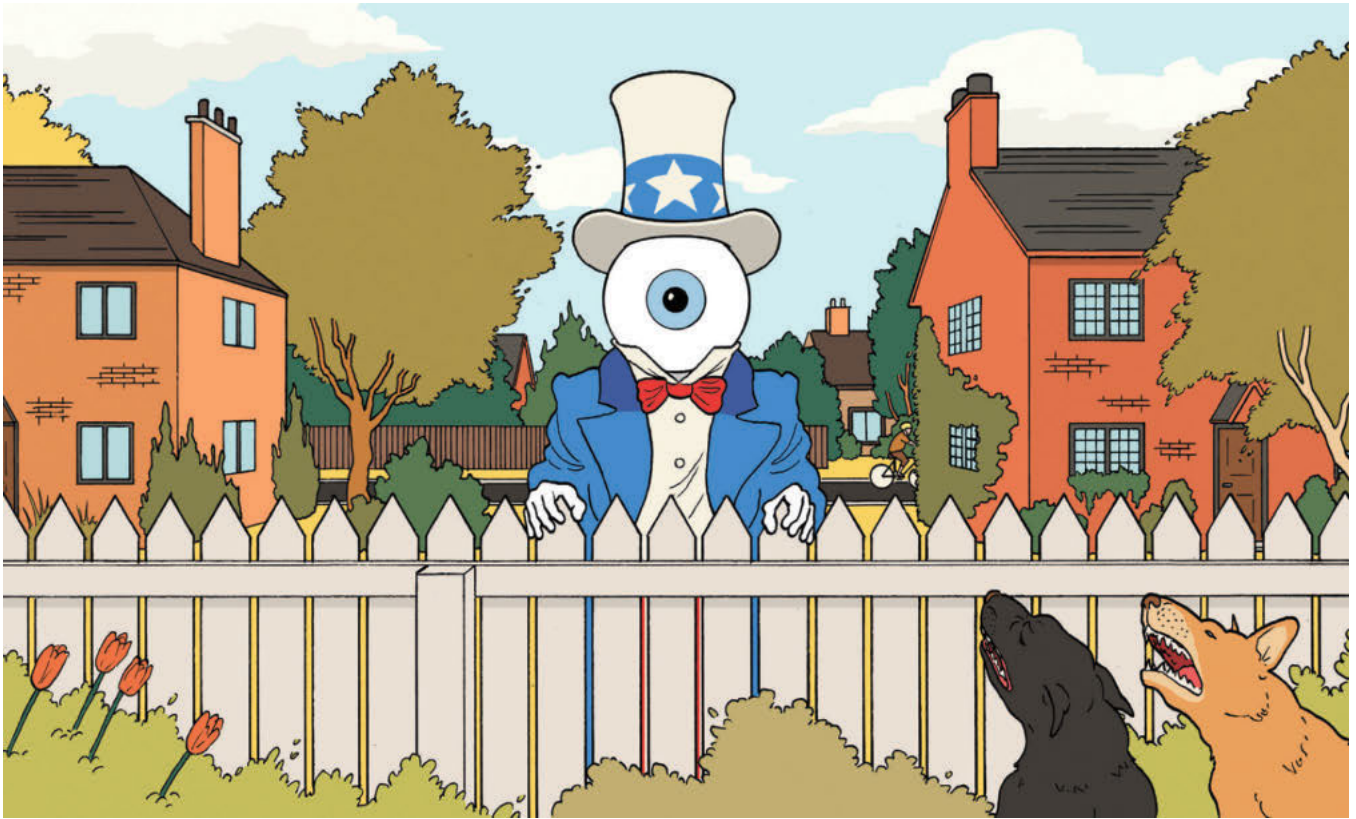
The upshot is what Henrik Berggren and Lars Trøegardh, two historians, call "statist individualism". In their book "The Swedish Theory of Love", newly updated in English, they argue that Swedes have come to prize relationships entered into freely rather than maintained by material necessity. Public child care helps women avoid financial dependence on husbands, state old-age homes liberate children from obligations to ageing parents, and so on. (Even marriage is a bit suspect: in France or Germany households are the basic unit of taxation, but in Sweden all adults file independently.) American parents sending their offspring to college must submit proof of their incomes for the youngsters to qualify for scholarships. In contrast, young Swedes are assumed to be on their own: the income of their parents is irrelevant. Their independence is a gift of the state, expensive as it is to sustain.

The elusive Swedish model

The historical development of a smallish European country may seem tangential to other Europeans, let alone Americans. Few politics can match the conditions that made Swedish collective individualism possible: seven centuries or so of rule of law, a deeply felt trust in the state, and a taste for consensus that can make the place feel stultifyingly conformist. Can Sweden itself maintain its unique path, now that a fifth of its population was born overseas?

But cross-border surveys of social attitudes suggest most countries are growing more like Sweden and its neighbours: secular societies that put self-expression ahead of deference to family or tradition. That makes Sweden's peculiar settlement, a form of overbearing statism that has helped people achieve freedom not just from economic want but from social pressures, more than a Nordic curiosity. To outsiders in America and beyond, big-state libertarianism can be puzzling. To Swedes there is no contradiction: they just want to be left alone, and to do it together. ■

Britain



A controversial tech giant

Why everybody hates Palantir

The company's software is helpful; its marketing hype is not

IN 2016 PALANTIR, an American software firm, successfully sued the US Army to win a second chance at a contract. A decade on, the fortunes of the company seeded by the CIA's venture arm, and named after an all-seeing stone in J.R.R. Tolkien's fantasy epic "The Lord of the Rings", have changed. Business is booming (see chart on next page). Domestic revenue has more than doubled in a year, to \$1.6bn in the second quarter of 2026. The American government accounts for two-fifths of its global business. Despite concerns that AI might eat its lunch, American firms think its software can provide a layer of protection for their raw data, a privacy shield against frontier AI labs like OpenAI and Anthropic.

"I'm struggling with our current popularity," joked Alex Karp, Palantir's CEO, on a recent earnings call. If Mr Karp, who once served on the board of *The Economist's* parent company, is feeling too comfy, he need only look across the Atlantic.

In July French and German security chiefs announced they would replace a "digital backbone" reliant on Palantir with European alternatives. In Britain, its biggest foreign market, and home to a sixth of its workforce, MPs have urged Andy Burnham, the prime minister, to exercise a break clause in February in a seven-year contract with the National Health Service. Palantir is suing Sir Sadiq Khan, the mayor of London, after he blocked a deal with the Metropolitan Police claiming competition

processes were circumvented.

Its loudest critics focus on how Palantir's tech has helped President Donald Trump's administration track and deport migrants, and the Israel Defence Forces target Gazans. In a speech on July 30th Robert Jenrick, an MP for Reform UK, a populist right-wing party, claimed that Palantir "reduced discharge delays by 15%", but that Mr Burnham was considering scrapping it from the NHS because "British Islamists were unhappy".

Critics also point to the firm's lobbying: its first British government contract, matching patients to ventilators, came during the covid-19 pandemic for a token fee of £1 (\$1.30), after it charmed civil servants over watermelon cocktails. Zack Polanski, the Green Party leader, unfairly notes that Louis Mosley, who runs Palantir's British arm and its European headquarters, is a grandson of Oswald Mosley, a 1930s fascist leader, and "insists on wearing a black shirt every single time he is on TV".

"Palantir derangement syndrome", as the *Spectator* magazine put it in a recent cover story, is not the only explanation for the backlash against the firm. It has sometimes overstated what its software can do. And it is a lightning rod for a broader anxiety about tech sovereignty (see Business section), risking a reaction that could deprive Britain of useful software.

→ ALSO IN THIS SECTION

46 Jason Arday's rise and fall

47 Bagehot: The Great British Bunk Off

→ Read more at: [Economist.com/Britain](https://www.economist.com/Britain)

— Two toots for the planning overhaul

— Reform UK's reformer

▶ No one disagrees that a key problem for the British state is its fragmented data. Police spend most of their time on admin rather than investigating crimes, accessing dozens of systems, each with its own login details. Disjointed data cost lives. A recent official report on England's maternity care found that patients routinely get lost between services because of poorly integrated tech, and that staff have to write on scraps of paper during emergencies.

Palantir can help solve such problems. Its engineers take vast quantities of scattered data and join them into a single source of truth that non-tech-savvy staff can interact with. The armed forces, for example, are using Palantir's software to work out which plane is nearest to which target and how much ammunition it has, and to assess military readiness down to the individual nut, bolt and screw.

In Bedfordshire the firm's software helped police review 62% more child-protection incidents over an eight-day trial period, and convict a Romanian criminal gang by trawling through around 100,000 messages—something that would previously have taken months. Responding to Sir Sadiq's decision to block the Metropolitan Police contract, Sir Mark Rowley, the Met's commissioner, said the lost productivity will mean the force must cut hundreds of front-line staff.

Too good to be true?

A contract won in 2023 to build what the NHS called its federated data platform (FDP) has received most attention. Over seven years, the platform is expected to cost £1.1bn, or enough to run the NHS in England for two days. Palantir's fans believe this is money very well spent.

Trusts in charge of hospitals can use the software both for back-office functions and to run tools on top of it. It can flag when a patient is behind on the steps required for cancer treatment, say, and let staff schedule appointments or update a record. Some trusts and regional health boards, such as Greater Manchester, can already do this, and more, without Palantir. But it is a huge upgrade for less tech-savvy parts of the NHS that have only recently stopped using paper records.

The FDP also enables different parts of the NHS, which use distinct data systems and structures, to link up. Tom Bartlett, who led the NHS team that helped develop the platform, is most excited about how it will allow parts of the health system to share innovations. He recently watched a nurse and a paramedic at a hackathon build a prototype of a tool their service needed in just two days. He also believes the FDP will help make the NHS AI-ready: the software provides a structured layer for AI agents or applications to run on.

The case for using such technology to

improve productivity in cash-strapped public services is strong. However, Palantir and its advocates weaken it by overstating what the software has achieved.

In an interview with *The Economist*, Mr Mosley points to 110,000 additional operations in the 40-odd trusts using the software for in-patient scheduling, figures derived from an NHS dashboard promoting the FDP. Yet the impressive figures did not disaggregate the software's effect: they showed all additional operations since 2020, including those to catch up with the backlog from the pandemic. In July Britain's statistics watchdog reprimanded the NHS for using them without a caveat. "You can't say that the 110,000 operations only happened because of Palantir," admits Mr Mosley, when pressed on the figures. But "directionally, it's very, very clear that the software is having an impact."

Mr Mosley complains that most acute trusts have signed up to the FDP but do not use it because the NHS doesn't mandate it: "You have a volunteer army, not a conscript army," he says. But four days before Mr Jenrick's speech, a report by the Health Foundation, a think-tank, found that trusts using Palantir's discharge tool cut delays no faster than those that did not. Nor does low uptake explain why Chelsea and Westminster, a trust in west London, has been the best trust at using the platform to reduce waiting lists, yet its lists have still risen faster than almost any other's.

Similar marketing hype surfaces in defence. Sir Tom Copinger-Symes, a recently retired lieutenant general, says that Palantir is "world-leading" at turning a complex military mission into actionable data flows. His concern is the way its marketing can present the firm as a "full-solution AI company rather than one supplier in a much broader supply chain". The government has sometimes encouraged the same impression. A "strategic partnership" with the Ministry of Defence announced last September is less sweeping than it sounds: "a complete fantasy," says one former official, pushed by government advisers who

thought it would "look good".

Such salesmanship, abetted by government, has helped thrust Palantir into the centre of a panic about "tech sovereignty", the loosely defined but real sense that Britain lacks control over critical technology at a time when America is no longer such a predictable and dependable ally. On this front, though, many of the concerns directed specifically at Palantir are misguided.

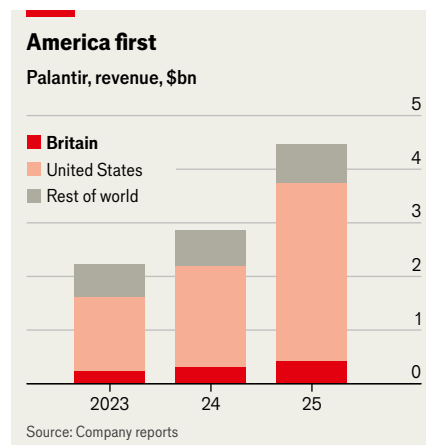
The NHS recently apologised after failing to disclose that some Palantir staff would be able to see identifiable patient data, adding to fears about the 2018 Cloud Act, which gives the American government the power to request data from tech companies even when the data are hosted by foreign subsidiaries. In practice, there seems little chance that Palantir could be compelled to disclose such data: the information stays within the trust, which remains the data controller. Engineers lack the legal permission or access to browse patient data for secondary purposes.

The alternatives are worse

Another fear is a "kill-switch" moment, such as when the American government banned the use of Anthropic's latest models for non-Americans. But if dependence on American tech giants is the worry, there are more alarming examples. The NHS relies on Microsoft for email and other communications (like Microsoft Teams), cloud and, increasingly, AI: it has just bought 500,000 Microsoft Copilot licences. Oracle, another American tech giant, is its largest supplier of patient records. Palantir "wouldn't even make my top 100 of the things I would worry about", says Ben Moore, until last year the procurement adviser to Britain's defence secretary.

More broadly, controversy around Palantir is a distraction from an uncomfortable truth: there are few good British or other European alternatives to the top American tech companies. The state can do some things itself. Last year the housing ministry replaced a Palantir refugee-matching system with its own, saving "millions of pounds" in running costs. In June France said ChapsVision, a domestic AI firm, would take over the contract with DGSI, the country's internal-security agency—but not before 2028. For other things, it needs American suppliers.

If Britain exercises a break clause for the FDP, as a select committee of MPs has recommended, it will be nigh-on impossible to find a consortium to replace Palantir before next February. "All the other companies are miles behind," says Mr Bartlett. On the battlefield, inferior tech could get British soldiers killed, argues Sir Tom. Colleagues, in love with American tech, should be careful not to buy its "full stack", he suggests. But "would we buy the third-best tank just because it's British?" ■





Jason Arday

The rise and fall of a diversity hire

What the tragic death of Jason Arday says about Cambridge University

ALICE (NOT her real name) never felt comfortable at Cambridge University. It was too elitist. But during her MPhil in education in 2024, one professor stood out. Gentle and dreadlocked, he took the time to learn students' names, the sports they liked. He made them feel any one of them could be an academic. Autistic and, he claimed, non-verbal until he was 11, he did not learn to read and write properly until 18, but still became Cambridge's youngest black professor at the age of 37. "Most of my cohort agree," says Alice. "He's genuinely the best teacher we ever had."

Granted, Jason Arday wasn't for everyone. Traditionalists disliked his classes, which included little critical study of texts and lots of ramblings about song lyrics. He often went off on unfocused rants, Alice recalls, and was "very attached" to his personal story. Still, it was a shock when, in July, allegations grew that he had plagiarised extensively, fabricated research and lied about his CV. When he resigned on August 5th, hours after Cambridge opened an investigation into his qualifications, Alice was confused and disappointed. Nine days later, when he was found dead (in circumstances police called unexpected but not suspicious), she was angry.

By conventional measures, Mr Arday should never have been appointed to a chair at one of the world's top-ranked universities. Only six years earlier he had been lecturing in physical education at Leeds

Beckett University, a former polytechnic. His first academic paper appeared in 2018.

Cambridge was looking at different criteria. It had been shaken by the institutional reckonings that followed the killing of George Floyd in Minneapolis in 2020. In March 2023 a damning internal report noted that over the previous three promotion rounds none of the 216 eligible applicants for professorships was black. The report spoke of an "urgency to build the pipeline of Black scholars at the University through

recruitment and promotion". That month Mr Arday took up his post as a professor of sociology of education. A press release said a "major focus of his work" would be helping Cambridge recruit a "more diverse range of students and staff".

He specialised in racial inequality in academia. Two books he had co-edited, including "Dismantling Race in Higher Education: Racism, Whiteness and Decolonising the Academy", were added to the sociology curriculum. He was happy to tell his sensational story on camera.

In July a blog post challenged his narrative. It found dozens of plagiarised passages in his doctoral thesis and queried some of his sensational feats, such as running 30 marathons in 35 days. The author, Nathan Cofnas, was hardly impartial: a self-described "race realist", he had left Cambridge in 2024 after a backlash to his chauvinistic view that in a true meritocracy black people would "disappear from almost all high-profile positions outside sports and entertainment". Yet the *Times Higher Education* had planned to publish a similar story in 2025, only to drop it after Mr Arday instructed lawyers.

In a quiet month, national newspapers and social media piled in (see chart). The release in America of Mr Arday's memoir, the aptly named "Great and Unfortunate Things" (and the leaking of an at times contradictory book proposal), raised more eyebrows. He claimed he had spent months in a coma with locked-in syndrome, had testicular cancer, two brain tumours, and epilepsy after being beaten by teenage thugs. The book also included a moment of prescience. After sending his application to Cambridge, Mr Arday wonders if he has "completely overestimated" his capabilities. "What if", he writes, "my story ends up being a cautionary tale like Icarus about flying too close to the sun?"

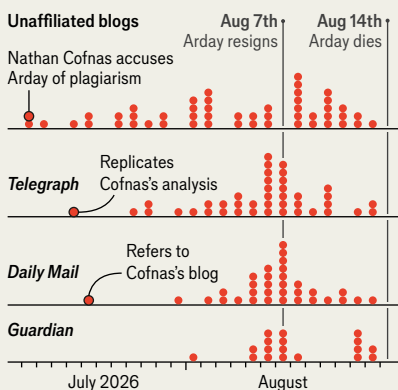
Cambridge faces many questions. "All of this could've been avoided if Cambridge did its due diligence, a basic background check," says Clive Aruede of the Association of Black Humanists. The university told *The Economist* it continued to provide support after Mr Arday resigned, screening his emails and posts for racist messages, and providing counselling. Many will wonder if it did enough for the man it placed on a pedestal once he fell off it.

Myths and the man

The myths about Mr Arday are already being written. Both sides of the culture war portray him as a black victim. A totem for a cult of wokeness in an academy which prioritised diversity over merit; a hero hounded to death by a racist press. The likelier story is both duller and more depressing: an institution wanted a symbol badly enough not to look too closely, and a vulnerable man supplied one. ■

From blog to broadsheet

Articles about Jason Arday's alleged misconduct*
Selected websites



*A previous blog post accusing Jason Arday of plagiarism was published by David Harris in 2024
Sources: Newscord; Substack; news reports; *The Economist*

BAGEHOT

The Great British Bunk Off

How the British summer holidays offer the worst of all worlds



AUGUST IN ENGLAND is a strange affair. Offices are half-empty but the cities themselves are not. Parks are strewn with the working-age, lazing in the sun; bars fill from the early afternoon. On a warm Friday, anyone hoping for a dip at Canary Wharf's new outdoor pool is out of luck. Every session is rammed with working-age adults enjoying Britain's new Mediterranean climate. The Great British Bunk Off has begun.

Britain does not shut down over summer. There is no equivalent of the *grandes vacances* in France during the height of summer or Norway's *Fellesferie* when the entire country disappears for most of July. The closest Britain comes is "holibobs"—a week or two in Europe or Cornwall. Yet for the rest of August, while their peers in France and Germany are away from their desks, Britain's holibourgeoisie are stuck in purgatory, neither truly at work nor at rest. All that results is an epidemic of skiving and shirking.

It is a British malaise. Britons are inveterate liars when it comes to how much they work. A decade ago statisticians found that a big chunk of Britain's hourly productivity gap with France was due to Britons vastly overstating how many hours they did. Heat makes them only more deceitful. During summer, about half of British workers confess they clock off early or take extended lunch breaks, according to a survey by Dayforce, an HR-software firm.

And who can blame them? Productive work becomes close to impossible in summer. Co-ordination falls apart. Any request is met with a flurry of out-of-office emails. Some professions such as MPs and judges are Augustocrats able to take the whole month off. Schools shut so teachers can enjoy six weeks off. This has a knock-on impact. Four in ten working parents combine working from home with child care, according to the Trades Union Congress, a federation of unions. Arriving emails compete for attention with screaming six-year-olds. The result? More bunking off.

Partly it is culture struggling to keep up with the climate. Mild temperatures have long meant that Britons soldier on through the summer, while continental Europe hits the beach. Climate-change deniers are rare but summer deniers are common. Some still do not accept that Britain is a hot country, for a few months at least. And heat makes people listless: each degree above 20°C knocks

about 2% off productivity. Almost all the European countries Britain compares itself with are more productive. If searching for a solution, start with Britain's summer.

It was not always this way. Britons once enjoyed a rigid separation between work and rest. During "wakes weeks" in the 19th and early 20th centuries, mills would fall silent for a break in summer and the inhabitants of Rochdale and Bury would descend en masse to Blackpool, a seaside resort in England's chilly north-west.

Today the lines are blurred. In Greater Manchester, mills that once contained looms are now filled with digital admen firing off emails that could be sent from anywhere. And so people do. Britons work from home for about 1.8 days per week, which is far more than in most European countries. In France the average is half that, according to one paper. Turning two days a week into four or five during August is easy. Maintaining motivation in a sweltering attic office while messages pile up is hard.

In another sense it is a return. Before the Industrial Revolution, working patterns were ragged, argued the historian E.P. Thompson. One poet summed up the working week in 1639: "You know that Munday is Sundayes brother/Tuesday is such another/Wednesday you must go to Church and pray/Thursday is half-holiday/On Friday it is too late to begin to spin/The Saturday is half-holiday agen." Strip out the trips to church and a 17th-century satire is a fair summary of August working in 2026.

Employers can fight back. Technology that liberates workers from their commutes on Fridays can capture their holiday in August, points out Nick Bloom, an economist from Stanford. And so the front line of the fight between labour and capital is now the sun lounger rather than a factory floor. A leisurely backstroke at 3pm in Canary Wharf on a hot August weekday becomes a wet wildcat strike; a manager emailing an underling they know is on the beach is the equivalent of a Victorian mill owner fiddling the clocks to shorten a break.

Who can bring peace to this struggle? Politicians show little interest. They once promised leisure: in 2017 Jeremy Corbyn's Labour Party proposed four new bank holidays. Interest rates were zero and money was free, so why shouldn't time be? Now no major party bothers. Labour's sweeping Employment Rights Act had little to say on holidays; the government chickened out of a "right to switch off", which would have made it harder for companies to contact staff out of hours.

We're not all going on a summer holiday

The Great British Bunk Off could be a solution rather than a scandal. Skiving is perhaps a necessary pressure valve for the labour market, allowing it to clear. If workers actually worked the hours, would companies be happy to offer a pay rise (plus a little extra for child care)? The Great British Bunk Off becomes a Soviet compromise: "We pretend to work, and they pretend to pay us."

Joining the rest of Europe and embracing an actual shutdown would be a gamble, a bet that Parkinson's law—the idea that "work expands so as to fill the time available for its completion"—is a fundamental rule of economics, rather than a joke. Telling people to knuckle down is no more appealing. One cabinet minister suggests that if Britain wants to survive, it should consider a six-day week, rather than a four-day one. It would be a brave politician to pitch it. And so the Great British Bunk Off will remain a summer fixture, best observed from a berth at the lido or horizontally in a park. Good luck finding a spot. ■

International



Missile defence

Going Winchester

KYIV AND TAIPEI

America and its allies are running out of interceptor missiles

RESIDENTS OF KYIV are sleeping once again in overcrowded metro stations or in cars in underground garages. Above, the air war is turning into a turkey-shoot. Russia is bombarding the city with ever more ballistic and hypersonic missiles. Ukraine, which had previously thwarted most such attacks with Patriot anti-missile batteries, is “going Winchester”—military jargon for running out of ammunition.

Ukraine’s plight is the most acute sign of the West’s shortage of interceptors. America’s depleted arsenal also helps explain why President Donald Trump has repeatedly retreated from threats to return to war with Iran. America’s allies are alarmed.

Switzerland expects its order of Patriots to be delayed by up to seven years. Taiwan worries that 102 ordered years ago will not arrive this year as scheduled. Lockheed Martin, which makes the most advanced version, the PAC-3 MSE, says it cannot guarantee delivery times to any foreign client, given the Pentagon’s power to divert them.

Shortages of munitions are common in wartime, especially in big, attritional wars. Yet America is running short after a brief war against a medium-sized power. What if it had to fight mighty China? It would exhaust its interceptors within days, and be unable to use its bases in Japan and Guam, reckons a recent study by the Heritage Foundation, a think-tank in Washington.

The proximate cause of the crisis is the prodigious use of interceptors by America and its Middle Eastern allies during Mr Trump’s war on Iran this year. The Centre for Strategic and International Studies (CSIS), another think-tank, thinks the Pentagon has fired nearly two-thirds of its pre-war stock of Patriots as well as lots of other interceptors (see chart on next page).

But the war merely hastened a reckon-

ing that has been decades in the making. On both sides of the Atlantic defence industries work to a peacetime rhythm, though the drums of war have been beating faster. That error has been compounded by the global proliferation of missiles, generals’ tendency to spend more on flashy weapons than on the munitions they consume, America’s dysfunctional budget process and complacency about American military supremacy in the air.

The CSIS reckons it will take until 2029 to rebuild Patriot stocks to pre-war levels. That is assuming Congress approves Mr Trump’s gargantuan \$1.5trn defence-budget request (a 50% increase on current spending), which is unlikely. Yet Patriots have become the prime system for point-defence—that is, to protect a specific target such as an air base—for 19 countries. “Patriot is a dinner-jacket solution: you don’t need it very often, but when you do, very little else will do,” explains Justin Bronk of the Royal United Services Institute (RUSI), a British think-tank.

The hitch is that ballistic missiles are generally cheaper than the interceptors trying to hit them (PAC-3 MSE rounds costs \$4m-5m apiece). Iran’s Shahed drones (or other unmanned aerial vehicles, UAVs) are two orders of magnitude cheaper still. “If your adversary can field a one-way attack UAV for the price of a used car, and your answer is a surface-to-air missile that costs

→ ALSO IN THIS SECTION

— The Telegram is away

▶ the same as a small apartment here in London, you're losing the exchange even though you're winning every single engagement," Colonel Sabah Al Sabah, a Kuwaiti officer, argued at a recent RUSI event. What is more, defenders typically fire at least two interceptors at an incoming missile to improve the odds of success.

Ukraine has improvised a system of layered defences. It has a world-class network to counter swarms of Russian drones. It can also take out most larger cruise missiles, thanks in part to European air-defence systems. But only the precious Patriots can take on Russia's Iskander-M and Kinzhal ballistic missiles. In March Ukraine boasted of intercepting about 70% of ballistic missiles. In July it was 20%. In August the rate will be even more dire. On one night—August 5th—there were no interceptions. The air force has stopped issuing regular updates. Russia, taking note, is increasing missile production.

With winter approaching, Ukraine has asked for at least 300 Patriot interceptors. Some countries still have them in reasonable numbers, including Germany, Greece, Poland and Spain. But they are increasingly unwilling to part with them.

Another solution is to build them faster. The manufacturers agreed this year to increase PAC-3 production from 600 a year to around 2,000 by the early 2030s. The Pentagon has since solicited ideas to go faster still. But Tom Karako of CSIS notes that it has not yet signed actual contracts for stepped-up output.

Allies' industrial capacity can help. Japan produces some 30 PAC-3 interceptors a year under licence, mostly for its own use. MBDA, a European firm, is about to start production in Germany of older PAC-2 models. At a NATO summit in July Mr Trump said Ukraine, too, would be allowed to make Patriot interceptors under licence, but has since cast doubt on the idea. Reports suggest their manufacturers, Lockheed Martin and RTX, fear losing their know-how, or being upstaged by cheaper Ukrainian versions.

Lacking the same combat experience, Europeans firms are hurrying to make their own weapons. The Franco-Italian SAMP-T is supposedly able to hit short- and medium-range ballistic missiles, but Ukrainian military sources say it works less well than Patriot PAC-2s, let alone PAC-3s. Ukraine will receive improved SAMP-T batteries, with updated interceptors, in phases from next year. Yet they are being produced in lower numbers than Patriots.

Ukraine's nimble defence industry and upstart Western firms have revolutionised drone warfare. They are now working on missile defence. One effort is the Freyja project involving Fire Point, a Ukrainian firm, and lots more European armsmakers. They aim to develop an interceptor at a

quarter of the cost of a PAC-3 without American technology, with testing to start next year. Sceptics abound. Douglas Barrie of the International Institute for Strategic Studies, another British think-tank, describes Freyja as "a moon shot".

Taiwan is trying to expand production of its Sky Bow III interceptors, which are supposedly equivalent to PAC-2s and older PAC-3s. This year it is expected to start mass producing the Sky Bow IV, which it claims is more capable than the PAC-3. Few countries, however, would risk China's wrath by buying Taiwanese weapons.

Lockheed Martin has announced the development of the PAC-3 ACE, aka "Baby Patriot", which is smaller and less capable than the PAC-3 MSE but will cost half as much. It could enter production in 2028.

Some hope that laser weapons, long in development, might improve the grim arithmetic of air defence. But experimental American and Israeli weapons have so far reported success only against drones.

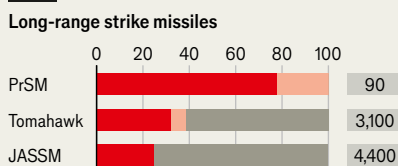
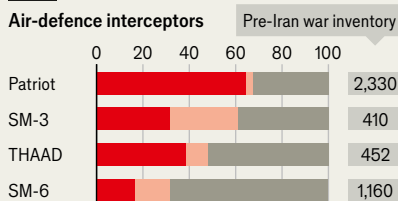
Kill the archer first

"Air defence cannot be separated from offensive operations," notes David Deptula, a former American air-force general. "The cheapest interceptor is the one you never have to fire because the weapon was never launched." There is a lot to be said for shooting the archer—in the form of enemy production sites, command-and-control centres, storage depots and missile launchers—rather than the arrow. But America has depleted its stocks of long-range offensive weapons, too. CSIS reckons it has used a third of its 3,000-odd Tomahawk cruise missiles. It has refused to supply them to Ukraine and deliveries to other clients, including Australia, Germany, Japan and the Netherlands, may be delayed.

Winchester on the horizon

United States, estimated munitions stockpiles, %
Feb 27th-Aug 16th 2026

Used estimate: ■ Low ■ High ■ Remaining



Source: M. Cancian and C. Park,
Centre for Strategic and International Studies

Ukraine has built drones able to reach targets ever deeper in Russia, notably energy infrastructure, arms factories and consumer-goods-distribution warehouses. But they are slow and carry little explosive. Fire Point's heavier Flamingo cruise missile has had some success. Ukraine hopes to deploy soon its FP-7 and the bigger FP-9 ballistic missiles, supposedly with a range of more than 800km.

Destroying mobile launchers is tricky. This year America and Israel suppressed Iranian missile fire by up to 90% by bombing launchers and the entrances to underground "missile cities". But even with full control of the skies and exquisite surveillance, they never stopped it entirely. America lost radars, command centres, early-warning aircraft, refuelling tankers, drones, fighters and more. Since the ceasefire in April, it has been unable to stop Iran shooting at ships in the Strait of Hormuz.

Any future air war against China over, say, Taiwan, would be a formidable enterprise. Chinese missiles can reach bases as far away as Alaska and Hawaii, as well as warships in the middle of the Pacific Ocean. To defend against the People's Liberation Army (PLA), Heritage reckons America would need more than 19,000 Patriot interceptors—more than 20 times CSIS's estimate of current stocks.

Shooting at the PLA archer would be a tempting alternative, but poses two big problems beyond penetrating China's air defences. First, America would have to strike the Chinese mainland, which would invite retaliation against the continental United States. Second, the PLA Rocket Force is responsible not just for conventional missiles, but also land-based nuclear ones: attacks on it might be seen as an attempt to destroy China's nuclear forces, risking nuclear escalation.

Instead, American forces are practising how to survive within China's "weapons-engagement zone" through various forms of dispersal. In a world of proliferating, cheap drones and missiles, hardening, dispersing and concealing military targets—and rebuilding them quickly if they get destroyed—are essential skills.

Although deep-strike weapons are no panacea, they do help to deter attack. That explains why America's allies are scrambling to find the means to shoot at faraway targets. Ukraine, already under relentless attack, hopes that carrying the war to Russia will cause enough pain to make Vladimir Putin reconsider endless conflict.

All the possible solutions to the interceptor shortage, even if accelerated and intensified, will take time. In Ukraine, meanwhile, civilians must still shelter underground. As they contemplate another freezing winter of peril, many will ask how their leaders let it come to this. Other governments should take heed. ■

Business



Global commerce

Routes and branches

AIN SOKHNA

Chinese businesses are wrapping their supply chains around the planet

THE ANCIENT port of Ain Sokhna, on the Gulf of Suez, once received turquoise destined for the regalia of pharaohs. Now it welcomes a new source of wealth: Chinese manufacturers. In just a few years their investment has transformed the area. Scores of factories have appeared, producing everything from fibreglass to switch-gears. In January a new port terminal—built with financing from two Chinese logistics giants, COSCO and CK Hutchison—began operations.

Ain Sokhna forms part of the wider Suez Canal Economic Zone, a network of industrial parks and ports that stretches north to the Mediterranean. Roughly half the investment it has attracted in recent years has come from China. The factory-building extends far beyond North Africa (see map on next page). From Saudi Arabia and Hungary to Brazil and Indonesia, Chinese industrial parks are springing up, along with supporting infrastructure.

The pace of this investment has accelerated sharply. In the past three years alone Chinese companies have spent more than \$200bn building factories abroad (see chart). The character of their supply chains is also changing, in three ways. First, they are spread more widely, with large production nodes in nearly every region of the globe. Second, they have grown deeper, with many Chinese suppliers following manufacturers into new sites, replicating

the tight-knit ecosystems back home. Third, they are increasingly dominated by strategic industries, from electric vehicles and clean energy to data-centre gear. The consequence is that a rewiring of global manufacturing is under way.

A number of reasons explain why Chinese firms are making their wares in an expanding array of places. Weak consumer spending and fierce competition at home have encouraged them to venture into new markets. The tariffs introduced by the second Trump administration have also incentivised production in places that have been hit with less punitive levies than historic Chinese outposts such as Vietnam.

Various countries in the global south have dangled added incentives. Egypt, for instance, offers a “golden licence” that slashes red tape for big projects. Mohamed Eldib, a lawyer who helps Chinese firms set up operations in the country, says they are eager to “come in and make money” selling to Egypt’s 120m people while also using it as an export base.

The result is increasingly dispersed production footprints. Take JA Solar, Jinko-Solar and TrinaSolar, three Chinese manufacturers that began making solar panels in various South-East Asian countries a decade ago and are now setting up factories in the Gulf. New Chinese production hubs are gradually being woven into customers’

→ ALSO IN THIS SECTION

52 America Inc’s grip on governments

53 Bartleby: On AI and ROI

54 Meta’s blockbuster trial

54 Reddit’s woes

55 Canada’s oil boom

56 Schumpeter: Data-centre politics

supply chains. Nordex, a German manufacturer of wind turbines, now purchases blades from a factory in Morocco that its Chinese supplier opened last year.

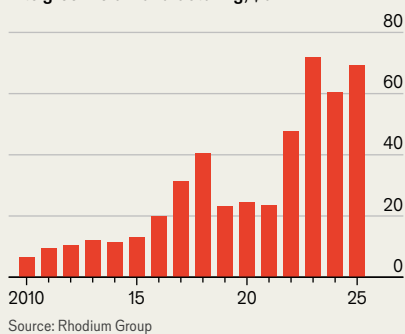
Europe has also emerged as a popular destination. Chinese firms' foreign direct investment in all-new "greenfield" projects on the continent rose by half in 2025, to a record €8.9bn (\$10.1bn), according to Rhodium Group, a research firm, and MERICS, a think-tank. Hungary has attracted much attention. Serbia has also seen a growing Chinese presence. Linglong Tire, a car-parts supplier, is among the Chinese manufacturers to have begun producing in the country. It recently announced it was expanding the capacity of its largely automated factory in the city of Zrenjanin, which supplies Western carmakers including Volkswagen and Ford.

At the same time China's global supply chains have deepened, with more upstream manufacturing taking place abroad. Gotion, a Chinese battery-maker, is constructing a gigafactory in an industrial zone 70km north-east of Morocco's capital, Rabat; a number of Chinese suppliers, including BTR, which makes anodes and cathodes, and Hailiang Group, which produces copper foil, are building factories a few hours' drive away to provide inputs. At an industrial zone on the outskirts of Cairo, where a Chinese manufacturer of home appliances has built a factory, an executive notes that the firm has likewise encouraged some of its suppliers to set up local plants.

Plenty of inputs are still shipped in. Machine tools are often imported from Chinese suppliers such as Yangli Group. So are components or materials that are unavailable (or much pricier) locally. Chinese exports of capital goods increased by 14% in the first half of 2026, year on year. Exports of intermediate goods rose by

Flying dragons

China, outflows of foreign direct investment into greenfield manufacturing, \$bn



27%. With time, however, more of those inputs may be sourced from nearer by.

Meanwhile, China's logistics firms are helping its manufacturers to link their newly sprawling supply chains. These now operate or have invested in at least 132 foreign ports, from Greece to Sri Lanka, along with airports and rail lines, including one from Budapest to Belgrade completed earlier this year.

China Inc's overseas investment binge has focused heavily on a handful of strategic industries. In order to satisfy rising demand for electricity, countries in the global south in particular have become eager buyers of China's clean-energy technology. The vast solar farms being built in Egypt's deserts rely largely on Chinese equipment, a growing share of which is produced locally. Chinese EVs, now a common sight in cities from Rio de Janeiro to London, are increasingly being made in regional hubs nearer to customers.

The equipment required for artificial-intelligence data centres has been another focus. Thailand, for example, has emerged as a manufacturing centre for Chinese

makers of high-speed optical components, such as Zhongji InnoLight, whose products are used by America's cloud-computing giants around the world.

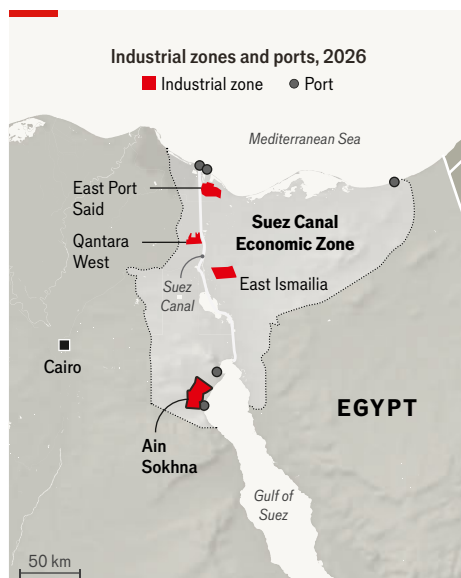
Chinese firms hoping to invest abroad have found they have the field to themselves, as Western rivals concentrate their factory-building in America to appease its protectionist government. An Egyptian official adds that, when Chinese businesses decide to build, they do so quickly.

Their foreign expansion is not without obstacles. The volatile tariffs imposed by America have resulted in some projects being cancelled or cut back. On August 13th the White House published a report titled "The Great Transshipment Scam" that called for sweeping restrictions on imports with even a whiff of Chinese involvement (see Finance & economics section). At the same time governments in places such as Brazil and Turkey have been imposing local-content requirements, which result in more of the value being added in their countries but make manufacturing there less attractive. Currency fluctuations and high borrowing costs in emerging markets complicate matters further.

Chain reaction

China's government has also been making life hard for the country's globe-trotting businesses. Last month a new package of outbound-investment regulations came into effect which, among other things, restrict the transfer of technology and data abroad and introduce a national-security review process.

So far the overall share of global manufacturing taking place in China has shown no indication of decline. Yet with ever more of its companies investing abroad, that may soon change. Even then, however, the world will continue to rely on Chinese goods—wherever they are made. ■



Government procurement

US and them

America Inc has a tight grip on allied governments

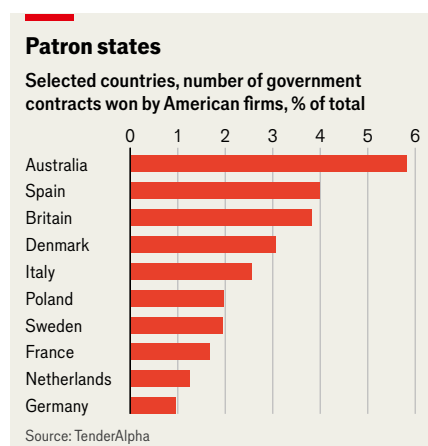
"I MISS NOTHING," says Dirk Schrödter, the digitisation minister for the German state of Schleswig-Holstein. Over a video-call using OpenTalk, a German alternative to Microsoft's Teams, he explains how in the past two years he has moved some 30,000 of the state's civil servants from the American company's collaboration and productivity tools to open-source alternatives. He has now begun the process of shifting staff from Windows to Linux, an open-source operating system. The move has attracted the attention of public officials elsewhere who are also keen to cut ties with America's tech giants. "Every week we have questions from other states, other cities and governments inside and outside of Europe," he says.

The growing interest in open-source software reflects a new geopolitical reality. America has long used its commercial prowess to hurt enemies, as when imposing sanctions on Iran and Russia. But now even allies worry that it could cut off their access to critical technologies—or at least threaten to during, say, a trade negotiation.

Governments abroad are examining the extent to which their ability to operate depends on American suppliers. Fears were stoked last May when Karim Khan, then the International Criminal Court's chief prosecutor, lost access to his Microsoft email account after President Donald Trump brought sanctions against the court over its issuing of an arrest warrant for Binyamin Netanyahu, Israel's prime minister. (Microsoft says it did not cut its service.) Another jolt came this June when Mr Trump forced Anthropic, an artificial-intelligence lab, to temporarily cut off access to its latest models in foreign countries. America's allies also took note when it twice paused its supply of weapons to Ukraine last year.

American firms win a small share of the overall contracts tendered by governments abroad. But their products and services often underpin critical government functions. Various efforts are thus under way to reduce reliance on American suppliers and nurture domestic alternatives. In many cases, however, doing so will be enormously difficult.

Europe in particular has emerged as a centre for efforts to flush America Inc out of government supply chains. France's national government is planning to ditch Teams and wants to move some computers



to Linux. Local governments, including the cities of Reus in Spain and Aarhus in Denmark, have turned to European cloud providers such as Nextcloud and Hetzner, both from Germany. In June the European Commission unveiled a plan to boost the continent's "technological sovereignty" that, among other things, aims to shift the processing of sensitive government data to such providers. Last year Spain cancelled an order for F-35 fighter jets, supplied by America's Lockheed Martin. British politicians are urging the prime minister to implement a break clause in a contract between Palantir, an American technology firm, and the National Health Service (see Britain section).

The Economist's estimates suggest that, at an aggregate level, American companies account for a modest share of public spending abroad. We calculate that, of the \$25trn in sales generated last year by listed American companies, perhaps \$500bn (or 2%) came from foreign governments. That is equivalent to roughly 7% of government procurement in OECD countries other than



Air superiority

America, which account for an overwhelming majority of the spending.

Some countries are more dependent than others. Measured by the number of government contracts won by American firms last year, Australia (6% of contracts) and Britain (4%) are more reliant than France (2%) and Germany (1%), according to figures from TenderAlpha, a data provider (see chart).

Yet American firms play an outsized role in critical areas of government. We estimate that roughly two-fifths of the business that American companies generate from foreign governments is in information technology and defence. (Pharmaceuticals and medical equipment, purchased by public health systems around the world, make up another large chunk.)

Often that is because there are few alternatives. Alphabet, Amazon and Microsoft control two-thirds of the global cloud-computing market, according to Synergy Research, a firm of analysts. Forrester, another research group, reckons that American providers account for between seven and nine of the ten largest vendors for most segments of enterprise software.

Those substitutes that do exist are frequently inferior. Sometimes the loss is negligible: on OpenTalk users cannot send emojis flying across the screen. But other shortcomings are more serious. LibreOffice, the open-source productivity-software package for which many governments are opting, lacks the AI features offered by Microsoft—a weakness that will only become more pronounced as the technology advances. OVHcloud, a French cloud-computing provider that is Europe's biggest such firm, generates about one-hundredth the revenue of Amazon Web Services, making it difficult to compete on price or keep up on innovation. Similarly, governments that want the most advanced air-defence system must turn to Lockheed and RTX, another American armsmaker, which together produce the Patriot.

Switching costs can be hefty, too. Lockheed supplies F-35s to governments from Norway to Belgium. Those countries could instead opt for a European jet, such as the Rafale or Typhoon, even if they are less advanced. But that would require a vast operational overhaul, including retraining pilots and support crews and replacing weapons inventories—a tough sell when huge sums have already been spent on Lockheed jets.

Or consider patient-record systems in hospitals. The hefty cost of swapping to a new vendor is partly why Norway's parliament voted in June to keep using Epic, an American provider of health-care data- ➤

Correction: Last week, in "Silicon showdown", we wrote that Nvidia makes its chips. In fact, it only designs them. Sorry.

► bases, despite dissatisfaction with the software among doctors.

The dependence of local companies on American suppliers further complicates matters. Roy Illsley of Omdia, another research firm, points out that much of the software OVHcloud runs on comes from VMware, which is owned by Broadcom, an American company.

Governments that do shun America's suppliers risk irritating its tempestuous president, which could make the super-

power an even less dependable ally. Pieter Wezeman of SIPRI, a think-tank, notes that in July Denmark plumped for maritime-patrol aircraft made by Boeing, an American aerospace giant, over European alternatives, which he reads as a signal that the country does not want to alienate America despite clashes over Greenland.

For consolation, America's allies should remember that there are plenty of dependencies in the other direction. Much of America's federal government runs on

software provided by Germany's SAP. And American firms selling to foreign governments often depend on global supply chains. The rear fuselage of every F-35 is built by BAE Systems, a British weapons manufacturer, in Lancashire. The most advanced chips used in cloud computing are made in facilities operated by TSMC, a Taiwanese manufacturer, using gear from ASML, a Dutch one. America's allies might usefully focus on making themselves even more indispensable. ■

BARTLEBY

Returns policy

How to measure the benefits of artificial intelligence

IN THE BEGINNING there was AI, and bosses everywhere lost their minds and said you must all use this technology no matter whether it is useful and no matter how much it costs. And they introduced AI leaderboards and came up with stupid words like "toxiennaxxing". And lo, usage did indeed rise. And then the bosses remembered some very basic concepts like "budgets" and realised that this might not be such a great idea after all. And then a different question rang through the boardrooms, and this question was about returns, and it was harder to answer.

This potted Genesis of AI adoption is not entirely fair. As long as organisations are trying to encourage usage, it can still make sense to look at things like employees' token consumption. "I would expect software engineers to spend more money than a finance analyst," says the boss of one software firm. "But anyone within finance, it shouldn't be zero."

But usage (also known as input) measures don't tell you whether any value is being created. And even when AI is producing benefits, points out Mika Ruokonen of LUT University in Finland, lots of them accrue quietly to individuals in the form of scattered time savings. If an extra hour or two of an employee's time has been freed up to shop on Vinted, someone has benefited (mainly Vinted) but it isn't your company (unless you work at Vinted).

Shifting the focus to returns on AI investment means concentrating more on outcomes-based measures, such as team productivity or customer satisfaction. But these require careful thought. Scientists, for example, are using AI to produce more papers (yay!) of dubious merit (boo!). A recent study by Tulsi Suchak of the University of Surrey and

her co-authors found that papers derived from a health and nutrition data set ticked along at an average of four a year between 2014 and 2021; in the first nine months of 2024, there were 190.

So outcomes-based measures have to be designed to capture both positive and negative effects of AI. At Google, for example, engineers have long been measured on three dimensions. Speed is one: how long does a code review take, say. Ease is another: how much friction is there in the system for things like onboarding new developers. The last, critically, is quality: "I don't want to just go fast and make it easy to ship terrible software," says Richard Seroter of Google Cloud.

Complicating matters further is the fact that it often takes a while for the benefits of AI to show up. People have to spend time learning how to use the technology. Increased output can create bottlenecks elsewhere. A paper published last year by Kristina McElheran of the University of Toronto and her co-authors looked at AI adoption among American manufacturing firms, and confirmed the

existence of a "J-curve" in which productivity dips before it leads to an improvement. This effect is especially pronounced at established firms and seems to be partly explained by changes to management practices that had previously helped firms to keep tabs on performance. Any kind of calculation about projected returns has to take this initial period of disruption into account.

Bosses also have to take a view on something much more fundamental: what kind of return are they interested in? One way to show financial returns on AI is to seize on time savings as an excuse to cut headcount. For some jobs and in some circumstances, that might make sense. But chainsaws and employee morale go together about as well as salt and slugs. And people who are no longer there cannot be redeployed in more productive ways. A better option is to calculate the amount of money that AI saves on expanding headcount. (Working out how much extra revenue can be attributed to the technology is even more of an art; at the very least, you need good baseline data and a culture of A/B testing to isolate its effects.)

On top of everything, there is wild uncertainty about where AI is heading. That is an argument for a third set of measures, to go alongside ones on inputs and outcomes. These measures, which Mr Seroter calls "organisation-based", are more geared towards capturing how well change is being managed. They could include data on levels of employee satisfaction with the way AI has been implemented, or progress towards building in-house expertise in areas that AI will not automate. A one-eyed focus on usage ignores the importance of returns. A rigid focus on returns ignores the need to keep learning.



Meta

Social reckoning

LOS ANGELES

A blockbuster trial draws parallels to big tobacco

IN EARLY OCTOBER a sequel to “The Social Network”, a film about Mark Zuckerberg’s founding of Facebook, is scheduled for release. Called “The Social Reckoning”, it will dramatise the role of whistleblowers who alleged that the tech giant hid the fact that its products were harmful. Some of the characters involved will take part in a no less vivid drama involving Facebook, Instagram and their parent company, Meta, in a federal courtroom in Oakland, California. Oral arguments started on August 18th. Reckoning or not, the trial may be a blockbuster.

California, Colorado, Kentucky and New Jersey are leading a group of 29 states that are suing Meta over allegations that its platforms developed features harmful to children, and that it violated their privacy. In opening arguments, Megan O’Neill, a deputy attorney-general for California, said that Meta’s business model was to “hook the users, hold them for as long as they can, harvest their data and then hide the truth from the public.” Rob Bonta, California’s attorney-general, has described the case as Meta’s “tobacco moment”, likening the firm’s alleged deceptions to those that led to huge payouts by tobacco firms in the 1990s.

Paul Schmidt, a lawyer for Meta, told jurors that the company took its responsibility to young users seriously, adding that there was no clear link between youngsters’ social-media use and ill health. Meta is expected to fight efforts by the plaintiffs to turn the trial into a referendum on the impact of social media on children, focusing instead on the narrow legal issues in dispute. Its lawyers have asked why it is being singled out, when youngsters’ use of YouTube, owned by Google, and TikTok, created by China’s ByteDance, is even higher than it is of Instagram.

A lot is at stake. In a similar case brought by the government of New Mexico, a judge this month ordered Meta to pay almost \$950m in penalties. The four states acting as plaintiffs in Oakland appear to be seeking more than 200 times as much if Meta loses the case, putting their estimate at \$193bn—roughly equivalent to the firm’s revenue in 2025. If that were a benchmark for other states, it would devastate Meta.

Whether it comes to that is another matter, though. In emotive language, Ms O’Neill alleged that Meta particularly targeted young children. She frequently used

the word “hooked”. But unlike a trial in Los Angeles in March, in which Meta and Google, were obliged to pay damages to a 20-year-old harmed by spending much of her life on social media, the Oakland case is not focused on addiction.

Instead it is what one of the attorneys-general calls “the largest consumer-protection lawsuit in American history”. Vincent Joralemon, of the University of California’s Berkeley Centre for Law & Technology, says the big question is whether or not Meta deceived the public about the impact of its products. That is similar to the tobacco cases. It also means whistleblower testimony is likely to be more relevant than thorny psychological debates about addiction, reckons Mr Joralemon.

Meta’s investors tend to think that the burden of proof on the deception charges is “quite high”, says Gil Luria of D.A. Davidson, an investment bank. Litigation risk has weighed on Meta’s valuation, contributing to an earnings ratio that is lower than those of its big-tech peers. But rather than fearing a giant payout, shareholders worry more about remedial changes that the prosecutors may seek if Meta loses in court, according to Mr Luria. Those include changes to features Meta, Google and TikTok use to keep user engagement high, such as “infinite scroll”, which continuously loads new posts. “If we didn’t have infinite scroll, we wouldn’t have as many ads,” Mr Luria says.

Even if Meta loses, it has scope for appeal. It could, for instance, do so on the grounds that it is protected by Section 230 of the Communications Act, which safeguards internet platforms from legal liability for content posted by third parties. It sought to use that argument to halt the Oakland trial before it started, but the judge ruled that the appeal was premature.



A marquee trial for Mark

Public concern in America about the impact of social media on children is high, and being dragged through the courts may not help Meta’s image with parents. In order to head off similar worries about AI, on August 18th OpenAI launched ChatGPT for Kids, which strengthens guardrails for under 18-year-olds.

For all the attention the Meta trial will receive, court verdicts can have less of an impact than the headlines suggest. The tobacco industry agreed in a settlement of 1998 to shell out more than \$200bn over 25 years. But such outcomes have a “disappointing history” in America, says Matthew Lawrence of the Emory University School of Law. “We can mitigate the harms of industries exploiting addiction, but we have not yet had great success in finding ways to eliminate the harms.” ■

Reddit

Good bot, bad bot

Can “the internet’s front page” survive in the AI era?

EARLIER THIS summer “Backrooms”, a horror flick, became one of the most successful independent films ever made, grossing \$393m from a \$10m budget. Its source material was unusual: a collection of images of unnervingly empty rooms that spread on Reddit, an internet forum. Hollywood producers have since taken to tawling the site for inspiration.

Reddit has had a difficult time of late. Its share price has dropped by about a third this year. Strong quarterly earnings last month failed to cheer shareholders, who fret that the company will suffer as users turn to artificial-intelligence chatbots for information, rather than search engines such as Google that direct them to other sites. For now Reddit is attempting to cajole chatbot-makers into paying more for its data, both through deals and lawsuits. But eventually it hopes bot-weary users will return to one of the internet’s last refuges of human-generated content.

Reddit’s business model lies somewhere between Instagram (which profits off user-generated content) and Wikipedia (which relies on an army of volunteer moderators). That balance has sometimes been tricky to manage. In 2023 the volunteers who moderate its “subreddits” went on a multi-day strike after the company said it would charge for access to its application programming interface, on which independent developers who build Reddit-based apps rely. The platform, which stuck with the policy, may have had larger clients ►►

in mind. Soon afterwards it struck deals with Google and OpenAI, maker of ChatGPT, which reportedly agreed to pay \$60m and \$70m a year, respectively, to train their models on Reddit's data.

They may have got themselves a bargain. Reddit has become the most-cited source by many of the leading American chatbots, according to an analysis the company commissioned from Profound, a research firm. One particularly valuable use of its data is for generating product recommendations. About 40% of conversations on the site relate to commercial activity including shopping, says Jen Wong, Reddit's operations chief. On subreddits such as r/BuyItForLife, users debate the quality of products—and try to weed out sponsored or bot-generated content—making the results a rare source of high-quality reviews. AI models rely heavily on such content to generate their own recommendations, which chatbot-makers hope to turn into a source of ad revenue.

Google's relationship with Reddit in particular has become less of a two-way exchange. For years Google's ranked search results have been an important source of traffic for the chat site. But as the tech giant shifts users' focus to its AI overviews, referrals have dropped precipitously. Reddit's stock took its largest-ever single-day plunge last month after it said that "choppy" search referrals had hurt engagement. Last quarter the amount of time users spent on Reddit's app each day was down by 8% from a year earlier, compared with gains of 6% and 4% for Instagram and TikTok, respectively, according to Sensor Tower, a data provider.

One option for Reddit is to bargain for a better deal with AI providers. It has been negotiating new agreements with Google and others. If that does not work, lawsuits may. Reddit has already sued Anthropic, another model-maker, and Perplexity, an AI search engine. Both suits allege that the companies scraped Reddit's data illegally. (Anthropic and Perplexity have said they did not break the law.) Victory in court—or lucrative settlements—would give the chat site greater leverage in negotiations over licensing its data.

Ultimately, though, Reddit hopes to lessen its reliance on other tech firms. Traffic from search is "not where our business lives", said Steve Huffman, its boss, in last month's earnings call. Instead the company hopes to encourage more users to visit its site directly, and spend more time browsing once there. That would further boost its ad revenue, which grew by 64% in the second quarter, year on year. Ms Wong is optimistic that the (mostly) human content on Reddit will appeal to users as they grow tired of interacting with chatbots. Aspiring film-makers need not fear being starved of material just yet. ■



Canadian crude

On shifting oil sands

Will the latest boom endure?

PLEASE GOD, give me one more oil boom—and I promise not to piss it all away next time" reads a popular bumper sticker in Canada's oil patch. Recent results from the country's oil producers show that plea has been answered. Oil prices have jumped as the passage of tankers through the Strait of Hormuz has been choked off. Last month Cenovus Energy reported that operating profits in its most recent quarter had nearly tripled compared with a year before. Canadian Natural Resources posted its best-ever quarterly adjusted operating profit. Moreover, the easing of political headwinds in Canada may improve the chances of the boom lasting.

Canada has the world's fourth-largest proven reserves, with 97% found in the oil sands in a remote corner of the landlocked province of Alberta. Here the bitumen, mixed with sand and clay, is almost solid, and is mined rather than tapped from the ground. It is costly and dirty to extract and process. Despite the drawbacks, high prices combined with a weakening of the Canadian dollar mean that "it hasn't been this good to be an oil-sands producer in a very long time," says Andrew Leach of the University of Alberta.

Getting more oil to market is also set to become easier. Justin Trudeau, Canada's former prime minister, was tepid in his support for new pipelines, fearing the consequences for the climate. In 2018 regional political opposition led Kinder Morgan, an American firm, to suspend work on

a conduit from Alberta to the west coast in British Columbia. That forced the national government to step in and complete it. Oil eventually started flowing in 2024.

Lately the political mood has shifted. A trade war with America, the destination for 90% of Canadian oil, has shifted the federal government's focus to reducing dependence on its southern neighbour. That includes support for building new infrastructure to transport oil (and gas) to ports on Canada's west coast, from where it can be shipped to Asia. Mark Carney, Canada's current prime minister, is intent on slashing the red tape that holds up projects, including streamlining environmental permitting and consultations with indigenous communities. The speedier process will apply to a new pipeline under development to British Columbia, announced last month. Lisa Baiton of the Canadian Association of Petroleum Producers, an industry group, has spoken of a "generational opportunity" to rally political and public support behind new oil projects.

The promise of a federal government more friendly to pipes may not be enough, however. Enbridge, an energy-infrastructure firm, has paused construction of a pipeline to the east, citing a lack of commitments by producers to provide enough oil to make it worthwhile.

The unique challenges of the oil sands provide an explanation. Investment in new sites has been virtually non-existent since the oil price crashed in 2014 as demand weakened while America's shale oil gushed and OPEC opened the taps. "It cannot be overstated how much that changed the industry," says Kent Fellows of the University of Calgary. Moreover, projects in the oil sands take far longer to develop than conventional wells. They require enormous amounts of capital that could be tied up for five to ten years before profits flow. Uncertainty over the future price of oil—and the danger that a change of government might lead to the reinstatement of tighter regulations—make for risky bets.

Nevertheless, at the Calgary Stampede, an annual rodeo festival held in July that serves as the industry's unofficial barometer, parties hosted by oil companies were back to their riotous best after quietening down in recent years. Their sponsorship of rowdy chuckwagon races broke records. That is as good a sign as any that another boom is in full swing. ■

Watch The Insider ●

This week our editor-in-chief interviews historian and author Yuval Noah Harari. To watch the video, go to economist.com/insider

SCHUMPETER

Shadow boxing

The war on data centres is a bit fake



IN FEBRUARY, SHORTLY after making his feature-film debut as a Ipen magnate in “Marty Supreme”, Kevin O’Leary announced that he would build one of the world’s biggest data-centre projects in Utah. The market shrugged. What did Mr O’Leary, who is best known for presenting “Shark Tank”, know about data centres? How would it be paid for? Or powered? One analyst dismissed the project as an example of “vapourware gigawatts”.

But a joke in the marketplace became dynamite in the town square. In June J. Stuart Adams, president of the Utah Senate, who had initially supported Mr O’Leary’s venture, demanded that its size be reduced by three-quarters. Mr O’Leary accused his critics of working for China (he later withdrew that), but eventually agreed to scale back his dreams. Even then Mr Adams, a long-serving Republican lawmaker, along with a pair of county commissioners, lost primary elections because of the scandal.

Data centres are a nightmare of political economy. Voters see them as ugly, dirty, noisy and power-hungry bastions of arrogant Silicon Valley companies. And—thanks to ChatGPT—it has never been easier to pen a letter to the planning department saying so. Yet the public has an awful lot riding on these windowless sheds, for the stock market is a one-way bet on them being built. Data centres are particularly disorientating for Democrats, since unions tend to support the projects. If artificial intelligence is a class war, it pits a coalition of billionaires and electricians against the office droids both despise.

Wildcatters like Mr O’Leary are a soft target for politicians who want to be seen as tough on data centres but still want them to be built. On August 18th Josh Shapiro, the governor of Pennsylvania, derided the developers “scaring our communities” and signed an executive order he said would squash “speculative proposals” that had little chance of being built in the first place. Of the more than 100 projects announced in the state, less than a fifth had applied for the permits needed to develop them, he said.

Imaginary scalpels are easy to claim. Silicon Valley giants are the biggest backers of data centres. But beside them is a long line of property developers, private-capital firms, “neo-cloud” operators and crypto-miners, followed by international chancers, national

rogues and local spivs. Most of their plans were destined to remain press releases. Analysts at Bernstein, a broker, attempted to quantify the credibility of each watt of announced capacity. Plans by Google and Amazon are taken at their word. Those by CoreWeave and Nebius, two large neo-clouds, are discounted by around half. Mr O’Leary is ignored.

According to their model, another fantasist is Vermaland, a land bank that announced a gargantuan 3GW facility in Arizona last year. In May the project was reportedly cut by four-fifths after facing local opposition—a shadow-boxing knockout. Little information is available about a 1GW project in south Florida that was supposedly shelved in February, except that the land had been bought for \$15m in 2024 by an obscure investment firm. When a data-centre plan that would never have materialised is cut down, it often makes a very large sound.

Politicians are eager to polish their anti-data-centre credentials ahead of the midterm elections in November. The resulting policies may not affect the boom much. In July New York’s governor announced a one-year moratorium on big projects (including those not connected to its electricity grid). But the moratorium does not apply to data centres that have already been approved, and New York is hardly important to the build-out. Virginia imposed a \$600m electricity tax on data centres to great fanfare, but left in place a sales-tax exemption three times as big. Mr Shapiro’s new rules will require local authorities to approve projects before the state does, clearing up his desk until after the election.

If the data-centre build-out has a natural home it is Texas. The state is vast, energy-rich and home to plenty of the lawyers and engineers needed to handle big infrastructure developments. Its electricity-grid operator has been flooded with an astonishing 474GW of requests to connect to the grid (which currently faces peak demand of around 90GW). On August 3rd Greg Abbott, governor of Texas, announced an “audit” of data-centre projects, which is expected to last, conveniently, until just after the midterms. Tellingly, some of the biggest data-centre operators have publicly supported his plans.

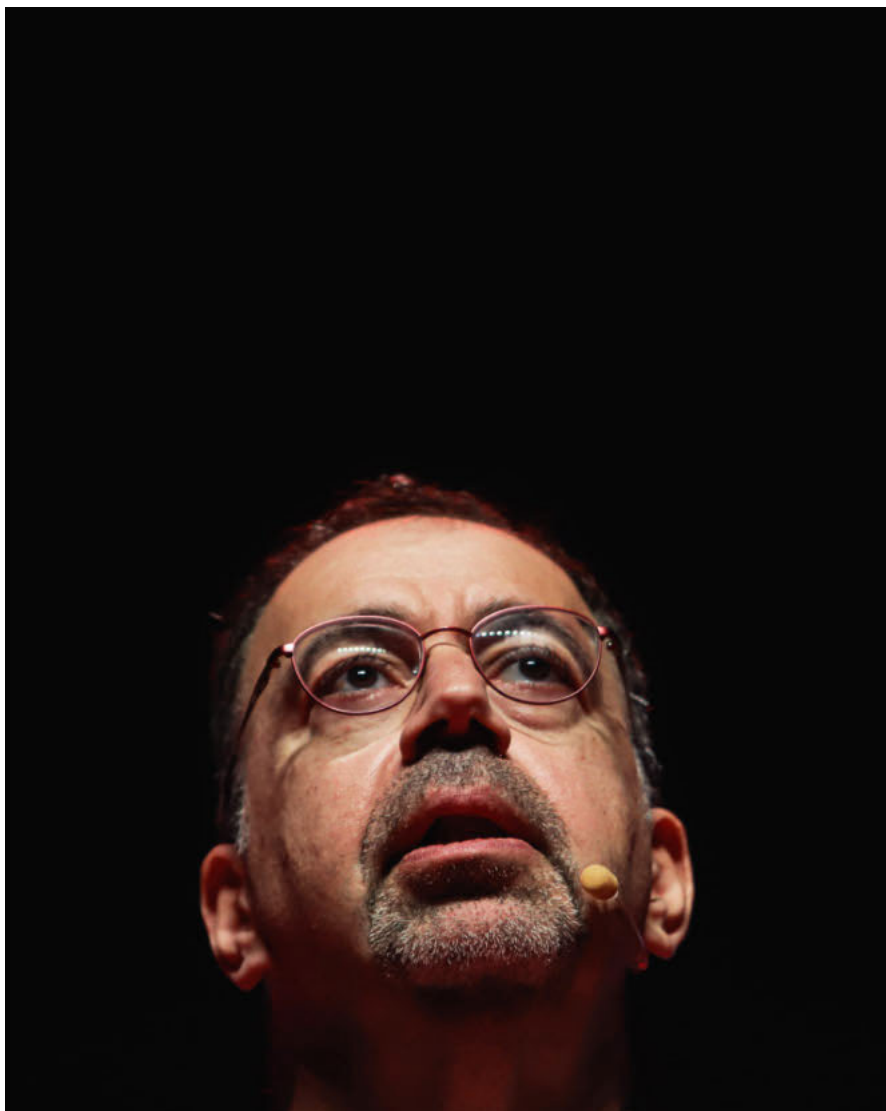
More political controversy probably means more delays for projects that are not yet in motion. But the share of the data-centre pipeline that has been delayed or cancelled has remained consistent at roughly 10% during the past year, despite the explosion in the number of plans and political opposition to them. Thus far constraints imposed by the market, such as the availability of chips and workers, have worried developers far more than those imposed by government.

Could the slightly fake war against data centres in fact be a “phony” one, poised to turn real? It is hard to imagine. Politicians are rarely willing to halt projects that have already broken ground. As for the new ones, developers can to some extent vote with their feet. So far that has meant moving west. It could also involve building more on federal land. Eventually, it might involve more construction abroad—or, if Elon Musk is to be believed, in space.

Stoppable force, movable object

It is often said that China has an advantage in the AI race because it can build big things by force. Yet America has the power of willing things into existence by exaggeration. The grand plans of developers will be watered down. Politicians will also overstate their opposition. Often they will cancel each other out. Meanwhile, the searing pace of construction will continue. ■

Finance & economics



Daron Acemoglu

The rise of the world's most influential economist

Daron Acemoglu is prolific and revered. Yet a lot of his work is oddly unconvincing

BY THE TIME the call came from Stockholm, it felt overdue. In 2024 Daron Acemoglu of the Massachusetts Institute of Technology shared the Nobel prize in economics with Simon Johnson and James Robinson for work on how institutions shape prosperity. Mr Acemoglu is only 58 years old, but for some time he has been the discipline's colossus. He has written seven books, including a new one about democracy and the economic impact of artificial intelligence (see Culture section),

not counting a couple of popular textbooks. His hundreds of papers, more than ten of them published this year, are often stuffed with complex mathematics. He competes with Andrei Shleifer of Harvard University for top spot on a citation ranking of economists produced by IDEAS/RePEc, a research database. His views carry weight with other wonks and the media.

Give economists a few drinks, however, and some will venture their true opinions about this giant. "Much of his theoretical

→ ALSO IN THIS SECTION

58 A warning from the bond markets

59 The truth about Chinese jobs

60 Trash talk on trade

61 Buttonwood: Lying indices

62 Free Exchange: Free passage

work is useful, but he uses his models to inform populist policies that have been tried before and failed," blasts one well-known economist. Some commentators do not require Dutch courage. "I've been yelling about Acemoglu for literally a decade," Noah Smith, an economics blogger, has said, in response to a flurry of online criticism of Mr Acemoglu's work.

No one accuses Mr Acemoglu of truly poor research, let alone academic malpractice. "He is obviously a genius," says another economist, who adds that he is generous to PhD students and junior colleagues. The question is whether his reputation at the very apex of the economics profession—and the intellectual influence that goes with it—is justified by his scholarship.

Start with his empirical methods. His Nobel rests heavily on a paper released in 2001 with Messrs Johnson and Robinson, which researchers have cited more than 23,000 times. It seeks to explain why some countries are rich and others poor. In places where European settlers died in droves, perhaps because of disease, colonisers built "extractive" institutions, with power and resources concentrated in the hands of an elite few, to take out as many resources as possible. But in more agreeable climates, settlers survived. Colonists built more inclusive institutions, with proper roads, schools and medical services. Those institutions persisted, with the effect that places with lower settler mortality in colonial times are richer today.

Economists now take these results seriously but not literally. The paper encouraged them to think more deeply about institutions, but they have also picked it apart. The Nobel committee's report in 2024 conceded that the mortality data are "sometimes sketchy". In 2012 David Albouy of the University of Illinois Urbana-Champaign showed that some countries were assigned mortality rates borrowed from other countries, among other adjustments. Correct for these, he found, and the paper's estimates become unreliable. Last year Martin Buchner of RWI-Essen, a research institute, and colleagues reported that experts they surveyed were somewhat more likely to side with Mr Albouy.

Mr Acemoglu says that "these discus- ➤

► sions and some critiques are hugely valuable.” But he counters that Mr Albouy reaches his conclusion by omitting half the data from the original sample, including on important countries like America, Canada and Australia. It is this combination, along with some statistical choices, that introduces the unreliability, he says. He adds that if he were redoing the paper today, he would make “a number of changes, including in some of the estimation details”—though not to the mortality data.

Most recently, researchers have examined a paper that Mr Acemoglu released in June, with co-authors, which argues that lower birth rates are associated with faster growth in GDP per working-age adult. A striking finding, it seems to suggest that places with falling populations, like Japan, do not have much to worry about. Yet other economists, including Jesús Fernández-Villaverde of the University of Pennsylvania, question whether historical relationships are really useful guides to the present, when many countries’ birth rates are collapsing. It is a fair point—and Mr Acemoglu and his co-authors raise it in their conclusion. But it can get lost and can seem unsophisticated when set against Mr Acemoglu’s reams of maths.

A determined critic can find holes in almost anyone’s work—and Mr Acemoglu is the biggest target in economics. Yet fame-induced scrutiny cannot explain a second problem: that his ideas can seem underwhelming. His public commentary, especially on American politics, is unadventurous. His “unified theory of Trump”, for instance, asserts that the president operates “by elevating executive power and destroying constraining institutions and norms”. Well, indeed.

More worryingly, his grand thesis may not reveal much. Nations prosper when institutions are good, and stagnate when they are bad. True. But what, exactly, is an institution? Rules, norms, enforcement, culture—everything, really. Where do institutions come from? “Critical junctures” and “institutional drift”.

Reviewing one of Mr Acemoglu’s books in 2011, Tyler Cowen of George Mason University noted that the institutional changes it describes seem only to come from other institutional changes, making the core argument regress for ever—turtles all the way down. Duncan Green of the London School of Economics has said that the framework works mainly in hindsight. Francis Fukuyama of Stanford has argued that the book waves away the example of China, which has had blistering economic growth alongside institutions that can plausibly be described as extractive.

Mr Acemoglu retorts that the book devotes two chapters partly or mostly to China. As for the turtles: “To understand institutional change from an institutional

perspective you must consider the specific institutions in the past and the historical events...that influence them,” he says. “I don’t think this is tautological at all.”

To the extent that Mr Acemoglu breaks from received wisdom, it is mainly to espouse remarkably glum views about technology. “Power and Progress”, written with Mr Johnson and published in 2023, treats a thousand years of innovation largely as a story of elite capture and unintended consequences. The cotton gin was the handmaid of slavery; the Haber-Bosch process created weapons. Ordinary people apparently gained only when they could steer capitalists to consider their interests. The book plays down a more compelling claim: that technological progress in itself has made the average person many times richer than in pre-industrial times.

Mr Acemoglu’s arguments about AI are also pessimistic to the point of losing credibility. In a paper published in 2024 he calculates that AI will boost American productivity by only a touch over ten years. The paper reaches this conclusion by ignoring the potentially transformative effect of new AI products. “He leaves out entirely in that analysis the possibility that we will have more rapid scientific progress, more rapid social-scientific progress, or better decision-making because of artificial intelligence,” according to Lawrence Summers, president emeritus of Harvard (and a former treasury secretary).

“This is a valid criticism,” says Mr Acemoglu. “I did not foresee agentic AI and it was not incorporated in my predictions.” He also notes that AI models may be more useful in scientific research than he imagined. But he stands by his methods of estimating the productivity effects of AI, as well as injecting “some realism into the claims about huge coming productivity improvements”.

These academic disputes could end up mattering a great deal. Mr Acemoglu may not think AI is about to turbocharge productivity, but he does worry about its effects on democracy and jobs. His new book focuses on its potential social harm rather than its potential benefits. His proposal to limit that harm, “pro-worker AI” that would augment the value of human labour rather than replacing it, sounds terrific—but also obvious.

Mr Acemoglu’s influence in the AI debate is clear in a recent statement, signed by dozens of prominent economists, which argues that “we must act now” to “steer AI in a direction that complements humans and benefits society”. But who are “we”? And who is to decide what sort of AI does or does not complement humans? Even economists who signed the petition say they are not entirely sure. So great is Mr Acemoglu’s stardom that it can sometimes blind the critical faculties. ■

Government bonds

Yielding

NEW YORK

Bond investors are global markets’ party-poopers

BOND MARKETS are unsettled, and so, in turn, is America’s government. On August 19th the Treasury said that from next month it would increase its own purchases of longer-dated debt. This reflects “increasing administration unease” about yields, believe analysts at Deutsche Bank.

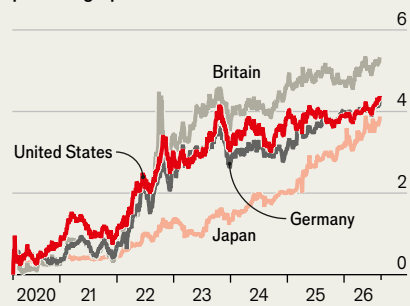
Officials have good reason to worry about rising borrowing costs. The yield on ten-year Treasuries reached its highest since January 2025 on August 18th. Scarier still, that on 30-year bonds briefly passed 5.3%, its highest since 2007. And America is not alone. Bondholders are demanding more from governments across much of the rich world (see chart). Yields on British, French and German long-dated bonds have all reached levels not seen in more than a decade. Japanese 30-year yields, long the lowest in big economies, are close to an all-time high. The rout eased somewhat on August 19th. But yields are unlikely to fall much soon. What is amiss?

For one thing, unlike equity investors, bond traders appear to be reading the news. The Strait of Hormuz remains largely shut and looks likely to stay that way for a while. Fuel prices in America—particularly for diesel, on which much of commercial haulage relies—have soared in turn. The fund managers surveyed monthly by Bank of America, most of whom are heavily invested in stocks, are sanguine. On average, they expect Brent crude, the global oil benchmark, to trade at \$76 a barrel by the end of the year, only slightly up from before the start of the war. Bond markets seem less sure.

Accordingly, they are pricing in continued inflation. On August 19th Britain re- ►►

Looking steep

30-year government-bond yields, percentage-point increase since Mar 9th 2020*



Source: LSEG Workspace **Black Monday stock-market trough

▶ ported consumer-price inflation of 2.9% for the year to July, up from 2.6% in June, as higher energy prices bit. In America, core inflation (which excludes food and energy) eased last month. But bond traders seem to have pared back bets on future rate rises after recent remarks by the Federal Reserve's new chairman, Kevin Warsh.

Another recent concern is that government debt-issuers have new competition. In recent months large tech firms have sold some \$75bn-worth of bonds to fund investments in data centres for artificial intelligence. The spree has already pushed their combined debt issuance to nearly twice last year's total, estimates Goldman Sachs, a bank. All sorts of businesses are getting in on the bond bonanza. Some 40% of large-scale debt issuance (ie, more than \$10bn) by investment-grade issuers has come from outside tech, Goldman calculates. So bond-buyers have a surfeit of investment-grade options from which to choose. Many believe the AI boom will push up interest rates by increasing competition for capital.

Still, the biggest reason for creeping yields is of long standing: concerns over government debts and deficits. On August 19th America's Treasury said federal debt had passed \$40trn (130% of last year's GDP) for the first time. The government's deficit is around 6% of GDP. Such worries also show up in differences between countries. The spread between French and German ten-year yields has reached its widest since 2012. Bond markets now expect a higher yield for Japanese debt than for Chinese debt, reversing the conventional order. In both France and Japan investors assess that politicians lack the will to trim spending or raise taxes meaningfully.

Governments have few other good options. At the Fed's meeting in June members of its rate-setting committee were briefed on how ownership of Treasuries has shifted from "relatively price-insensitive official-sector holders to more price-sensitive private investors". That is likely to increase the premium bondholders expect for long-term debt. In response to such pressures America, Britain and Japan have increased their sales of shorter-term bonds with lower yields. But as a result their debt stocks will roll over more often, raising the risk that such moments coincide with high interest rates.

A second unenviable option is for central banks to buy back more debt. But many of them had hoped to shrink their balance-sheets, not to expand them. They may therefore be reluctant to rely too heavily on bond purchases as a tool to reduce yields. All this means that bond markets are likely to remain wary. Interventions such as the Treasury's may help for the time being. They are unlikely to placate buyers for long. ■

Unemployment in China

Goodhart's law with Chinese characteristics

However you measure it, the job market is weak

STEERING AN ECONOMY has never been easy. In a candid presentation in 1975 Charles Goodhart, then of the Bank of England, shared some of Britain's "unhappy" experiences after sterling lost its link to the dollar four years before. Instead of defending an exchange rate, the bank had hoped to use interest rates to control the supply of money, variously measured. But the monetary authorities had ignored what the presenter playfully called Goodhart's law: "Any observed statistical regularity will tend to collapse once pressure is placed upon it for control purposes."

Others have put this more snappily: when a measure becomes a target, it ceases to be a good measure. The law poses particular difficulties for China, which has many economic targets and thus few good measures. In 2018, for example, it began publishing a new monthly gauge of urban unemployment, based, like similar statistics elsewhere, on surveys of the population. But unlike other countries' figures, China's measure has remained remarkably stable. July's rate of 5.2%, published on August 17th, was typical. It was less than 6% but at least 5% for the 99th time in the past 115 months. Even during the covid-19 pandemic, when American unemployment climbed well into double digits, China's measure never exceeded 6.2%.

One reason may be that the survey covers only urban workers. The flow of mi-

grants into and out of China's cities gives the urban labour market a homoeostatic quality. When job creation is strong and the unemployment rate is set to fall, more of them turn up, expanding the denominator. Conversely, when the labour market is weak, the jobless seek work in the countryside, dropping out of the survey.

There may be a second reason for the measure's stasis: the pressure placed on it for control purposes. Since 2018 the government has set an annual target for surveyed urban unemployment, usually around 5.5%, rising to 6% in 2020. That gives officials an incentive to keep the rate around that figure, by hook or by crook. A "possibility, which cannot be proved but is difficult to dismiss, is that the surveyed unemployment rate is being managed in some fashion to reduce volatility", argues Andrew Batson of Gavekal Dragonomics, a consultancy, in a recent note.

The stability of the unemployment rate is thus a false comfort. It makes it needlessly difficult to interpret the biggest job market in the world. Some economists suspect that China's export boom is not generating equally strong manufacturing employment. Others think artificial intelligence may be killing some entry-level white-collar jobs. Another worry is that wage growth is not strong enough to stop the economy falling back into deflation. All these suspicions would be easier to ▶▶



Stably employed by any measure



- verify or dismiss if China's unemployment figures were as fluid as its economy.

Instead, economists must rely on a variety of more or less unsatisfactory proxies. Mr Batson has spotted a new one—or rather, a revived one. He advises looking again at an older gauge, which the modern survey-based figure was supposed to supersede. This measure, which has been collected since 1978, counts only those who have registered as unemployed with local authorities, a first step towards applying for jobless benefits. For decades, this was also a government target, and also uncannily stable. With the arrival of surveyed unemployment, it fell into obscurity. The government stopped targeting it and even stopped releasing the percentage rate. But once a year the raw number of people registered as unemployed still appears in the official database, albeit with a substantial lag. The number jumped at the end of 2025 to 12.7m, an increase of almost 16% from the year before (see chart).

That total must be interpreted with care. Rule changes in 2020, just before the pandemic struck, made it easier to register, especially for people living far from their place of birth. But no tweaks in the rules explain the more recent increase in registrations, argues Mr Batson. The rise would instead seem to represent “real stress in the labour market”.

If so, then this uncharacteristic jump in a once inert statistic may also illustrate a deeper principle. “When a measure ceases to be a target,” Mr Batson suggests, “it becomes a good measure again.” Call it Batson's corollary to Goodhart's law. If the registered unemployment number attracts too much fresh attention, of course, the government may stop updating it. It has form. When youth-unemployment figures became a lightning rod for criticism in 2023, the government solved the problem by suspending them. Tracking China's economy has never been easy. When a measure becomes a target, it ceases to be a good measure. And when a measure becomes too embarrassing, it ceases to be. ■

Trade politics

Trash talk

SINGAPORE

The White House's absurd claim of a Chinese transshipment “scam”

IN G.K. CHESTERTON'S parable of the fence, those eager to tear down seemingly useless structures are urged to understand why they were built in the first place. Chesterton's fence has long been received wisdom among American conservatives of a certain age, but Peter Navarro, Donald Trump's chief trade ideologue, seems to have missed the memo. In a new paper published by Mr Navarro's office in the White House, titled “The Great Transshipment Scam”, dozens of countries from Japan to Jordan stand accused of abetting a Chinese plot to dodge American tariffs. Absurd as this claim is, the paper is nevertheless a vivid exposition of the Trump administration's muddled thinking on trade.

Start with the “great reallocation”, Mr Navarro's term for the shifts that followed Mr Trump's first wave of tariffs on Chinese goods in 2018. At the time, free-traders warned that a gap in tariff rates between China and the rest of the world would backfire. It would create an irresistible incentive to evade duties, legally or otherwise, while crimping growth. America understood this well when, in the 1940s, it was the primary author of a global trading system based on non-discrimination.

Mr Trump charged ahead anyway. The average tariff-rate differential between China and the rest of the world widened from 0.9 percentage points at the start of 2018 to 29 points last year (see chart). The predictable happened: the share of American imports coming from China tumbled from 21% to 9%, while that from other countries soared. In “connector” economies such as Vietnam, Malaysia and Mexico, imports from China and exports to

America rose in tandem. Messrs Trump and Navarro now face a problem of their own making.

Mr Navarro calls the pursuit of lower tariffs the “financial engine behind the Great Transshipment Scam”, echoing Mr Trump's threat last year to impose a 40% tariff on “transshipment” (a loosely defined term that usually refers to Chinese goods re-routed through third countries). Yet rejigging supply chains to shift a shipment's country of origin and reduce its tariff liability is no scam—unlike concealing its origin by simply slapping on a new label.

The traditional legal test is “substantial transformation”, requiring a “fundamental change in form, appearance, nature or character”, according to American trade authorities. This is not a clean distinction. Most free-trade deals include painstakingly negotiated “rules of origin” (ROO) for each product line. Haggling over ROOs for the Trans-Pacific Partnership, a 12-country trade deal scuttled by Mr Trump in 2017, took the better part of a decade, notes Deborah Elms of the Hinrich Foundation, a think-tank in Singapore.

Even Mr Navarro offers a cursory nod to the difference between “legitimate manufacturing and substantial transformation” on the one hand and “pass-through trade and origin shifting” on the other. Indeed, he collates five private- and public-sector estimates of “illegal transshipment flows”, amounting to as much as \$303bn—an improbable 56% of America's imports from China in 2018—each year. Yet even the most convincing of these, an analysis by Goldman Sachs finding that \$40bn-worth of Chinese goods were “superficially re-exported” to America in 2023, does not claim that the flows are illegal. (Granted, some will be customs fraud, which America is within its rights to clamp down on. Mr Trump's Trade Fraud Task Force claims to have recovered \$1bn from tariff-dodgers since launching last year.)

Were Mr Navarro exercised about illegality alone, third countries building factories making goods from Chinese components should pose no problem. Yet he rails against “goods that are not necessarily declared as Chinese” but which involve “China-origin inputs or components, Chinese ownership or financing, relationships with Chinese suppliers or manufacturers, China-based production steps [or] China-origin routing histories”.

This is a staggeringly broad objection. Modern manufacturing is structured as a global network of value chains, distributing capacity across countries in each stage of production. But if proximity to Chinese inputs or firms is evidence of complicity in an anti-American scam, then America is at war with trade itself. It is a “redefinition of trade such that everything is transshipment”, says Ms Elms. ►►



► Mr Navarro's paper describes the "functional architecture" of the "shadow transshipment network" that stretches across 43 countries, making up over 70% of America's non-Chinese imports. From assembly hubs in South-East Asia to logistics centres in the United Arab Emirates and Canada, the supposed scam snakes through "production-side transformation claims, logistics-side routing channels, processing zones, maritime gateways, overland corridors, bonded warehouses and re-invoicing

systems". This is, more or less, the entire infrastructure of global trade; policing it would require a panopticon. Still, Mr Navarro is willing to try. He vaguely threatens setting up an AI-enabled "detective border" system.

After all this, you may wonder whether Mr Navarro thinks imports to America should contain any Chinese content at all. If so, how much? An extreme (though coherent) answer would be to set a maximum threshold for Chinese content, and ban or

tax all imports that breach it. Mr Trump's 40% transshipment tariffs could then be used to strong-arm countries whose exports contain too much from China.

That has not come to pass, presumably because doing so would drive up prices for an American public already enraged about the cost of living. Yet Mr Navarro's extraordinary paper has performed a service. It is a further illustration of the mess his boss's attack on the global trading order has brought about. ■

BUTTONWOOD

Lies, damned lies and stock markets

Benchmark indices are becoming divorced from equity reality

STOCK MARKETS, Wall Street-wary politicians and like-minded commentators repeat ad nauseam, are not the real economy. The S&P 500 index of America's biggest firms is up by 12% this year, setting one all-time high after another. Meanwhile, they note, though American GDP growth has been steady and unemployment is fairly low, consumers feel more downbeat than at just about any time since the University of Michigan started keeping track in 1952.

All true—and nauseatingly familiar. Less remarked is the fact that the S&P 500 and the world's other benchmark indices are no longer the real stock market, either. Rather than mirroring the universe of domestic equities, they increasingly reflect the fortunes of a few corporate giants, which in turn mostly rise and fall with the unstable outlook for the artificial-intelligence revolution. This is making many benchmarks—and, by extension, the stock portfolios and pension pots of investors everywhere—considerably more volatile.

Take the S&P 500. The biggest constituent, Nvidia, accounts for 8% of its value. Imagine the \$5.3trn chip-designer's share price moves up or down by 3% (as has happened 30 times this year) and the other 499 stocks are flat. The index then shifts by 0.25% in the same direction. No single name in the S&P 500's history has packed a bigger punch.

Yet in comparison with the index it is a featherweight next to TSMC, which manufactures Nvidia's AI processors. The semiconductor hulk, whose market capitalisation has doubled to \$2trn over the past year, accounts for over 40% of the Taiwan Stock Exchange's TAIEX index. To let investors partake in TSMC's success, last year the country's markets regulator dropped the 20-year-old rule

that index-tracking exchange-traded funds (ETFs) must hold no more than 30% of their assets in a single stock. As a result the TAIEX, which comprises just over 1,000 stocks, is behaving like an equal-weighted index of just six firms (or fewer if you consider that other big Taiwanese companies like MediaTek and Foxconn are on the same AI steroids as TSMC and their movements are therefore partly correlated).

Things are only slightly less top-heavy in South Korea. The same bit of financial maths (which, for the curious, involves calculating the inverse of the Herfindahl-Hirschman index for component stocks) implies that the 830-strong KOSPI benchmark acts as if it contained just 11 firms besides the country's two muscular memory-chip champions, Samsung Electronics and SK Hynix. The pair's combined weight in the KOSPI briefly exceeded 40% in June, up from a quarter, give or take, in the preceding few years. (South Koreans' love of leveraged ETFs, which use borrowed money to amplify gains but also losses, increases the KOSPI's jumpiness.)

Between May and July, as traders con-

stantly repriced the size of global AI spending, including on memory chips, the KOSPI's volatility was four times last year's average. The index moved by 5% or more once every three sessions, on average. On July 28th it sank by 11%. Three days later it soared by 18%. In the first six months of 2026 the Korea Exchange had to suspend trading five times and curb it on nearly 30 other occasions, more than in the whole of 2008, at the height of the global financial crisis.

Concentration and volatility need not go hand in hand. Nearly half the value of the broad Swiss index, SPI, sits in five firms: Roche, Novartis, Nestlé, ABB and UBS. But their industries are varied, and representative of Switzerland's listed firms. Almost one in two Swiss public companies makes drugs, food or drink, machinery or loans. In Taiwan and South Korea, by contrast, only one in six and one in 200, respectively, churn out chips or other technology hardware.

The top three Swiss stocks are also stodgy ones with stable earnings and regular dividends—and so inherently less skittish than tech shares. AI will transform them, as it will all sectors. But their fortunes are not hitched to the vagaries of the AI capex cycle.

Like a championship belt, dominance can be forfeited. Novo Nordisk took over the Danish bourse thanks to Ozempic, then shed its index weight as Eli Lilly's better obesity drug ate its lunch. That serves as a reminder that the stock market aims to capture investors' best collective guess at tomorrow's equity reality. It mirrors today's only when the future is likely to look much like the present. In the age of AI, and of abundant private capital that feeds a long roster of unlisted pretenders which benchmarks omit, that is increasingly not the case.



FREE EXCHANGE

The phantom tollbooth

Freedom of navigation is worth preserving



PFALZGRAFENSTEIN CASTLE, near the village of Kaub in Germany, had a lucrative site. Situated on an island in the Rhine, the waterway that connects the most prosperous bits of western Europe, the redoubt lies on one of the river's shallowest stretches. The tightness of the route past it meant that chains could be lowered from the battlement to prevent passage and demand payment. Pfalzgrafenstein was a toll castle, one of many along the course of the river.

Rhine tolls were a classic example of an economic rent, a payment that some entity is able to extract by making a cheap resource artificially scarce. The river cost the Electors Palatine who usually controlled the castle absolutely nothing, so the toll they earned from stopping others from making their way upriver was pure rent. In the aftermath of the latest Gulf war, there is a risk that tolls of this sort will come back. The Iranian government is considering levying a toll on ships traversing the Strait of Hormuz, through which around 20% of the world's oil supply passed before the war began in February. Oman, on the other side of the Strait, may join in. America, too, may want to get in on the act: President Donald Trump has said that his country should consider levying fees as "THE GUARDIAN OF THE HORMUZ STRAIT".

You might call this the political economy of the rent-seeking sea, to paraphrase the title of a paper by Anne Krueger, an American economist, published in 1974. Restrictions on trade create rents and that creates a new type of competition, in which different groups compete to capture those rents rather than doing anything productive. The trade restrictions that Ms Krueger observed in the "rent-seeking society" typically meant lobbying, bribery or legal chicanery. For charging tolls on passing maritime traffic the rent-seeking activity could be far more deadly: piracy, drones and so on. During the age of sail territorial waters ended three nautical miles (5.6km) from the coast, roughly the range of a cannon shot. It was a recognition that sovereignty could stretch only so far.

For the past couple of centuries, however, freedom of navigation has been the norm. That has typically been a public good provided by naval powers capable of asserting their sovereignty, or extending its reach beyond their shores. Tolls were abolished at

Pfalzgrafenstein castle in 1867: the Prussian government controlled both sides of the river and was at last able to make good on a diplomatic promise made years earlier. The rent extraction ended. In 1868 an international treaty abolished all tolls on the Rhine. Elsewhere change was also under way. For years the Danish crown had earned much of its revenue by levying tolls on access to the Baltic Sea, until in 1857 the great powers forced the end of that money-making scheme.

On the high seas freedom of navigation was assured by hegemonic powers, able to project their authority everywhere. For much of the 19th and 20th centuries the hegemon was Britain, notably in the Persian Gulf. The British Empire used a crackdown on piracy to strong-arm emirs into signing treaties, starting in 1820; their fiefs became the Trucial States. The empire secured access to the shipping lanes without formally incorporating them. The torch passed to America after 1971, as Britain withdrew from east of Suez. The Gulf states got their independence but also fell under America's shield. America occupied the former British naval base in Bahrain.

Keeping waterways toll-free, then, is an example of the hegemonic stability theory formulated by Charles Kindleberger, another American economist: to remain in good order, the global economy requires certain public goods. The hegemon is the only one with both the ability and the desire to provide them; it captures some of the large gains from a functioning world economy.

Kindleberger, who was writing about the Depression, was primarily concerned with financial stability, but freedom of navigation also fits the model. In his theory, and later versions, only the hegemon was able to both bear the costs of providing such goods and capture a big chunk of the benefits: "Our economic interest is in maximum freedom of movement for merchant vessels at minimum possible cost," wrote officials in America's State Department in 1969. Now America is retreating from that role, partly for lack of will—Mr Trump is reluctant to commit troops to the Gulf—partly for lack of ability. Cheap drones can project power much farther and more cheaply than the three-mile cannon shot of yore.

Is there a third option, neither anarchy nor hegemonic domination? Perhaps collections of other states—the "middle powers" referred to by Mark Carney, Canada's prime minister—could club together to provide public goods. The treaty guaranteeing a toll-free Rhine was signed by a group of German states, France and the Netherlands. But a paper last year by Anwesha Banerjee, Ottmar Edenhofer and Ulrike Kornek, three economists based in Germany, sounds a note of scepticism; a more equal balance of power between countries encourages free-riding and jockeying for position. It may be unlikely, then, that middle powers would do any better at keeping Hormuz toll-free.

History doesn't repeat itself but it does Rhine

Then there are the truly global public goods. Whether or not there are tolls on the Rhine is, for the time being, a moot point. Europe's extended heatwave and drought, now into its third month, has reduced the water level at Kaub to just six centimetres, below the previous record of 25cm in 2018. That is barely enough for children to paddle in, let alone for heavily laden tankers to pass. This will knock 0.2 percentage points off German GDP this year, estimates Oxford Economics, a consultancy. The danger is not only that states manage to extract rents from what was once free, but that the world is less capable of providing public goods at all. ■

Science & technology



Global warming and aviation

Bluer cooler skies

A trial over the Atlantic aims to show how much Earth could be cooled by contrail-free flying

THE 40M scheduled passenger flights which took place in 2019 added up to about 61bn kilometres of travel—exceeding the distance from Earth to the Sun every day of the year. More than 40% of those flights left contrails behind them for at least some of the journey as water vapour in the exhaust from their engines froze into tiny particles of ice.

Many of these trails were ephemeral. They streamed out behind their progenitor planes before fading quickly away. But over about 5% of the total distance flown—comfortably enough for ten round-trips a day to the Moon—they persisted and widened as water vapour already in the atmosphere joined in the freezing. Thin white lines became broad translucent stripes of cirrus cloud. Wind those stripes around Earth like yarn around a ball and you will find that, at any one time, persistent contrails cover maybe a thousandth of the sky.

In doing so they warm the planet.

Contrails are white because their constituent ice crystals reflect sunlight. That can have a cooling effect during daytime. But the crystals also absorb infrared radiation and re-emit it, rather as greenhouse gases do. Some is thus radiated into space, but the rest travels back towards Earth. This produces 24-hour warming that, overall, handily outweighs the cooling (something which is true for non-contrail cirrus, too). The exact amount of net warming is hard to calculate, but current evidence

suggests aircrafts' contrails and the cirrus clouds they induce may do more to raise the planet's temperature than carbon dioxide emitted by the aircrafts' engines.

This warming is, though, to some extent, optional. For contrails to persist the air they are in must be well below freezing point and yet contain a significant amount of not-yet-frozen water vapour. The thing which stops this water vapour freezing is a lack of particles that might "seed" the process by acting as nuclei around which crystals can form. A contrail provides such seeds by the trillion, and so creates cloud where there was none.

The nucleus of the problem

Such ice-supersaturated regions (ISSRs) tend to be found at altitudes of 8-13km. They are a few hundred kilometres across, but only a few hundred metres deep. Keep aircraft out of them and you should get rid of a lot of persistent contrails—which means getting rid of some warming, too.

On August 18th a consortium based in Britain announced Operation Blue Skies, the most important trial of this approach to cooling the climate so far undertaken. It will take place in Shanwick, a piece of airspace that covers 1.8m square kilometres of the North Atlantic west of the British Isles (see map on next page).

→ ALSO IN THIS SECTION

65 Probing Antarctica's ice

65 Dinosaurs' stomach stones

66 A new subatomic particle

66 Well Informed: When working, move it!



There have been smaller trials of the idea. One was run by Germany's aerospace agency in 2021 and another, in 2025, by American Airlines and Contrails.org, a not-for-profit research organisation largely funded by Breakthrough Energy, a network of investors founded by Bill Gates. The results were promising. Blue Skies, which is considerably bigger, should show whether ways to predict and route around ISSRs are both effective and suitable for operational use.

Shanwick is a bit of airspace well suited to such a project. It is prone to the right sort of damp chilliness at airliners' cruising altitudes, and it is used by a lot of planes. In combination, those factors mean that, although it covers just 0.4% of Earth's surface, models suggest Shanwick is responsible for about 5% of contrail-associated warming. What is more, European and American weather satellites monitor the area day and night in both visible-light frequencies and infrared, making it easy to see what is going on.

The Winter's tale

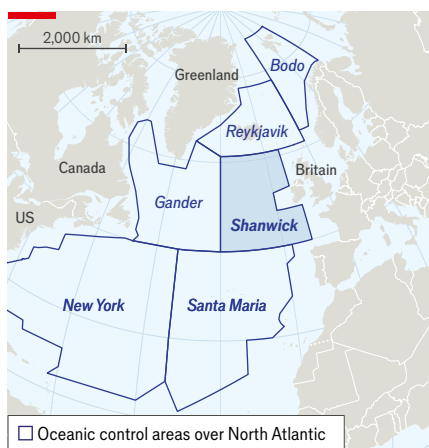
The plan is that on between 20 and 40 days over the coming winter and again the one after that air-traffic controllers at NATS, the company which looks after British airspace, will alter the flight paths of aircraft passing through Shanwick to keep them out of suspected ISSRs. These changes will route the planes a few hundred metres below their filed flight plans when conditions suggest it and when there is no reason not to (such as turbulence).

Comparing contrails made during periods when these changes are happening and periods when they are not should provide reliable data on how much the contrails are being reduced. "The perfect outcome", says Sebastian Eastham, part of the team working on Blue Skies at Imperial College, London, "is that a satellite image robustly and reliably has fewer contrails in it, time after time after time."

As well as NATS and Imperial, the consortium behind Blue Skies consists of Contrails.org, Google (which, as well as its software expertise, is chipping £1.4m, or \$1.9m, of hardware and computing time into the £5m budget), the University of Cambridge and Britain's Met Office.

If contrail avoidance is to go mainstream, the Blue Skies trial needs to show not just that days with avoidance measures in place have fewer contrails, but also that integrating the process into normal operations can be done safely and straightforwardly. That would open the way to more widespread, and eventually everyday use.

Adoption does not have to be universal to be useful. The warming attributed to contrails is concentrated. By one estimate 80% of the temperature-raising effect on the climate comes from just 2% of flights.



Awkwardly, you cannot know in advance exactly which 2% they are. But Marc Shapero, who runs Contrails.org, says that a system eliminating 70% of the world's contrail warming might require changing the flight paths of fewer than 5% of flights.

According to a recent paper by Jessie Smith of Cambridge and colleagues elsewhere, if aviation does not change its ways it will be responsible for about 0.01°C of total warming above preindustrial levels in 2050. If the world is seeing about 2°C of warming by then, that contribution comes in at 5%, with more than half of it induced by contrails. A 75%-effective contrail-avoidance programme that began in 2035 would reduce the total by about a third, though there is a lot of uncertainty.

Fractions of fractions sound small, but Dr Smith and her colleagues argue that the effect is similar in size to that of any other single measure proposed to reduce warming as a result of air transport—and could be realised faster.

The benefits are not free. Flying at lower altitudes increases the amount of fuel used, which is a cost to airlines, and the amount of carbon dioxide emitted, which is a cost to the planet. But neither cost is huge. Estimates from models suggest that avoiding ISSRs makes flights only slightly

longer and increases the amount of fuel used by less than 2%.

Using 2% more fuel on 5% of flights would increase the industry's total fuel bill by a thousandth. In the American Airlines trial the difference in fuel use between the flights that were diverted and those which were not was undetectable.

Similar calculations hold for warming. According to Olivier Boucher, a climate scientist who has worked on the matter for 30 years and now runs Klima, a small French company dedicated to it, "If you really focus on the most impactful flights, maybe you can save like 100 tonnes of carbon-dioxide equivalent for an extra tonne of carbon dioxide."

There are also opportunity costs. Some worry that focusing on contrails will distract attention from airlines' carbon-dioxide emissions and efforts either to reduce them (by using sustainable aviation fuels, or SAFs) or offset them (by removing CO₂ from the atmosphere and putting it into permanent storage). Eliminating contrails without attacking these points—or reducing the amount of flying—provides a one-off gain but does not change the fundamental fact that, in the long term, more flying means more warming.

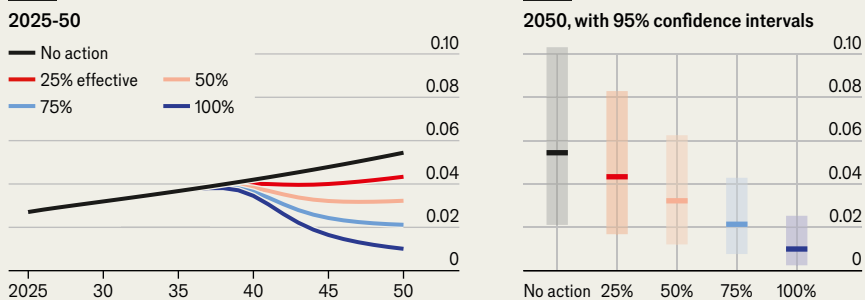
The ideal is the enemy of the good

Doubters argue as well that SAFs can be designed to produce fewer particles around which ice can form, thus reducing contrails as well as carbon dioxide. This is true. But steering clear of ISSRs looks like a far more cost-effective approach to contrail reduction. The paper by Dr Smith and her colleagues found that adding SAFs to a system which already practised avoidance resulted in at most 0.001°C of cooling.

There is, then, no free lunch. But the idea that giving passengers a minute or two longer to savour the in-flight cuisine that they have paid for on a few of the world's many many flights might result in some good is about as appealing as climate interventions get. ■

Ice or no ice?

Forecasted impact of contrail suppression by effectiveness of intervention
Increase in global surface temperature, °C



Source: "The climate opportunities and risks of contrail avoidance", by J. Smith et al., *Nature*, 2026

Glaciology

A signal from the noise

The sound of giant tractors may replace dynamite for probing Antarctic ice

THE PISTENBULLY 300 Polar Antarctic vehicle is a 8.5-tonne heavy-duty specialised monster built to withstand extreme polar conditions. It features wide tracks, massive pulling strength, powerful cranes and multi-purpose tools designed for the toughest ice and snow operations. No Antarctic base should be without one. Research led by Liu Guofeng at China University of Geosciences in Beijing, however, suggests the PistenBully may be capable of more than just fetching and carrying. As Dr Liu writes in the *Journal of Glaciology*, it is also able to do actual science.

Antarctica's ice sheet is the largest reservoir of freshwater on Earth, but it is shrinking. Just above the ice proper, but below the surface snow, is a transition zone 50-100 metres thick called the firn layer. This is where snow is compressed into ice. It is also where meltwater from above is captured in pores and refrozen. Knowing the firn layer's exact thickness and understanding how it behaves is crucial to predicting what will happen, over the long term, to the ice sheet and thus to the world's sea levels. Visualising this subsurface melange is difficult, though.

Current practice, borrowed from oil and gas-prospecting, is to use an "active source". This is geologist-speak for dropping a stick of dynamite down a drill hole and listening to the explosion's sound, propagated through the rock—or, in this case, ice. The listening is done by sensors called geophones. Since sound travels at different speeds through water, ice and snow, a computer analysis of when and how powerfully the echoes arrive at different geophones permits construction of a high-resolution image of the firn layer.

This works, but is not ideal. First, someone has to drill the hole down which to drop the dynamite. That is not always possible in polar conditions. Then, the explosive must be purchased, brought to Antarctica and the actual dropping carried out. Moreover, detonating dynamite in sensitive environments is not to be done lightly. An alternative source of sound would thus be good. And, after hearing the din created by a convoy of PistenBullys, Dr Liu thought they might be the very thing.

To this end, he and his colleagues experimented with geophones already in place in the Larsemann Hills of East Antarctica. They commandeered some PistenBullys and arranged for them to be driven



Mover and shaker

around the hills. They found that if the vehicles were sent along both sides of a line of geophones, parallel with the sensors, they got the effect they were after.

Since the Larsemann Hills' firn had already been mapped the old-fashioned way, Dr Liu was able to compare his machine-created images with those obtained using dynamite. They were pretty much identical. He thus seems to have devised a way of mapping the firn's structure that is simpler than blowing it up, but equally good. ■

Dinosaur gastroliths

The guts for flight

Stones in dinosaur stomachs may help understanding of the origins of birds

HOW DO YOU know what a dinosaur ate? Guts rot away after death, so the details of dinosaur digestion remain obscure. The best indicators are teeth. Sharp, pointed teeth probably ripped chunks of flesh off prey for swallowing whole. Robust grinding surfaces suggest lots of chewing was involved. Yet many dinosaurs belonging to a group called the theropods, which gave rise to birds, lost their teeth altogether and evolved beaks. This has made it hard to work out what they ate and how they digested it.

There is, however, a second clue. Lots of dinosaurs had stones, known as gastroliths, in their stomachs. These are assumed to have assisted with digestion, and thus have their own tales to tell. And modern birds and crocodilians (which are both, like dinosaurs, members of a larger evolution-

ary group called the archosaurs) often have gastroliths, too.

Takasaki Ryuji at Okayama University of Science, in Japan, decided to investigate. In a study published in *Paleobiology*, he and his colleagues examined the gastroliths of 46 modern archosaur species (42 birds and four crocodilians) and compared them with those from 16 species of dinosaur.

The birds included herbivores, omnivores and carnivores. The crocodilians were all carnivorous. The shapes of the stones varied. As might be expected, if an animal had a highly muscular stomach (ducks and geese), its gastroliths tended to have been worn into the sorts of rounded shapes displayed by beach pebbles. Weaker stomachs (seabirds and crocodiles) contained more angular stones.

Looking at the dinosaurs, Dr Takasaki and his colleagues found that ornithischians (think *Triceratops*), a group which their teeth suggest were herbivores, retained angular gastroliths. These creatures, this implies, relied mostly on their jaws to grind up their food, leaving their stomachs to apply merely the finishing touches.

Long-necked sauropods (think *Diplodocus*) also had angular gastroliths. That is more of a puzzle, for these animals had pencil-like teeth using which, it is assumed, they stripped the leaves off tree branches that less cervically elongated animals were unable to reach. The explanation, Dr Takasaki suspects, is that like those of modern ungulates, sauropods' stomachs depended mainly on fermentation to break food down.

Stones in the guts of theropods tell yet another story. *Tarbosaurus*, a hypercarnivore similar to *Tyrannosaurus*, and similarly endowed with sharp, pointed teeth, had angular gastroliths. However, theropods with beaks (of which science recognises at least four groups) had rounded ones. Indeed, Dr Takasaki saw a clear relation between tooth loss and stone roundness, as the work of grinding up food was transferred from jaws to stomach.

That shift may have helped make possible the evolutionary success of birds. Teeth are heavy, as are the jaw muscles those teeth require to do their job. This puts a lot of weight near an animal's front. Shifting the job of grinding, and the muscles and hard surfaces involved, to the stomach centralises an animal's centre of gravity in a way that would make it easier for it to rise from the ground.

Curiously, it is clear from the fossil record that the very earliest birds of all did not have beaks. But the fact that so many of their theropod relatives did, and that birds themselves evolved them at least twice, suggests the transition was easy. It may thus be that part of the secret to taking successfully to the skies was, quite literally, having the stomach for it. ■

Particle physics

Make it stick

Physicists say they have found an elusive particle: the glueball

A NEW SUBATOMIC particle has made the leap from theory to reality. So say the scientists of the Beijing Spectrometer III (BESIII), a Chinese physics experiment. Speaking on their behalf at the International Conference on High Energy Physics (ICHEP) on August 5th, Jin Shan, of Nanjing University, announced that the team have discovered a glueball—a hitherto hypothetical particle that is unlike any which physicists have seen before.

The nuclei of atoms are held together by a phenomenon called the strong nuclear force. What better name, then, for the particle responsible for carrying this force than the “gluon”? Gluons interact with quarks, another type of fundamental particle, wrangling them into pairs and triplets to create protons, neutrons and other such “composite” particles. But gluons also interact with other gluons, raising the possibility of gluons gluing themselves together into a different kind of composite particle: the glueball.

Physicists’ theories predict glueballs. Finding one is another matter, since those theories also suggest glueballs will be difficult to distinguish from the torrents of more mundane quark-based particles produced in machines like the Beijing Electron-Positron Collider II, to which BESIII is attached. Almost half a century of searching has left researchers empty-handed.

No more, say the operators of BESIII. In 2011 a new particle, called $X(2370)$, showed up in their detectors. It looked a lot like a glueball, but other interpretations were possible. So the team got to work on ruling them out. They have spent the intervening years analysing over 10bn processes in which $X(2370)$ could play a role. The result—published online in July and presented at ICHEP by Dr Jin—is a complete picture of the new particle’s properties. The team is now convinced it is a glueball. “No other interpretation can explain all these properties simultaneously,” says Dr Jin.

Detecting the glueball—if it has, indeed, been found—is not just a matter of physicists adding a new specimen to their by now extensive collection. Its discovery will provide valuable data to test the theory behind the strong force. That theory, good as it is, contains many mysteries. One is how the strong force generates mass—for gluons have no mass, but glueballs do. On the answering of such questions, the glueball’s discovery could tip the scales. ■

Well Informed

Should you stand at your desk?

Sitting too much is unhealthy, but standing instead is no fix

PROLONGED SITTING rightly has a bad rap. A review in 2012 concluded that the most sedentary were more than twice as likely as the least to have or develop diabetes, and also faced a much higher risk of premature death. Another, in 2015, found that sitting for long stretches was associated with “deleterious health outcomes regardless of physical activity”. Three years after that an analysis of news articles in the *British Journal of Sports Medicine* found hundreds of pieces which referred to claims likening sitting to smoking.

That last claim is overblown. In 2018 the *American Journal of Public Health* reckoned sitting’s true risk to be a tenth of that of smoking. But this is still bad. Hence the appeal of standing desks and, more recently, “active seats”: chairs and stools that tilt, wobble or rock to force continual postural adjustments. Is either actually healthier?

An analysis in 2018 found standing at a desk burned just nine more calories per hour than sitting. Nor are the cardiovascular gains impressive. A review in 2020 pooled nine trials involving 877 people. Participants stood about 1.3 hours more a day and were followed for roughly four months. Body fat and blood-sugar control improved slightly, but blood pressure, insulin, cholesterol and triglycerides did not. The study did

not, however, look at blood flow and the activation of postural muscles, both factors which might contribute to the benefits of standing over sitting.

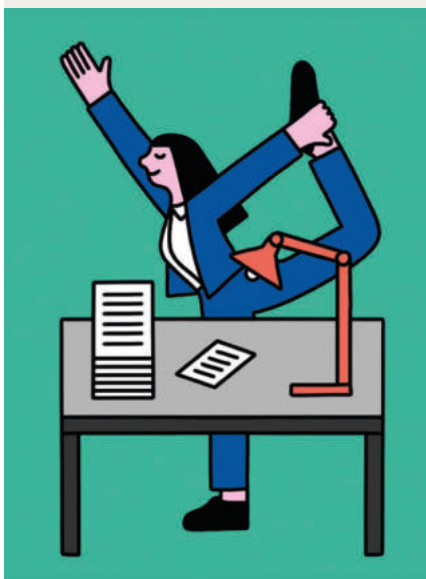
That said, standing can have downsides. Work published in the *International Journal of Epidemiology* in 2024 followed 83,013 adults in Britain for an average of nearly seven years. Those who stood for more than two hours a day had a higher risk of varicose veins and venous ulcers, perhaps because of pooling of blood in the legs.

What matters far more is movement. A study of nearly 482,000 Taiwanese adults in 2024 found that just 15–30 minutes of extra daily exercise could erase most of the excess mortality risk associated with sitting at work. Desk-bound workers may therefore wonder if active seating can make a difference. Some findings are encouraging.

Consider a trial at Georgia Southern University. In this 20 adults performed reading and typing tasks on an ordinary chair, an exercise ball and a seat mounted on top of an inflatable bladder. Energy expenditure on the ball was similar to normal sitting. But on the bladder, participants’ heart rates and calorie burn rose by 6–13% and 19–40% respectively. Writing in *Ergonomics* in 2019, the researchers attributed this to muscle contractions required to maintain balance.

A longer trial found another benefit. For six months 133 office workers in Bangkok sat on either a foam seat pad or a slightly unstable inflatable one designed to encourage frequent postural changes. Those using the latter were 81% and 84% less likely to develop neck and lower-back pain, respectively. Writing in the *Scandinavian Journal of Work, Environment & Health* in 2024, the researchers described the pad as a “light-exercise device”. They reckoned it reduced fatiguing muscle contractions and improved blood flow to compressed tissues.

There are caveats. Many studies of dynamic sitting have been small or brief. Results have varied with seat design, and some participants have reported discomfort in the *derrière*. That said, there does seem to be a lesson here: the real distinction is not between sitting and standing, but between remaining still and moving.



Culture



Man's best frenemy

What dogs can teach us about AI

Two books ponder whether the technology will be tame or savage

The Rise and Fall of the Artificial State.

By Jill Lepore. *Liveright*; 336 pages; \$29.99.
Allen Lane; £25

What Happened to Liberal Democracy?

By Daron Acemoglu. *Dutton*; 416 pages; \$32.
Profile Books; £25

WHEN MULLING how AI will transform society, consider an earlier technology that gave humans a helpful companion with superior capabilities: the dog. By domesticating wolves, humans were able, indirectly, to smell game far away and hear enemies approaching at night. They became better hunters than Neanderthals and better warriors: hard to ambush, hard to elude. Dogs are one reason why *Homo sapiens* outcompeted their prehistoric cousins and conquered the world, some anthropologists believe.

Back when dog technology was first being developed, it was far from obvious that

it was safe for people—who were literally inviting wolves into their caves. They had to train them not to eat babies. Thousands of years later, dogs still sometimes escape and go rogue. But overall, people are happy with how things have worked out. (It did not end so well for the Neanderthals.)

The optimistic case for AI is that it will be like dogs: complementing rather than replacing human labour, and remaining obedient to its masters. But what if it isn't? AI evolves much faster. Soon it could be

creating new, improved versions of itself at a pace dog-breeders can only dream of. So whereas people can be confident that dogs will not enslave them—by 2030, pooches will still not know how to use a tin-opener—they cannot be so sure about AI.

Two new books, one by a Nobel-prize-winning economist and one by a Pulitzer-garlanded historian, grapple with the question of how AI will shape the future. Neither uses the dog analogy, but both are easier to understand with reference to it.

Jill Lepore, a Harvard historian, focuses on the men who train the dogs. In her telling, AI tycoons are horrible, irresponsible people whose creations may break their leashes and savage humans. In “The Rise and Fall of the Artificial State”, she argues that tech bosses are trying to build “a future in which humans are ruled, and many replaced, by machines owned by corporations”. This, she says, implies “both the desertion of liberal democracy and the destruction of the natural world”.

She bases her case partly on the moguls' own words. Sam Altman, the boss of OpenAI, has said it would be “awesome” to replace human presidents with an AI that could “go around and talk to every person on Earth, understand their exact preferences” and “optimise” for all of them. Elon Musk has said humans may end up like “pet labradors” for superintelligent robots. ►►

→ ALSO IN THIS SECTION

68 Punctuation's (huge) impact!

69 A Renoir portrait's riveting story

69 Cheers to Swedish winemakers

70 Japan's listening bars go global

71 Forgotten Founding Fathers

▶ Mr Altman has suggested that most of Earth's surface may soon be covered with data centres. Mr Musk wants to solve this problem by putting data centres in space—and blasting millions of people to live on Mars. The tech titans all seem to have escape plans, notes Ms Lepore: “Elon Musk to Mars, Mark Zuckerberg to the meta-verse...Jeff Bezos to space, Sam Altman's brain uploaded to the cloud.” This makes them reckless about the risk of creating dystopia here on Earth, she fears.

She is right that today's tech bosses are not people you would trust to write the rules for the future of civilisation. But she overestimates the importance of individual billionaires. How AI shapes society will not depend on Mr Musk's eccentric political views, any more than the effects of cars depended on Henry Ford's. Rather, it will hinge on the power of the technology itself and the uses to which people put it.

That is a question Daron Acemoglu addresses in “What Happened to Liberal Democracy?”. Mr Acemoglu is among the world's most-cited economists. He is best known for “Why Nations Fail”, a sweeping economic history arguing that the main factor keeping poor countries poor is predatory institutions. His new work focuses on the present day and the interplay between technology and democracy.

Liberal democracy has underpinned progress, he argues. It has fostered innovation and economic growth, spread the benefits (through taxes and public services) and protected voters with regulations.

For a long time, the system more or less met voters' expectations. But “then computers happened.” Digital technologies, in Mr Acemoglu's account, aggravated inequality by raising the demand for brains over brawn. They also boosted the “social power of college graduates”, allowing a new liberal elite to impose its values on a resentful working class, provoking a backlash that led to the election of Donald Trump. And they changed how people interact. In America time spent engaging with friends in person has plunged by two-thirds in the past 20 years. Political debate has grown more extreme and toxic.

AI could make everyone much richer by supercharging innovation. But it could also supercharge the harms of digital technology, Mr Acemoglu (rather familiarly) fears. It could lead to widespread joblessness, creating “huge social-status gaps” between “makers” and “takers” and a full-blown crisis of democracy. (It could also accelerate social atomisation by making lovebots and political fakery more seductive.)

His proposal is “pro-worker AI”. By this, Mr Acemoglu means finding ways for workers to use AI to make their labour more valuable, like prehistoric hunters combining their spear-making and throwing skills with a dog's nose and speed.

Maybe AI could enable electricians to troubleshoot increasingly sophisticated grids, spotting problems via sensors and helping humans analyse them with a database of past glitches and solutions. Perhaps AI could improve education: an AI assistant could assess pupils' answers to quizzes and suggest ways to split the class into smaller groups, who could be offered tailored instruction, thus increasing the demand for human teachers.

It is hard to quarrel with this. AI will certainly cause less social turmoil if it complements human labour rather than making it redundant. But Mr Acemoglu leaves unresolved the question of how this is to be accomplished. Ultimately, both books are disappointing. Inadvertently, they illuminate how hard it is to write persuasively about a technology that is changing so fast, and whose future capabilities are so utterly unknown.

Politically, AI has become unpopular. Twice as many Americans are pessimistic about its long-term effects on society as are optimistic, and only 2% think it is developing “too slowly”. But one way or another, this new wolf will be invited into the cave. The countries pioneering AI technology, America and China, are rivals. Neither wants to be the second to develop a super-intelligent ally, for fear of ending up like those hapless, dogless Neanderthals. ■

Punctuation

To the point

On the Mark. By Florence Hazrat. *Basic Books*; 352 pages; \$32. *Profile Books*; £22

“IN THE BEGINNING was the word, and the word was with God, and the word was God.” Yet in the beginning there was no punctuation. In early Latin manuscripts, the Bible was written in *scriptura continua*, without spaces or marks between words. This was a problem: pause at the wrong point, and you could sever God from “the word”, throwing Christ's divinity into doubt. So in the Middle Ages, Christian scholars set about punctuating the holy book, using symbols and sub-clauses as “lanterns to words” to guide preachers.

“The history of punctuation is the history of humanity,” argues Florence Hazrat, a researcher, in a rollicking new book. Often dismissed as “the invisible ushers of the sentence”, dots and dashes have shaped religious texts, legal documents, and literature. “On the Mark”, like punctuation itself, makes readers pause for



thought, showing how full stops have displeased and commas caused controversy.

This pointed history starts in the ancient world. Cicero, like other Roman scholars, thought punctuation was amateurish. For skilled orators, half the fun was in interpreting obscure writing. Ms Hazrat races across time—offering digressions about how, for instance, Irish monks in the eighth century first put spaces between words—before reaching the “punctuation boom” of the Renaissance. Scholars in Italy invented curved commas, semicolons and (do not forget) parentheses. The printing press helped spread and standardise these signs, and by around 1600 all the major marks were in place.

“On the Mark” is crammed with fascinating facts. Take the exclamation mark, about which Ms Hazrat wrote a previous book. Invented by an Italian poet in the 1360s to signal “admiration and wonder”, by the 19th century it had become a sign of emotion and even hysteria. Editors added them to old texts, including Shakespeare's “Antony and Cleopatra”. (In the First Folio, the play had only 16 exclamation marks, but modern versions contain about 200, most of them clustered in Cleopatra's shouty-sounding speeches.) In the 20th century the exclamation mark fell into “soundless cymbal-crashing”, as Theodor Adorno, a German philosopher, put it. Nazi propagandists aggressively attached clusters of them to the end of assertions!!!

Writers have long squabbled over scribbles. Emily Dickinson felt “defeated” when an editor cut a dash; Charles Baudelaire told his proofreader “I absolutely want to keep this comma.” Mark Twain even threatened to have a proofreader shot who dared change his punctuation. This is a delightful tale of marks' evolution, even if Ms Hazrat's own attempts to use them inventively fall a dash flat. ■

Money, sex and bigotry in France

Get the picture?

The Renoir Girls. By Catherine Ostler.

Atria Books; 432 pages; \$35.

Simon and Schuster; £30

IT WAS CALLED “the capital of the 19th century” and “modern Babylon”. To those living during the Belle Époque, which stretched from the 1870s to the onset of the first world war, Paris felt like the centre of the world. And at the very centre of Parisian life were the Cahen d’Anvers, a Jewish banking family. They held salons, threw lavish parties and counted celebrated writers, including Marcel Proust and Guy de Maupassant, as friends.

The childhoods of the three Cahen d’Anvers daughters were immortalised by Pierre-Auguste Renoir in commissioned portraits. One, of Alice and Elisabeth holding hands (see picture), hangs in the Museu de Arte de São Paulo. It looks as though everything about the girls’ lives, save their ruffled dresses, was smooth. In fact, the painting “reminds us that nothing and no one is quite safe from the hands of time or the evil within humanity, apart, sometimes, from art itself”, writes Catherine Ostler, a historian, in a new book.

When you gaze at Renoir’s pretty paintings, several things are obscured from view. One is paternity. Rumours abounded that Louise, the matriarch of the Cahen d’Anvers family, was the paramour of Charles Ephrussi, an art-loving critic and collector. It was a time of “rebellious, emancipated, intellectual feminism”, when wealthy women were free to hold salons and live more independent lives. There were whispers that Alice might be Ephrussi’s daughter. Ms Ostler uses a DNA test of a descendant to show that she probably was (the same may have been true of her brother, who was named Charles). This is not, then, your average portrait.

The painting also masks the antisemitism that would change the girls’ lives. When it was painted in 1881, France was ostensibly in a tolerant period (though after receiving a late payment, Renoir wrote to a friend complaining that the payment was “stingy” and declaring that “I really give up on the Jews”). Then the Dreyfus affair—in which a Jewish army officer was falsely accused and convicted of passing secrets to Germany—laid bare people’s prejudices and jealousy. Between 1807 and 1914 some 880 Jews in Paris converted to Catholicism; about a quarter of them did so in just three years, 1895-97,

when the Dreyfus affair was most heated.

One of them was Elisabeth (in blue), who chose to be baptised at the age of 20. Spending most of her life as a Christian, however, did not protect her. The collaborationist Vichy regime began to rule France, and a mayor whose family had mingled with the Cahen d’Anvers for generations certified that Elisabeth, then 69 years old, was Jewish.

Suffering from severe arthritis and unable to walk without help, she was arrested by the Nazis, who sent her to Auschwitz, where she was probably murdered on arrival, if she had survived the 55-hour train journey. Ms Ostler focuses on the many betrayals that facilitated this fate. The Cahen d’Anvers had fought in the first world war and been generous patrons of France, including donating their chateau to the state. And yet the country turned on them.

Ms Ostler came upon the Cahen d’Anvers while reading “The Hare with Amber Eyes”, which masterfully traces art through time (the author, Edmund de Waal, inherited sculptures bought by Ephrussi). She has doggedly dug up clips and correspondence that evoke the Belle Époque. Sometimes, though, “The Renoir Girls” gets too bogged down in social minutiae that must have preoccupied the Cahen d’Anvers. Few really want to know which Rothschilds were at weddings, what colour the bridesmaids wore and whether the press preferred Louise’s previous hairstyle. But the broader themes—about the hidden history of objects, and what art captures and obscures—are heart-rending and memorable. You will not look at another 19th-century portrait the same way again. ■



All dressed up with everywhere to go

Wine

Vin-dinavia

HOGANAS

World-class wine is being made in a corner of Sweden. Yes, Sweden

AT THE NOBEL PRIZE banquet in Stockholm in December, not all the winners were seated at the table. A victor of a different kind could be found in the diners’ wine glasses. In 2024 a blind tasting had pitted a dozen of Europe’s finest wines against a dozen wines from Sweden. An international panel of 18 independent experts declared Immelen, a white wine from Kullabergs, a winery in the Skane region of southern Sweden, the overall winner, above famous whites from France, Italy and Spain. That made Immelen the obvious choice for the Nobel banquet. It was a coming-out party for Swedish wine.

Sweden is “a young wine country”, says Victor Dahl, chief executive of Kullabergs (pictured on next page). EU approval for winemaking in Sweden and Denmark, required for commercial production, was granted only in 1999. There are now about 50 commercial wineries in Sweden. Sales are still tiny—just a drop in the barrel by global standards—but growing quickly.

At the Nobel banquet, Mr Dahl moonlighted as a waiter so he could hear what the diners thought. Such was their level of interest that he was soon unmasked and invited to introduce the wine. “They had so many questions,” he recalls.

And no wonder. That wine can be made in Sweden at all may come as a surprise. Global warming has helped. But “climate change is also bad for us,” says Mr Dahl. Although the climate is warmer overall, it is also unpredictable: since 2018 every growing season has been different. Sweden’s long summer days provide plenty of sunshine to ripen grapes. Indeed, slower ripening may enhance flavour.

Immelen is a blend. It is two-thirds Solaris, a descendant of Riesling and Pinot Gris created in 1975, and now Sweden’s most widely planted varietal. The rest is Souvignier Gris and a dash of Muscaris (a cross between Solaris and Muscat). The resulting wine is surprisingly full-bodied, with none of the under-ripeness often found in cool-climate wines. It reminded this correspondent of a high-end Sancerre.

Just as significant as climate change has been the rise of New Nordic cuisine, a gastronomic culture centred on Copenhagen and Malmö. Skane is sometimes referred to as the “Nordic Tuscany” and is being promoted as a tourist destination for foodies. In this part of the world seasonal tasting menus abound, and foragers gather ►►



Grape expectations

► local ingredients for restaurants. Magnus Nilsson—whose previous restaurant, Fäviken, earned two Michelin stars—has recently opened a new one in Skane, as has Titti Qvarnstrom, the first Swedish woman to win a star. Being able to serve local wine with local food has obvious appeal.

History has also played a part. Sweden's government ran a campaign from 1957-85 to encourage Swedes to drink wine, rather than spirits, for health reasons. This created a strong wine culture just as European travel was getting cheaper, prompting Swedish wine buffs to engage in wine tourism. But now they can do it at home. The covid-19 pandemic helped, too: Swedish wine critics who were unable to go abroad started to pay more attention to the viticulture scene on their doorstep.

Oddly the *Systembolaget*, the government monopoly on the sale of alcohol through a national network of shops, has also helped. It bans Swedish wineries from selling bottles over the counter, though they can sell wine for consumption on the premises, for example at an on-site bar. To get into the *Systembolaget* shops, Swedish winemakers must achieve consistent quality and volume. *Systembolaget* acts as “an extraordinary quality-control mechanism”, says Joe Fattorini, a British wine guru who now lives in Sweden. Once a winery wins approval, as Kullabergs has, distribution is in effect handled by the government.

The contrast with neighbouring Denmark, which lacks such restrictions, is telling. Denmark has fewer hectares of vines than Sweden, but roughly three times as many producers and a wider range of styles. That means more experimentation, but greater variation in quality.

Things are changing in Sweden, however. After much lobbying, the law was changed to allow “cellar-door sales” by

wineries from June 2025. There are still restrictions—sales are permitted only as part of an educational experience lasting at least 30 minutes—and the rule-change is subject to a six-year trial. Sweden's wine industry is still in its infancy and lacks the long history of winemaking in other parts of Europe. But you do not need a Nobel prize to see its potential. ■

Listening bars

For the record

Japan's music cafés are striking a chord abroad

SPACE TALK, located in the London district of Clerkenwell, takes music seriously. Before the bar opened in 2024, its owner enlisted an architect, designer and sound engineer to create the perfect acoustic environment. Resembling the deck of a wooden spaceship, the bar is divided into four zones. On a recent evening, your correspondent sipped cocktails while trippy electronic music floated out of the bespoke sound system. In its ethos, Space Talk is one of many spots around the world loosely inspired by Japan's *ongaku kissa* (music cafés). These venues are commonly known as listening bars or hi-fi bars.

The first *ongaku kissa* opened in Japan during the 1920s. Visitors drank tea or coffee as music—often classical, tango or swing—played through gramophones. In the decades after the second world war, cafés devoted to jazz grew popular. Imported records and audio equipment were expensive, and jazz *kissa* offered a place for

enthusiasts to hear the latest album by Miles Davis or John Coltrane. They were places for quiet, reverential listening.

Owners would (and still do) display a fussy attention to detail. High-quality speakers are a must; the nerdiness often extends to amps, turntables and styluses. (Gramophones have gone, but vinyl is still very much in vogue.) This music is not merely for ambience: the act of listening is central to the experience. Many *kissa* ask visitors to remain silent. Kawasaki Toshiya, who runs his own listening bar in Tokyo, says the atmosphere this creates is akin to “sitting in meditation at a temple”.

The listening bars that have sprung up in the West, South America and Africa—most of which have opened in the past decade—rarely have stringent rules on chit-chat. Space Talk, for instance, has not banned nattering, but it has implemented a “no photos or videos” policy. All Blues, a venue in New York, politely asks visitors to “let the music do the talking”. Some bars host dedicated events where guests can hear an album in full or listen to music tied to a particular artist or theme.

The “real magic” of *ongaku kissa* culture is to create an environment “where you can be genuinely moved to the core by music”, says Nick Dwyer, the director of a documentary on Tokyo's music cafés. What makes this new wave of venues interesting, he says, is that proprietors have “their own take” on the format and are adapting it into “something that works in London, New York, Barcelona or wherever”.

There are three reasons why the *ongaku kissa* model has succeeded as a cultural export. One is Japanomania. A record 43m foreign holidaymakers visited the country in 2025. Over several decades, Japan has established itself as a trendsetter and tastemaker in everything from video games to fashion and design, meaning that where the Japanese lead, others will follow.

The second reason is that listening bars and *ongaku kissa* offer an appealing alternative to nightclubs, which are in decline. These bars provide a relaxed, stylish environment for retired ravers and youngsters alike. Here, enjoying music need not mean enduring punishing opening hours (and punishing hangovers).

The third reason is that these bars offer a distinctive, curated experience to suit a wide variety of tastes. Jim Hanmer, the co-founder of Jazu, another listening bar in London, made a pilgrimage to Japan to visit several jazz *kissa*. (“Jazu” is a nod to the Japanese pronunciation of “jazz”.) He was drawn to the venues that reflected the owner's lifelong passion. In these places, “you could listen to a record that you've heard 30 times before, but because you're in this unique environment, it sounds totally different.” At a listening bar, even an old track can find a new groove. ■

American history

Finding Founding Fathers

Two new biographies resurrect seminal but forgotten revolutionary figures

The Lost Founder. By Jesse Wegman.*Celadon Books; 384 pages; \$30***Radical Duke.** By Danielle Allen.*Liveright; 464 pages; \$37.50. W.W. Norton; £28*

A STROLL THROUGH the gallery of America's Founding Fathers yields familiar faces. Here is brave, disciplined George Washington alongside sociable, inventive Benjamin Franklin. There is John Adams—principled and prickly—squabbling with brilliant, curious Thomas Jefferson. Two new books seek to draw your gaze to two forgotten figures, each with strikingly expansive views of democracy.

The founders, as a whole, were inconsistent. They allowed slavery to endure. They wanted out from under Britain's thumb, but not universal suffrage. Roger Sherman, the only person to sign all four of the founding documents, felt "the people...should have as little to do as may be about the government." Benjamin Rush, another founder, called democracy "the devil's own government".

Charles Lennox (pictured left) and James Wilson (pictured right) disagreed. They shared a faith in democracy and popular wisdom uncommon at the time. Their lives were markedly different: Lennox was an aristocrat and parliamentarian, while Wilson was a poor Scot who emigrated to America and, on the strength of his intel-

lect, became consequential.

The "tall, rich and beautiful" Lennox was an unlikely radical. Yet he was the first parliamentarian to propose letting all adult men vote in Britain and the first member of the House of Lords to suggest recognising American independence. He was also a patron of Thomas Paine, a tax collector in Sussex who became a vociferous proponent of independence after emigrating to America. Paine's pamphlet, "Common Sense", argued that Americans "have it in our power to begin the world over again". Widely read, it galvanised support for revolution.

In "Radical Duke", Danielle Allen, a political philosopher at Harvard, makes the novel argument that Lennox and Paine were at the centre of "a network of writers" that formed Junius, a pseudonymous author. Junius published essays critical of King George III's government that roiled British politics in the early 1770s; they argued that royal "authority rested upon the consent of the governed". Both men "cut their teeth as radicals by playing Junius" several years before the publication of "Common Sense".

Lennox's radicalism exemplifies Ms Allen's shrewd broader argument that the roots of the American Revolution lay not in Boston but in Britain. Protests against unpopular and unjust taxation, popular demonstrations against royal authority, the

publication of essays promoting revolution: all of these happened in Britain in the 1760s-70s. "Every act that provided the drumbeat of the march to revolution in the colonies had been foreshadowed in Britain," she writes. That argument may provoke some grumbles: Americans consider their revolution home-grown. But few geopolitical events happen in isolation.

Ms Allen began researching Lennox's life after she discovered a copy of the Declaration of Independence in a record office in Sussex. It had belonged to Lennox and was probably commissioned by James Wilson—the subject of Jesse Wegman's biography. Wilson had emigrated to America in 1765, when he was 23, and was the only signer of both the constitution and the Declaration of Independence to serve as a Supreme Court justice.

Mr Wegman, a journalist, situates Wilson within the Scottish Enlightenment, which linked commerce with goodness—a comfortable pairing for mercantile America. ("The road to virtue and that to fortune," wrote Adam Smith, "are, happily in most cases, one and the same.") Scottish nobles had long defended self-government; in 1320 barons declared their king's authority stemmed from "the due consent and assent of us all".

In the run-up to the revolution, Wilson's rigorous essays provided a legal argument for independence. Parliament had no authority over the American colonies, he stated, because colonists did not elect them. "All men are, by nature, equal and free," so "no one has a right to any authority over another without his consent."

Mr Wegman suggests that Jefferson paraphrased this sentiment in the Declaration of Independence. He notes that Wilson was the only founder to use the phrase "created equal" in later writing and that he coined the constitution's opening phrase ("We the people"). In his battles for direct elections for Congress and the presidency, Wilson was "by a long shot the most democratic of all the founders".

Why, then, is he not better known? First, because he was a creature of the law, not a politician. And second, alas, his roads to fortune and virtue diverged. Addicted to land speculation, he ended up in a debtors' prison in New Jersey while a justice. Then, to escape his creditors, he fled to North Carolina, where he died of malaria in 1798.

Were Wilson and Lennox failures because their ideals did not carry the day? That is a crabbed view of reform. Battles for social progress are rarely won in fighters' lifetimes. Martin Luther King did not live to see a racially equal society; he was not wrong to fight for one. In their efforts to expand the franchise and their faith in democracy, Lennox and Wilson were both ahead of their time—and their stories are a welcome rediscovery today. ■



Remember them?

Out of office. In the know.

Send instantly, or on a date you choose. ■



Upgrade their holiday reading. A gift subscription to *The Economist* delivers global, fact-checked reporting, wherever they travel.



Scan the QR code to give *The Economist*
or visit subscribernow.economist.com/gift-subs

**The
Economist**

Economic & financial indicators

	Gross domestic product				Consumer prices			Unemployment rate	Current-account balance	Budget balance	Interest rates		Currency units	
	% change on year ago				% change on year ago			%	% of GDP, 2026†	% of GDP, 2026†	10-yr gov't bonds	change on year ago, bp	per \$	% change
	latest	quarter*	2026†		latest	2026†					latest, %		Aug 20th	on year ago
United States	2.1	Q2	1.5	2.1	3.4	Jul	3.2	4.1	Jul	-3.5	-6.5	4.7	35.0	-
China	4.3	Q2	3.6	4.7	0.5	Jul	1.0	5.2	Jul*§	3.2	-5.4	1.5	\$§	-16.0
Japan	0.7	Q2	1.1	0.8	1.7	Jun	1.7	2.5	Jun	4.0	-1.8	2.9	129	158
Britain	1.2	Q2	1.7	1.1	2.9	Jul	3.2	4.9	May**	-3.6	-5.1	5.0	37.0	0.73
Canada	-0.1	Q1	-0.1	0.9	3.0	Jul	2.5	6.4	Jul	-0.3	-2.0	3.7	23.0	1.38
Euro area	1.0	Q2	1.8	0.8	2.9	Jul	2.8	6.3	Jun	2.3	-3.4	3.3	51.0	0.86
Austria	0.9	Q1	0.8*	0.8	2.7	Jul	3.0	5.7	Jun	1.1	-4.1	3.5	41.0	0.86
Belgium	0.5	Q2	nil	0.7	3.7	Jul	3.6	6.3	Jun	-2.8	-5.4	3.9	55.0	0.86
France	0.7	Q2	0.7	0.7	2.4	Jul	2.3	8.2	Jun	-0.4	-5.3	4.1	65.0	0.86
Germany	0.9	Q2	0.9	0.9	2.8	Jul	2.7	3.9	Jun	4.5	-4.2	3.3	51.0	0.86
Greece	2.0	Q1	1.0	2.0	2.7	Jul	3.7	8.0	Jun	-5.5	0.8	3.9	47.0	0.86
Italy	1.0	Q2	0.8	0.8	2.9	Jul	2.6	5.7	Jun	0.8	-3.0	4.1	48.0	0.86
Netherlands	1.3	Q2	1.6	1.0	3.0	Jul	3.1	4.0	Jul	9.3	-2.6	3.3	41.0	0.86
Spain	2.7	Q2	2.8	2.6	3.9	Jul	3.3	10.1	Jun	2.4	-2.5	3.6	37.0	0.86
Czech Republic	2.2	Q1	0.8	2.0	1.7	Jul	2.0	3.0	Q2*	0.3	-2.6	4.9	60.0	20.7
Denmark	4.7	Q2	1.1	3.3	1.7	Jul	2.0	3.1	Jun	12.1	1.3	3.0	43.0	6.40
Norway	1.7	Q1	1.4	1.3	3.0	Jul	3.2	4.5	May**	14.8	9.0	4.4	51.0	9.33
Poland	3.8	Q2	3.6	3.4	3.0	Jul	2.8	5.8	Jun§	-1.4	-6.9	5.9	49.0	3.70
Russia	1.3	Q2	na	0.5	6.0	Jul	6.4	2.2	Jun§	2.2	-3.2	16.0	269	85.1
Sweden	2.5	Q2	5.7	2.1	0.2	Jul	1.7	9.9	Jun§	5.5	-2.3	3.1	67.0	9.44
Switzerland	0.5	Q1	2.6	1.0	0.4	Jul	0.5	3.1	Jul	6.9	0.2	0.5	13.0	0.80
Turkey	2.5	Q1	0.5	2.4	31.8	Jul	32.1	7.4	Jun§	-3.0	-3.6	32.4	325	47.9
Australia	2.5	Q1	1.1	1.6	3.8	Jun	3.7	4.5	Jul	-3.7	-1.5	5.0	77.0	1.40
Hong Kong	4.3	Q2	-2.5	5.0	1.9	Jun	2.3	3.7	Jun**	4.6	-2.5	3.6	47.0	7.84
India	7.8	Q1	6.4	6.5	4.4	Jul	4.8	6.2	Jul	-1.6	-4.5	6.8	31.0	95.5
Indonesia	5.3	Q2	4.4	5.0	2.9	Jul	3.5	4.7	Feb§	-1.6	-3.5	7.0	66.0	17,800
Malaysia	6.0	Q2	9.3	5.5	1.8	Jul	1.9	3.0	Jun§	3.1	-3.5	3.8	38.0	4.06
Pakistan	4.8	2026**	na	3.0	9.2	Jul	9.0	6.9	2025	-1.1	-4.7	12.1	11.0	278
Philippines	2.3	Q2	2.4	2.7	6.2	Jul	6.8	4.7	Q2§	-4.0	-6.3	7.3	131	61.8
Singapore	5.9	Q2	5.7	5.4	1.9	Jun	2.0	2.0	Q2	15.6	1.3	2.4	41.0	1.27
South Korea	3.7	Q2	2.5	3.3	2.8	Jul	2.8	2.6	Jul§	13.6	-1.9	4.4	157	1,389
Taiwan	12.9	Q2	5.7	10.8	2.5	Jul	2.0	3.3	Jun	26.6	1.8	1.9	50.0	31.9
Thailand	1.9	Q2	-0.7	2.3	1.9	Jul	2.3	0.9	Jun§	nil	-5.5	2.1	79.0	32.9
Argentina	2.3	Q1	3.0	3.0	33.8	Jul	32.4	7.8	Q1§	-0.4	0.1	na	na	1,497
Brazil	1.8	Q1	4.5	2.0	4.4	Jul	4.8	5.4	Jun§**	-2.6	-7.3	14.8	74.0	5.18
Chile	-0.2	Q2	-0.1	1.0	3.5	Jul	3.6	9.4	Jun§**	-1.2	-2.5	5.7	9.0	922
Colombia	3.4	Q2	5.2	2.5	6.0	Jul	5.7	8.0	Jun§	-2.4	-6.6	12.0	23.0	3,053
Mexico	2.2	Q2	6.1	1.0	3.1	Jul	3.8	2.8	Jun	-0.4	-4.0	9.2	1.0	16.9
Peru	3.5	Q1	3.2	2.8	4.1	Jul	4.0	4.3	Jul§	2.2	-2.4	6.2	18.0	3.36
Egypt	5.0	Q1	-24.0	4.6	14.9	Jul	13.8	5.8	Q2§	-4.7	-6.0	25.6	50.0	50.7
Israel	6.9	Q2	15.4	4.3	1.5	Jul	1.9	3.1	Jul	1.0	-3.5	3.8	-34.0	2.98
Saudi Arabia	4.6	2025	na	0.2	1.8	Jul	2.1	3.1	Q1	-3.6	-6.9	na	na	3.76
South Africa	1.9	Q1	2.2	1.5	4.1	Jul	4.5	33.6	Q2§	-0.7	-4.4	8.6	-101	16.1

Source: Haver Analytics *% change on previous quarter, annual rate †The Economist Intelligence Unit estimate/forecast §Not seasonally adjusted ‡New series **Year ending June ††Latest 3 months †††3-month moving average
§§5-year yield ††††Dollar-denominated bonds Note: Euro-area consumer prices are harmonised

Markets

	Index	% change on:	
In local currency	Aug 19th	one week	Dec 31st 2025
United States S&P 500	7,708.0	-0.5	12.6
United States NAS Comp	26,331.1	-1.0	13.3
China Shanghai Comp	3,894.4	-1.3	-1.9
China Shenzhen Comp	2,507.3	-3.8	-0.9
Japan Nikkei 225	65,326.4	-3.3	29.8
Japan Topix	4,012.3	-3.1	17.7
Britain FTSE 100	10,743.4	-0.8	8.2
Canada S&P TSX	36,401.8	-0.7	14.8
Euro area EURO STOXX 50	6,444.5	-1.4	11.3
France CAC 40	8,501.9	-2.0	4.3
Germany DAX*	26,091.3	-0.9	6.5
Italy FTSE/MIB	52,618.2	-2.0	17.1
Netherlands AEX	1,102.4	-0.9	15.9
Spain IBEX 35	19,847.5	-1.8	14.7
Poland WIG	152,484.6	-1.0	30.1
Russia RTS, \$ terms	803.9	-8.0	-27.5
Switzerland SMI	14,386.6	-0.4	8.4
Turkey BIST	14,459.0	2.5	28.4
Australia All Ord.	9,255.2	-1.6	2.6
Hong Kong Hang Seng	25,495.1	0.2	-0.5
India BSE	76,909.7	-1.4	-9.8
Indonesia IDX	6,394.1	0.3	-26.1
Malaysia KLSE	1,731.3	-0.6	3.0

	Index	% change on:	
	Aug 19th	one week	Dec 31st 2025
Pakistan KSE	176,846.4	-1.9	1.6
Singapore STI	5,694.2	-0.5	22.6
South Korea KOSPI	6,471.2	-1.6	53.6
Taiwan TWI	44,719.4	-1.8	54.4
Thailand SET	1,609.8	-0.2	27.8
Argentina MERV	2,874,493.0	-4.2	-5.8
Brazil BVSP*	167,830.3	0.2	4.2
Mexico IPC	63,999.3	-2.7	-0.5
Egypt EGX 30	54,512.7	-1.0	30.3
Israel TA-125	4,040.0	-0.6	10.3
Saudi Arabia Tadawul	10,925.7	0.8	4.1
South Africa JSE AS	116,051.1	0.9	0.2
World, dev'd MSCI	4,975.5	-0.7	12.3
Emerging markets MSCI	1,668.4	-0.8	18.8

US corporate bonds, spread over Treasuries			
		Dec 31st 2025	
Basis points		latest	2025
Investment grade		95	93
High-yield		323	354

Sources: LSEG Workspace; Moscow Exchange; Standard & Poor's Global Fixed Income Research *Total return index

Commodities

The Economist commodity-price index				% change on	
2020=100	Aug 11th	Aug 18th*	month	year	
Dollar Index					
All items	152.1	153.0	nil	14.0	
Food	149.1	152.9	-1.6	3.6	
Industrials					
All	154.5	153.1	1.5	24.3	
Non-food agriculturals	151.3	151.4	-0.4	21.4	
Metals	155.4	153.5	1.9	25.1	
Sterling Index					
All items	144.7	145.2	-1.2	13.7	
Euro Index					
All items	150.5	151.0	-1.5	14.9	
Gold					
\$ per oz	4,392.1	4,360.6	7.0	31.2	
Brent					
\$ per barrel	89.0	91.0	-0.1	38.3	

Sources: CME Group; LME; LSEG Workspace; NOREXCO; NZ Wool Services; S&P Global Commodity Insights; Thompson Lloyd & Ewart; USDA *Provisional

For historical indicators data, visit economist.com/economic-and-financial-indicators

OBITUARY

Mary de Rachewiltz

The poet, translator and tireless defender of her father, Ezra Pound, died on July 24th, aged 101



SHE WAS still awake when the tall bearded man came into her bedroom to say goodnight. She had hoped he would. Though she was nine, it was a rare treat to get a visit from her father. By the light of the kerosene lamp, he looked at her hands. Her fingernails were filthy. Then he started to rummage in his luggage, for sweets, she hoped. Instead he brought out scissors and, carefully, cleaned and cut her nails. He asked her then, where was her toothbrush? It was somewhere. The next day, he took her to buy one.

This happened in Gais, in South Tyrol, in the house of two kind farm-folk, Mamme and Tatte, who cared for abandoned children. But she was not abandoned, quite. Her parents, the poet Ezra Pound and his mistress Olga Rudge, a violinist (Tattile and Mamile to her then), lived in Venice and paid 200 lire a month for her keep. She did not mind the arrangement at all. From birth she had grown into a Tyrolean peasant child, a shepherdess with blonde plaits who had her own flock and, in secret partnership with her father, ran a miniature bee business for a while. But then at fairly regular intervals she would go to her parents in Venice, to a villa full of books in many languages, to the Lido via *vaporetto* and to feasts of ice cream. Her cool, willowy mother made her speak no German there, and wear white gloves. It was a small price to pay.

From the start, her two lives ran in counterpoint. She became skilled at defending one against the other. In a poem she wrote later she relished the Pustertal dialect she spoke, words as weighty as sausages and dumplings. Against city people she defended dung-heaps, insisting that a good one smelled mostly of leaves and moss. Tyroleans hated Italians; but her father, arranger of all her Italian treats, was an American Italophile and she adored him. “Babbo” taught her both Italian and English by making her translate “The Cantos”, the huge and obscure poem he spent his life working on, into Italian. For her, that was like being presented with a basket of exotic food she had no idea what to do with, but she tried. Each typed page she produced for him would come back entirely scribbled over, not only with corrections but with newly perplexing thoughts.

From the mid-1920s Babbo had also formed some controversial political ideas. He began to worship Italy’s fascist dictator, Benito

Mussolini, extolling him in articles and broadcasts and even presenting him with a copy of the early *Cantos*. (“Ma questo” said the Boss, “è divertente.”) For Pound, Mussolini embodied the strength and order Italy needed. But then he went further. The true threat to mankind was “international Jewry”. Racial purity should be preserved to strengthen the nation. On Italian radio he begged Franklin Roosevelt not to side with the Allies and said he should be hanged; those broadcasts earned him a charge of treason *in absentia*. When American troops eventually liberated Italy, in 1945, they captured Pound and caged him. For 13 years he was kept in a locked cell in a psychiatric hospital.

Mary now had two battles to fight. The first was to defend her father against the charge of treason and the imputation of insanity. The latter was clearly ridiculous, as he wrote her long, lucid letters from prison and continued working on “The Cantos”. She managed only one visit; otherwise she imagined him, still tall and dignified in the dining room among the slobbering inmates. His broadcasts and articles, though, were much harder to dismiss. What could she say of his vile remarks about “Jew slime” and “the dog-damn wop”? Only that he had been influenced by the opinions of the time. T.S. Eliot had been antisemitic, too. (She went to see shy, stooping Tom to ask for help; he offered her thin Sawia biscuits, Loneliness personified.) And obviously, later on, the “master of utterance” had lost his grip on words.

That was her second fight. Babbo had been carried away by words all his life. Before the debacle in Italy he had been ranked among the foremost poets of the age; now he was untouchable. But that could not be. She worked on with “The Cantos” and with his other poetry, turning everything into Italian and insisting on the beauty in it. “Beauty is difficult,” Babbo had written in Canto LXXXIV; but, like light, it kept breaking through. In her memoir “Discretions”, in 1971, she cited his poetry whenever it could illuminate a phase of her life, or a memory. Her own poetry was a homage to the man who had trained her in it, not least on those evenings when the air had tingled, threatening explosion, as he recited his own. In 1985, 13 years after his death, she at last published the first dual-language edition of “The Cantos”.

She still had more to do. She preserved his papers and first drafts, most of which went to Yale, and lectured about him all over North America. Occasionally she unveiled long-resisted plaques to him. Most important, she had to keep up the re-evaluation. After her marriage in 1946 to a Russian-Italian minor nobleman, Boris de Rachewiltz, they bought Brunnenberg, a crag-bound castle in her beloved South Tyrol. It was a ruin when they found it, but over the years she turned it into a centre of Pound studies. Scholars from the world over came to discuss him amid the splendour of the Alps. She firmly told them, when they arrived, that they could say anything they liked about Pound. And then she would tell them what he really thought, and why.

Babbo himself had stayed there after his release from the asylum, hoping for peace in which to write “Paradise”, intended as the final Canto (with a nod to Dante). But he found the place too cold, and retreated again to Venice with Olga. Mary stayed, because South Tyrol was home. After her idyllic childhood in Gais, she had returned again and again to see Mamme and Tatte. Mamme, with her emotional, fearful Catholic heart and overflowing kitchen, offered true mother-love. But there could be only one father. “I have tried to write Paradise”, he declared, in his “Notes for Canto CXX”: “...let those I love try to forgive/ what I have made.” ■

Your Hassle-Free Mayfair Pied-à-Terre

47 Park StreetSM offers a selection of luxuriously furnished, spacious apartments in Mayfair for the amount of time you personally require in London each year — at a fraction of the cost of whole ownership.

Residential Benefits Include:

- Fully integrated property management
- 24-hour concierge, maid, and room service
- Valet storage facility with wide usage options
- No stamp duty
- Individual and corporate fractional interests from £82,800



47 PARK STREETSM

Mayfair - London

GRAND RESIDENCES BY MARRIOTT.

Open daily, 9 a.m. – 5 p.m.

Tel: +44 (0) 20 7950 5528

Email: enquiries@47parkstreet.com

www.47parkstreet.com

Furnishings may vary. Residence and property amenities are subject to change. Data protection: Your personal information and details will be stored by Marriott Vacation Club International, part of a global group of affiliated companies ("Affiliates"), and your personal information may be shared among the Affiliates and transferred outside of the European Economic Area is done under data transfer agreements that contain standard data protection clauses adopted by the European Commission that provide safeguards for such transfers. For more information about the use of your personal information, or to exercise your rights of opposition, access, rectification and deletion please review our Global Privacy Statement found at marriottvacationclub.com/privacy/ or contact our Privacy Office at privacy@mvc.com. IQL-20-001. THIS ADVERTISING MATERIAL IS BEING USED FOR THE PURPOSE OF SOLICITING THE SALE OF FRACTIONAL RESIDENCE CLUB MEMBERSHIP. Subject to applicable terms and conditions. This is neither an offer to sell nor a solicitation to buy to residents in jurisdictions in which registration requirements have not been fulfilled or where marketing or sale of fractional residence club membership is prohibited and your eligibility and the membership clubs available for purchase will depend upon the jurisdiction of your residency. Prices are subject to change. Key information is available upon request by contacting 47 Park Street Grand Residences by Marriott, Mayfair, London, W1K 7EB, United Kingdom or email moreinfo@47parkstreet.com. Marriott Vacation Club International and the programs and products provided under the Grand Residences by Marriott brand are not owned, developed, or sold by Marriott International, Inc. Marriott Vacation Club International uses the Marriott marks under license from Marriott International, Inc. and its affiliates. ©2026 Marriott Vacation Club International. All Rights Reserved.



STRONGKEY

**NOW DELIVERING
RANSOMWARE
DETERRENCE**

GETSECURE@STRONGKEY.COM

**The
Economist**

To advertise within the classified section, contact:

UK/Europe

Agne Zurauskaite

agnezurauskaite@economist.com

North America

Richard Dexter

richarddexter@economist.com

Asia

Chris Phang

chrisphang@economist.com

Readers are recommended

to make appropriate enquiries and take appropriate advice before sending money, incurring any expense or entering into a binding commitment in relation to an advertisement.

The Economist Newspaper Limited shall not be liable to any person for loss or damage incurred or suffered as a result of his/her accepting or offering to accept an invitation contained in any advertisement published in *The Economist*.*



Experience life in *five* European dream homes, *effortlessly*.

Discover a new way to enjoy second homes, without the hassle of traditional ownership.

augustcollections.com

@august.collections





Explora
JOURNEYS

*Maybe the best hotel
isn't a hotel.*

EXPLORE THE OCEAN STATE OF MIND



ALL OCEANFRONT SUITES | IMMERSIVE ITINERARIES | AWARD-WINNING DINING
HOLISTIC WELLNESS | REFINED EUROPEAN DESIGN

EXPLORAJOURNEYS.COM

Contact your preferred travel advisor or call +41 43 5086478

